Physiological State and Executive Function in ADHD

A framework for measuring attention and daily function across interventions

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Full integrated research edition
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Complete illustrated manuscript, evidence registers and proposed research methods

A targeted narrative research perspective. No original participant dataset or completed intervention is reported. The proposed studies are unregistered, and the paper has not been independently peer reviewed.

The numbered bibliography and U evidence identifiers retain their original meanings. Figures retain their original numbering. Additional gap, methods and correction sources use G, M and N identifiers.

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Main argument

Abstract

Attention and daily functioning can vary from one day to the next in the same person with attention-deficit/hyperactivity disorder (ADHD). I propose measuring four things separately: how much room the person has before missing a task requirement, how strongly their body responds to a change, how much cognition shifts for a comparable bodily response, and how long performance takes to recover.

The first application is a 14-day feasibility pilot for a proposed 28-day observational study in adults. The primary analysis asks whether sleeping one hour less than usual is associated with a larger absolute increase in next-day missed targets in adults with ADHD than in comparison adults. An optional secondary analysis asks whether accounting for time to spare improves prediction of missed deadlines on a recurring task the participant already does.

I keep attention, perceived effort and everyday task completion separate. A later intervention study could track all three alongside the treatment target and the burden of treatment. This paper describes the framework and the proposed study and reports no original participant data.

Keywords: ADHD, state regulation, within-person measurement, sleep, attention, executive function, daily functioning.

The idea in plain language

Here is a made-up example. Two people owe a short work check-in at noon. Their attention drops by the same amount, but one has five minutes to spare and the other has thirty. One could miss the deadline while the other still finishes on time. Figure 1 puts numbers on the same idea.

I call that room to spare functional margin. Moving the deadline gives someone more room. So can better attention or a task that takes less effort. A timestamp tells us when the check-in arrived, while those explanations remain open.

The main sleep study uses a brief attention test to count missed targets. In a separate, optional part, a participant could choose a task they already do, such as submitting a regular class check-in. We would record when it was due and whether they submitted it on time.

I'd start with sleep because it varies from night to night. We can compare each person's shorter nights with their usual sleep and test their attention the next morning.

Recovery is a separate question. To measure it, I would follow performance over time and see whether it returns toward the person's usual range.

Medication timing and the demands of the day may also explain why the same person performs differently on two mornings.

Reading the evidence

The main argument develops the hypothesis and explains the first study. The evidence modules trace the wider research. The appendices give the working definitions, proposed methods and source checks.

The hypothesis and the first study

Sleep and direct ADHD evidence

Supplement · histamine and immunity

Supplement · hormones and endometriosis

Supplement · migraine and autonomic evidence

Supplement · GLP-1 and other treatment evidence

ADHD medication and cognitive outcomes

Sleep study and interpretation rules

Supplement · mental health, gut and environment

What the study results would justify

Glossary and quantitative evidence

Supplement · four-intervention comparison

Pilot decision rules

Source audit

How the evidence is weighed

An intervention in people with ADHD addresses a different question from an association in ADHD, an experiment in another human population, or a mechanism isolated in mice. A finding can justify a test without confirming the whole framework. The recovery example, Figures 1, 2, 7, 13, 30, 31 and 40, and Visual guides A, C and D are hypothetical or conceptual. The remaining charts show attributed study results or explicitly labeled calculations; some describe recruitment or follow-up rather than health outcomes.

Literature approach and limits

I looked for studies that could help explain changes in cognition or suggest a test that could change my mind. I sought negative findings as well as positive ones. The search and checks for the illustrated edition ran through September 6, 2026. Targeted evidence and methods reviews continued through November 8, 2026. This narrative review covers selected diagnoses, exposures, peptides and treatment indications.

I gave priority to primary reports and original evidence syntheses. Where I could get them, I checked full texts and tables as well as original abstracts. Some methods and supplements remain inaccessible. The charts label the sample behind each estimate, distinguish clinical outcomes from biomarkers and identify

confidence or Bayesian credible intervals. They show published results or labeled calculations. The review used targeted source checks, with exhaustive screening, independent dual review, formal risk-of-bias assessment and GRADE outside its scope. It reports no original participant data or new meta-analysis.

How to read the charts

A dot marks an estimate; a line shows its uncertainty. The caption gives the population, comparison and outcome. A line crossing the no-difference value leaves more than one interpretation open. It does not prove that two treatments are equivalent. Check each axis: a lower score is better in some figures and worse in others. In Figure 16, waffle grids show response proportions per 100, not actual participant counts; the printed intervals describe uncertainty in the treatment differences.

Most intervals are 95% confidence intervals. The sleep-surgery figure also labels adjusted 97% intervals. A percentage, a percentage-point difference, a standardized score and a risk ratio answer different questions; they cannot be combined into one brain-health score.

The 42 numbered figures include published results, calculations from reported counts and explanatory diagrams. Figure numbers are retained across this revision. Figures in the supplement can be found by number as well as by topic.

1. Why attention varies from day to day

People with ADHD vary considerably in executive ability. A group average can hide that spread, and executive deficits are neither necessary nor sufficient to explain every case. Here I'm interested in what happens to a particular person doing a particular task under different conditions. [1]

Earlier work already asked how the same person's performance can vary. Sergeant's cognitive-energetic model placed arousal, activation and effort alongside information processing and executive management. Intensive everyday assessment has since measured cognitive dynamics directly: one study assessed 122 young adults five times daily for five days, finding that some inhibitory-control dynamics related to emotional difficulties where a baseline assessment did not. [2, 3, 4]

I want to know how much the body contributes to those changes. Sleep, hormones, pain, autonomic regulation and immune activity could each affect attention, and their importance may vary from person to person. The test is whether measuring a specific change helps explain how someone's performance varies after we account for simpler explanations.

It's easy to explain a difficult day in hindsight. Can we predict how someone will do on a task from a bodily change we can measure? If we can, a second study can ask what drove that relationship. A neural, immune, vascular or metabolic explanation would each need its own evidence.

1.1 Developmental risk and changing state

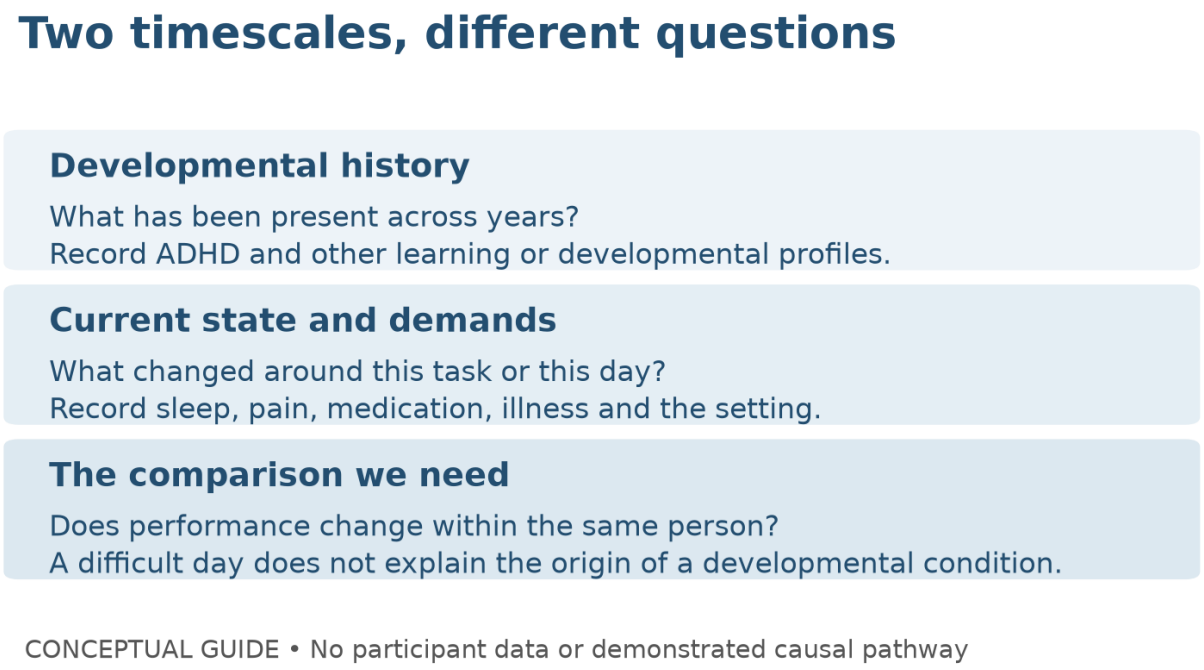
Day-to-day changes take place against a developmental history. A large genetic study compared 38,691 people with ADHD with 186,843 controls and identified 27 associated genomic regions. The results point to many small genetic contributions, with implicated genes enriched for expression during early brain development. I turn to repeated measurements within a person for a different question: why might a particular afternoon be harder than usual? A change in bodily state is one possibility. [150]

Symptoms also change over longer periods. In the Multimodal Treatment Study of ADHD, 558 participants with childhood ADHD were assessed eight times across 16 years. About 63.8% followed a fluctuating course; 9.1% met the study's definition of recovery, meaning sustained full remission through the end of follow-up. This is evidence about changes over years. It does not establish day-to-day fluctuation or identify a physiological cause. It also comes from the same research programme as the MTA evidence already discussed here, so it should not be counted as an independent replication. [151]

The date of diagnosis can come long after difficulties began. A repeated-assessment study followed 239 comparison participants without childhood ADHD. Many apparent late-onset cases identified by symptom checklists were excluded after closer review of impairment, timing, other psychiatric symptoms and substance use. That finding does not invalidate adult diagnosis or settle every case of late onset. For the proposed studies, I would distinguish longstanding difficulties recognized late from a newly acquired cognitive problem, and record how that distinction was assessed. [152]

A developmental history takes years to unfold. The first study here asks what changes from one day to the next.

Developmental history shapes where someone starts. Autism and dyslexia can also change what a task demands of the person doing it. Section 16.9 returns to those differences.



Visual guide A. Developmental history and current state answer different questions. This is a conceptual reading aid for Chapter 1, not a model estimated from participant data.

2. Four things to measure on a difficult day

I use Neurophysiological Resilience Hypothesis as a working name for several explanations of what can feel like the same kind of difficult day.

2.1 Starting point, response and recovery

Someone may start close to a task's limit. Their body may react more strongly to a change, or their thinking may shift more for a bodily response of the same size. They may also take longer to recover. Each possibility would leave a different pattern in the measurements.

Table 1. Four proposed quantities and the observations needed to distinguish them.

Proposed quantity	Plain-language question	What a study must distinguish
Functional executive margin	How much room is there before this task becomes unmanageable?	An equal absolute performance loss causes different functional consequences because starting margins differ.
Physiological reactivity	How strongly does the body respond?	The same challenge produces a larger measured bodily change.
Cognitive sensitivity	How much does thinking change for a comparable bodily shift?	The cognitive response is larger even when the measured physiological change is similar.
Recovery	How long does it take to return?	Cognitive recovery differs beyond what persistent physiology, medication, practice and context explain.

I use functional margin as a provisional measure for a particular task, with a requirement set independently of the group result. IQ and cognitive reserve in aging research are different constructs. If all we have is a baseline score or an arbitrary cutoff, report performance and everyday functioning separately. That gives us a clearer account of what was measured. [5]

Starting close to a deadline can make the same decline more costly. A larger decline would need a different explanation. When I call two bodily responses comparable, I mean they look similar on the measures we took.

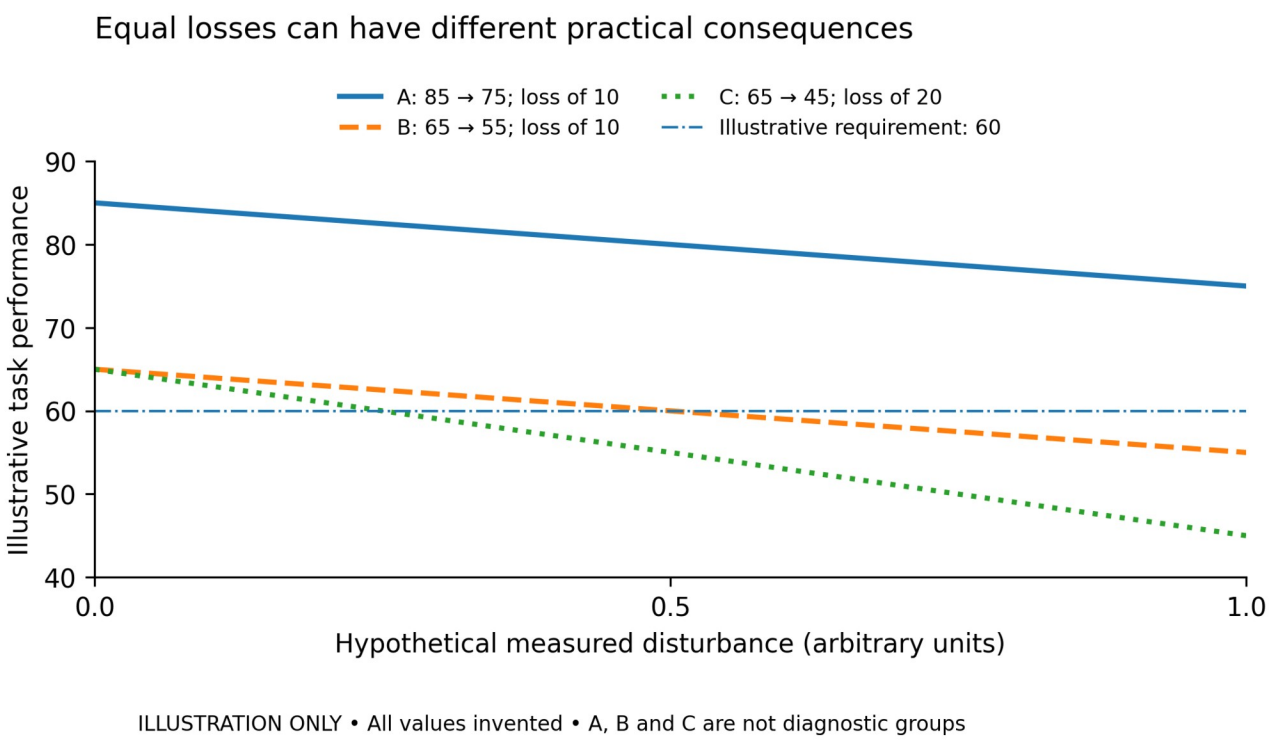


Figure 1. Margin and sensitivity are not the same claim. Hypothetical values, not research data. A and B each lose 10 points, but only B crosses the illustrative requirement of 60. C starts at B's value and loses 20: that illustrates greater sensitivity, not margin alone. No curve represents a diagnostic group or an IQ score.

Worked example 1 · A task requirement, not a ranking of people

In Figure 1, the invented requirement is fixed before comparing the curves. In a real study, it would need an independent functional anchor. Moving the cutoff afterwards to produce an ADHD difference would not test margin. A flexible deadline or external support could also change the practical requirement without changing the person's underlying task score.

2.2 Different outcomes deserve different measurements

When someone describes brain fog, they are reporting a loss of mental clarity. Sleepiness is the tendency to fall asleep. Fatigue is exhaustion or the cost someone feels as they keep going. A missed target is a lapse we can define before testing. Everyday impairment depends on what the person needs to do in their own setting. These are working definitions for measurement. A diagnosis takes a separate assessment.

Someone can do well on a brief task and still find it exhausting to think. A slow response likewise leaves open how clear-headed the person felt. Cognitive disengagement syndrome gives researchers a more specific construct to study within that range of experiences. [6, 7]

The proposed measurements also separate capacity, activation and sustainability. Capacity is performance under standardized conditions once someone is engaged. Activation is starting or choosing to invest effort at a given expected reward. Sustainability is how performance changes during continued demand and how quickly it recovers afterward. Candidate measures include accuracy and speed, initiation latency or effort

choice, and performance change during demand and recovery. The feasibility pilot checks the attention task. Separate activation and recovery tests need additional validation. [21, G22]

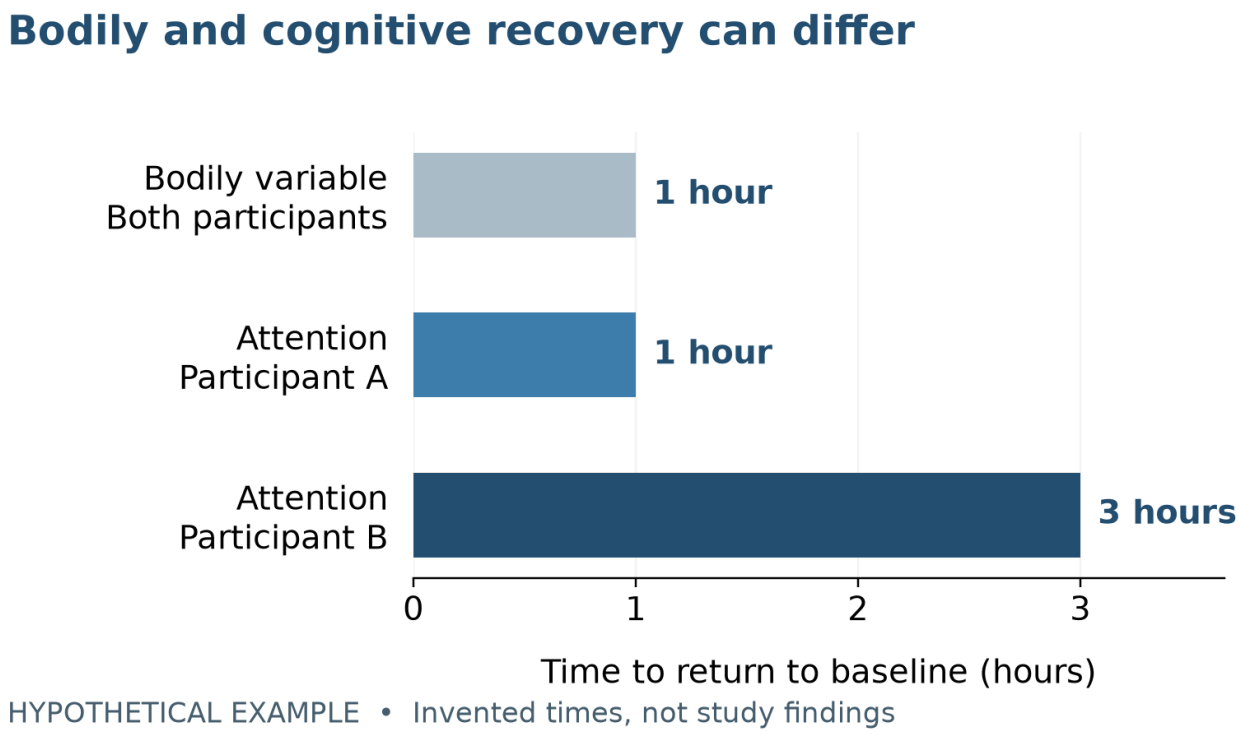
The model also needs context. In the MTA follow-up, remission periods were associated with greater environmental demands. Better functioning may have enabled greater responsibility; the study did not show that adding stress treats ADHD. It nevertheless cautions against treating structure, interest, urgency and reward as if they were simply more physiological burden. [8]

From proposed explanations to observations

Explanation or question	Observation	Role in the first study
Functional margin	Time available minus typical completion time. Observe the real deadline outcome separately.	Optional handoff feasibility and secondary prediction.
Bodily reactivity	Change in physiology after a specified challenge.	Sleep duration is an exposure. A physiological challenge requires a separate design.
Cognitive sensitivity	Change in task performance for a comparable measured physiological change.	Estimate a sleep–attention association first. A matched physiological response is a later question.
Recovery	Repeated performance after a defined disruption.	Requires a dedicated follow-up schedule.
Cost of performance	Accuracy or speed for capacity, initiation or effort choice for activation, and performance over continued demand for sustainability.	Record attention and perceived effort separately. Dedicated activation and endurance tests require validation.
Everyday outcome and burden	Independent task completion, adverse effects and the effort or inconvenience of participation.	Record completion where eligible and track missing sessions separately from adverse effects.

Worked example 2 · Recovery requires more than one follow-up

Suppose two hypothetical participants show the same initial change. A measured bodily variable returns to baseline for both after one hour, but attention returns after one hour in one participant and three hours in the other. That raises a recovery question. It does not yet show an ADHD-specific delay: unmeasured physiology, treatment timing and repeated-task fatigue remain alternatives. The times are invented to illustrate the design.



Worked example 2 chart. Both hypothetical participants have the same one-hour return of the measured bodily variable. Attention returns after one hour in A and three hours in B. Bars show only the invented return times; they do not show the depth or shape of either response.

The two-hour gap tells us when to look. Repeated measures of physiology, attention and context would help tell delayed recovery from something that is still affecting the person. An ADHD-specific explanation would need its own test.

2.3 The first study: sleep and next-day attention

A controlled adolescent sleep study gave me a starting point. [9] Chapter 3 reviews the evidence. The proposed study observes adults under usual care and measures missed targets. Appendix L gives the methods.

Compare one hour less sleep with usual sleep. I predict a larger increase in missed targets in the ADHD group. For example, 2% to 4% and 10% to 12% are both two-point increases. Their odds ratios differ, but the absolute contrast is zero. Testing sensitivity to comparable physiology would take a different design.

2.4 What this framework adds

Sergeant's cognitive-energetic model links arousal, activation and effort to performance. [2, 3] I ask what happens when that performance falls close to a real task requirement. In Figure 1, the same drop can have different consequences.

The sleep study measures changes in missed targets. The optional task asks whether estimated time to spare helps predict missed deadlines, using an existing task and its submission record. Both models get the same information. I would use the simpler model if the comparison rules out a worthwhile gain from adding the margin term. See Section 5.3.

3. What the direct ADHD evidence tells us

3.1 Symptoms and objective performance

A crossover study enrolled 72 adolescents with ADHD; 48 adherent participants with complete actigraphy contributed to the main analyses. The protocols allowed 6.5 versus 9.5 hours in bed, producing an average 1.6-hour difference in measured sleep. Parent-rated inattention worsened during restriction (Vanderbilt scale, d = 0.60), whereas adolescent-rated inattention and continuous-performance testing did not show the same effect. Participants stopped medication for the study. This supports effects on specified symptom outcomes, not universal executive decline or an ADHD-versus-control interaction. [9]

A smaller study tested 16 young men with ADHD and 18 controls before and after 25 hours awake. Both groups deteriorated and were described as similarly affected. The study did not establish statistical equivalence or independently validate functional thresholds. Its result is compatible with an additive effect from different baselines; it does not prove the margin model. [10]

A newer experiment adds a more specific positive interaction: 19 young men with ADHD and 14 controls completed an emotional-face task before and after 25 hours awake. Target-face omissions showed a time-by-group interaction ($\chi^2 = 4.59$, $p = .032$), while several shape and later EEG outcomes did not. The small male sample and multiple endpoints require replication. This qualifies the earlier continuous-performance result: sensitivity may depend on the task and outcome, rather than being either absent everywhere or universally greater in ADHD. [75]

Sleep recordings give us information beyond total hours. One study enrolled 62 medication-free adolescents—31 with ADHD and 31 controls—with usable ambulatory polysomnography in 56. In those recordings, sleep EEG features related to neurocognition despite no detected group differences in sleep duration, awakenings or latency. That supports measuring sleep physiology as well as duration, while leaving causality unresolved. [11]

In the sleep studies, ADHD ratings can improve while a cognitive task shows little change. Results also depend on who supplies the rating. Advancing the internal clock can leave cognition unchanged. I would measure whether sleep changed, whether symptoms improved and whether daily life got easier as separate outcomes. [9, 12, 19, 20, 65] Treatment for mild sleep-disordered breathing also improved some symptoms without a clear gain on the primary cognitive outcomes. [92]

3.2 When sleep treatment changes symptoms or performance

Hiscock and colleagues randomized 244 children with ADHD and sleep problems to a brief behavioural sleep intervention or usual care. The adjusted parent-rated symptom differences were −2.9 points (95% CI −5.5 to −0.3) at three months and −3.7 (−6.1 to −1.2) at six months, using multiple imputation. Teacher-rated total ADHD scores did not show the same benefit. Most children were already taking stimulants; families were not blinded. This tests an adjunct in a selected clinical population, not the claim that sleep explains ADHD. [12]

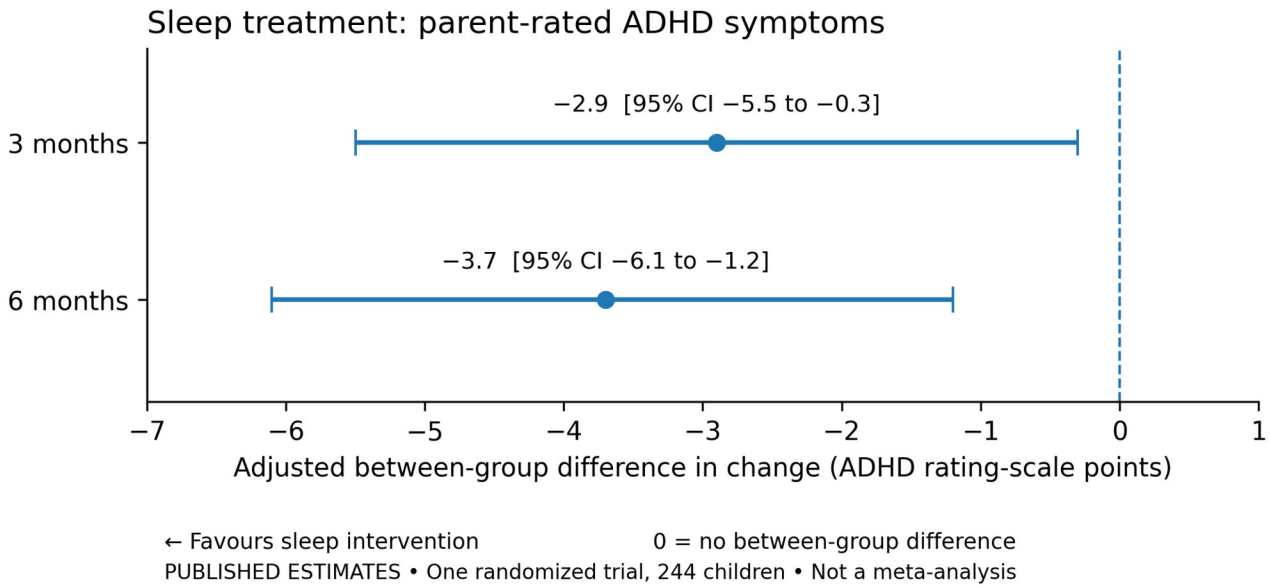


Figure 3. A defined intervention can help one outcome without improving every measure. Adjusted differences in parent-rated ADHD symptom change, with reported 95% confidence intervals, from Hiscock et al. The two follow-ups belong to the same randomized trial; they are not pooled independent studies. Estimates use the authors' imputed dataset. Families were not blinded, and teacher-rated total ADHD outcomes did not show the same benefit. [12]

Sleep-disordered breathing is another useful test. PATS randomized 459 children with habitual snoring and mild obstruction to surgery or watchful waiting; 458 entered analysis. At 12 months, neither primary executive-function nor attention outcome showed a clear between-group benefit. Sleepiness, symptoms and some health outcomes improved. Six children had a surgery-associated serious adverse event. This was not an ADHD trial. Improved sleep-related health cannot stand in for a measured cognitive gain. [92]

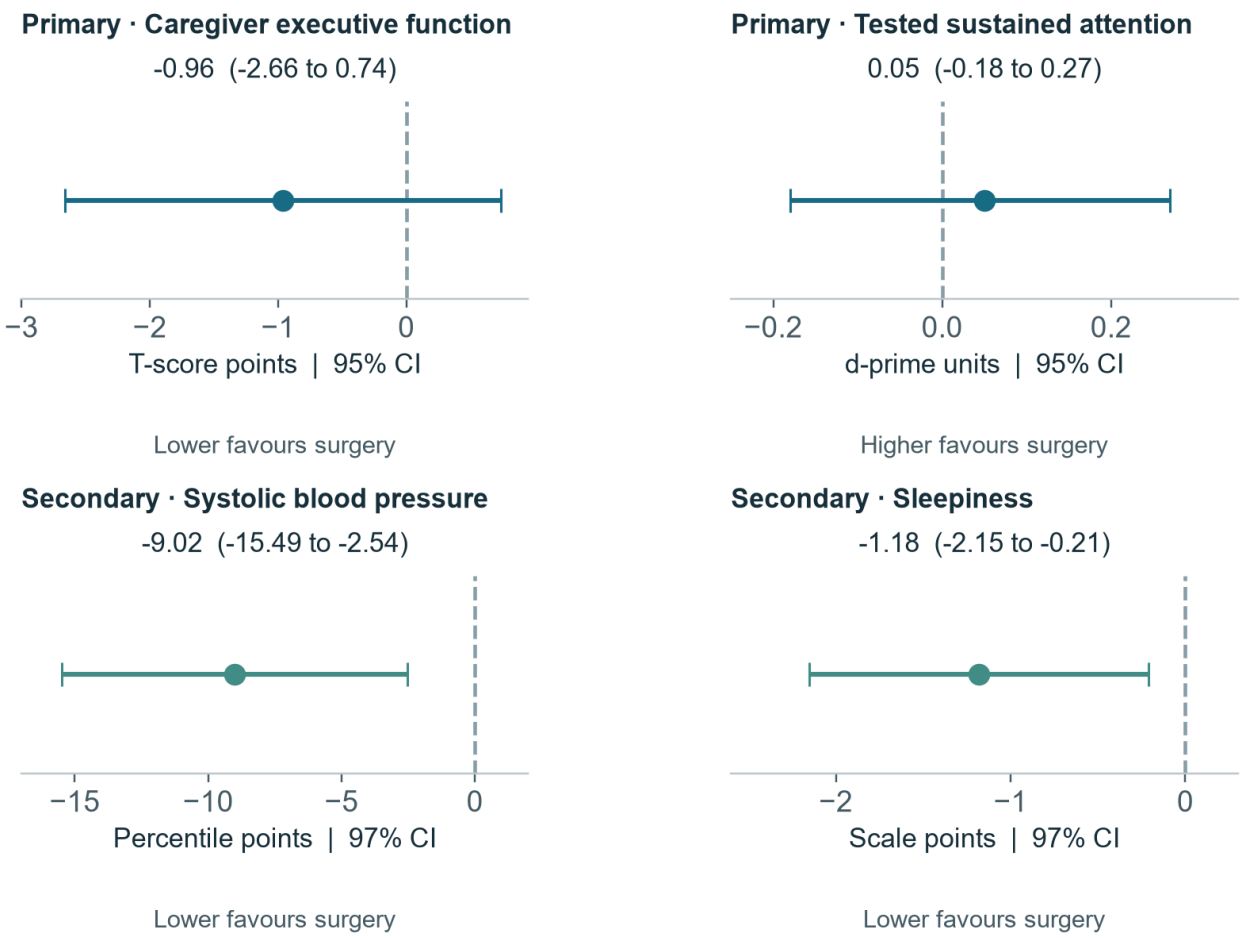


Figure 4. Sleep-related improvements can coexist with inconclusive cognitive outcomes. PATS, surgery minus watchful waiting: separate axes retain each outcome's units. Primary outcomes use 95% CIs; secondary outcomes use multiplicity-adjusted 97% CIs. Blood pressure is in percentile points, not mm Hg. Effect directions are labeled. These outcomes do not establish mediation or recommend surgery for ADHD. [92]

A preliminary open-label trial studied 12 weeks of treatment in adults with ADHD and a positive sleep-disorder screen. Adding sleep treatment to usual ADHD care did not significantly improve the primary self-reported ADHD outcome versus usual care alone (β −1.30, 95% CI −5.57 to 2.96). Secondary self-reports favored the add-on for sleep quality and fatigue. Performance-based outcomes were secondary. The abstract reports 70 randomized, but its three arm counts sum to 80. The primary comparison is interpretable as reported, while enrollment counts need confirmation. [187]

These studies support keeping symptom ratings, sleep outcomes and task performance separate. The first study therefore measures missed targets directly and records sleepiness and fatigue alongside them.

3.3 Vigilance instability and task performance

A poor average score can come from brief lapses in an otherwise steady performance. Doran and colleagues described sleep-deprivation state instability in 2001. Later, Andrillon and colleagues associated waking slow waves with attention lapses, mind wandering and mind blanking in healthy adults. These are antecedents, not ADHD-specific diagnostic findings. [13, 14]

A 2026 waking-EEG study compared 32 adults with ADHD and 31 controls. Sleep-like slow-wave density was associated with omissions, slower and more variable responses, and sleepiness; cross-sectional mediation linked these signals to group differences. A pediatric study reported presentation- and region-dependent findings. The final adult report's complete methods remain an access gap; neither study establishes a diagnostic marker or causal pathway. [15, 16]

Some difficult periods may involve more frequent lapses in attention. We would need to see whether a bodily change precedes those lapses and whether they make an everyday task harder to finish. The local-sleep findings give us a question to test about ADHD.

3.4 Timing, demand and effort

Among 43 evening-type students, including 19 with elevated ADHD symptoms, morning–evening testing produced a reaction-time interaction but not the same working-memory result. Morning always came first, leaving practice and order as alternatives. A larger study of 302 adolescents found that ADHD status did not strengthen associations between eveningness and poorer functioning. The designs differ, but together they argue against universal circadian hypersensitivity. [17, 18]

The PhASE trial randomized 51 adults with ADHD and delayed sleep phase to sleep education plus placebo, melatonin, or melatonin with morning bright light. In the late-onset subgroup (n = 36), melatonin and melatonin plus bright light advanced dim-light melatonin onset by 88 and 118 minutes versus placebo. The 14% symptom reduction with melatonin alone was nonsignificant in that subgroup ($p = .062$). Including participants with earlier onset yielded a 12% reduction ($p = .038$). The analysis did not detect mediation by the clock shift, and scores returned to baseline two weeks after treatment. [19]

An exploratory sleep analysis of the same trial, including 49 adults, found no corresponding advance in sleep timing or general sleep improvement. This is a second analysis of the same participants. Target engagement, behaviour and symptoms still need separate outcomes. [20]

A separate double-blind trial in 105 medication-free children with ADHD and sleep-onset insomnia makes the distinction especially clear. After four weeks, sleep onset advanced by 26.9 minutes with melatonin but became 10.5 minutes later with placebo. Sleep duration also improved; behaviour, cognitive performance and quality of life did not show significant benefit. Better sleep is worthwhile, but its measurement cannot stand in for demonstrated cognitive improvement. [65]

A 2026 study monitored 54 adults with ADHD and 47 controls with actigraphy for ten days. Activity rhythms were less stable and later in the ADHD group. Several sleep measures initially correlated with cognitive performance, but none of the cognitive correlations survived correction for multiple comparisons. The groups also differed substantially in age despite adjustment. The study gives us a reason to measure rhythms more carefully, while leaving their contribution to cognitive difficulty uncertain. [195]

How much effort someone is willing to invest is a separate question from how well they perform. In a study of 20 adults with ADHD and 24 comparison participants, amphetamine medication increased willingness to invest cognitive and physical effort in the ADHD group. This identifies an experimentally tractable process; it does not show that immune activity caused the difficulty. [21]

The effort result gives the study a reason to record perceived effort separately from task performance. An accurate response may still be more demanding on a difficult day.

3.5 Measuring attention as it varies

Paediatric pupillometry distinguished between-trial variance from within-trial fluctuations. The unmedicated ADHD group had greater between-trial variability in an adjusted model, while its participant-level comparison narrowly missed the FDR threshold; within-trial findings differed by analysis. Among 17 participants tested on and off methylphenidate, between-trial variability decreased, but accuracy did not significantly improve. Task-locked relative pupil responses are not a specific measure of tonic arousal, noradrenergic activity or inflammation [U01].

In a small adult imaging study, frontostriatal functional differences were more apparent during working-memory demand than at rest, while structural differences were less robust. Task engagement should therefore be specified when measuring a proposed state effect [U03].

3.6 Which aspects of sleep matter?

The primary analysis uses nightly duration. Sleep opportunity, fragmentation, intermittent hypoxia and circadian mismatch are distinct exposures. The daily study should record intended bedtime, sleep onset, waking, naps, light exposure, work demands and medication timing. A clinical subset can assess respiratory arousals, hypoxic burden and position-specific apnea with sleep-medicine instruments. The primary sleep-duration interaction and the secondary added-prediction question are specified in Section 5.2 and Appendix L.

4. ADHD medication and everyday function

Medication trials give us the closest comparison in ADHD. A person can feel better or work faster while learning about as much as before. The timing of a dose can also matter when we test attention at different points in the day.

The study records what people take, when they take it and any measured physiological changes. That gives us a way to check whether a difficult day coincided with a change in treatment. [76, 77]

Medication has the strongest direct ADHD treatment evidence considered here. Short-term symptom benefit is established for several agents; cognitive tasks, learning, sleep and cardiovascular outcomes do not all move together. Medications for overlapping conditions can also affect thinking. Any state-based explanation needs treatment timing and comedication in the model. [76, 77, 80, 82, 83]

4.1 Benefit depends on what we measure

A recent crossover experiment makes the outcome distinction concrete. The main analysis included 57 adults with ADHD and 60 controls. In participants with ADHD, methylphenidate reduced response-time variability on a cost–benefit task by 0.03 seconds (95% credible interval −0.05 to −0.01 for treatment minus placebo). The estimated accuracy gain was 1 percentage point, with an interval from −1 to 3. These are Bayesian credible intervals. Participants with ADHD were already prescribed methylphenidate; this acute task experiment does not establish durable improvement in learning or everyday function. [196]

A 2018 network meta-analysis included 133 double-blind trials. At roughly 12 weeks, clinician-rated symptoms improved with amphetamines, methylphenidate and atomoxetine in both age groups, with different average estimates in children and adults. A newer adult synthesis, spanning 113 trials across several treatment types, also supported short-term symptom benefit for stimulants and atomoxetine. Table C5 gives the estimates, populations and uncertainty. These reviews share some underlying trials; they are not independent replications. [76, 77]

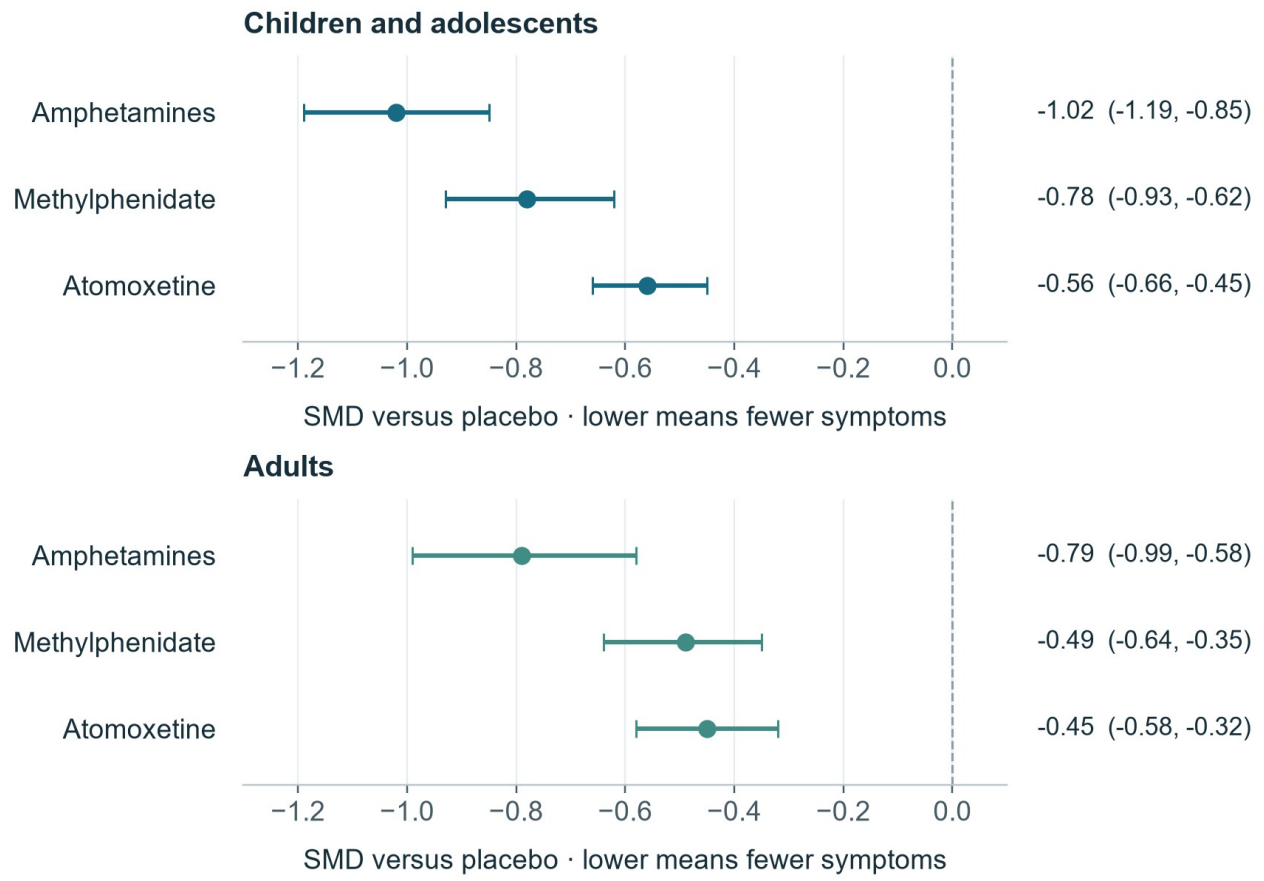


Figure 22. Direct ADHD symptom evidence provides a treatment benchmark. Selected clinician-rated placebo comparisons from Cortese et al., closest to 12 weeks; dots are standardized mean differences and bars are 95% CIs. Network estimates include indirect comparisons. Separate age panels are not a randomized test of age differences. These estimates neither measure learning nor rank treatment for an individual. [76]

Symptom relief, better task performance and learning are related, but separate. A metaanalysis of 31 placebo-controlled methylphenidate studies in 804 children found cognitive effects ranging from d = 0.20 to 0.73 across domains. Dose effects were not significant for executive memory, inhibition or flexibility; high-dose data were unavailable for executive memory and flexibility. A 2026 dose-response synthesis likewise found benefit plateaus for some drugs and outcomes, alongside increasing adverse-event withdrawals in some comparisons. Neither supplies a universal optimal dose for an individual. [78, 79]

Worked example 4. More work completed is not the same measurement as more learned

In a triple-masked crossover study of 173 children, extended-release (OROS) methylphenidate produced 37% more arithmetic work per minute and 53% fewer classroom rule violations than placebo. Yet it did not improve the amount of new science, social-studies or vocabulary material learned during the instructional periods. This does not erase the classroom benefit. It tells us that a child may need both effective symptom treatment and teaching that supports learning. The result comes from a structured summer programme, not every classroom or age group. [80]

That is why the optional task uses an existing deadline and records how much time was available. The outcome comes from the submission record itself.

4.2 Treatment can change more than attention

Sleep is one practical example. Across nine randomized studies with objective sleep measures in 246 young people, stimulants were associated with longer sleep latency, lower sleep efficiency and shorter total sleep time. Average results do not describe every person: better daytime regulation may help an individual routine even while a particular dose or timing disrupts sleep. Record the observed pattern rather than assume that either benefit or disruption must occur. [82]

4.3 How medication fits the proposed model

If someone usually takes medication, their treated performance is the baseline for the study. Record formulation, prescribed dose, actual dose time, missed doses and co-medications, alongside sleep, food intake, symptoms and the task. A late-day decline might involve waning drug effect, accumulated demand, poor sleep or pain. Calling it “inflammation” before comparing these explanations would make the theory less useful.

If a medication helps, I would ask what improved and whether a bodily change could account for it. A symptom improvement can stand on its own. Claims about a particular neurotransmitter or a route proposed in this paper would need evidence beyond a response or side effect.

People may change when they take a dose because their symptoms are worse, and that dose may in turn affect performance. The study keeps usual treatment in place. We’ll record actual dose times, identify stable treatment periods and decide in advance how to analyse changes.

5. The first application: sleep and next-day attention

The earlier ADHD studies give me two reasons to measure attention repeatedly. Symptoms and task scores can move differently, and performance can depend on when medication was taken. In a 28-day adolescent study, vigilance varied with how the previous night compared with each person's usual sleep. I want to know whether that pattern differs by ADHD status in adults. [173]

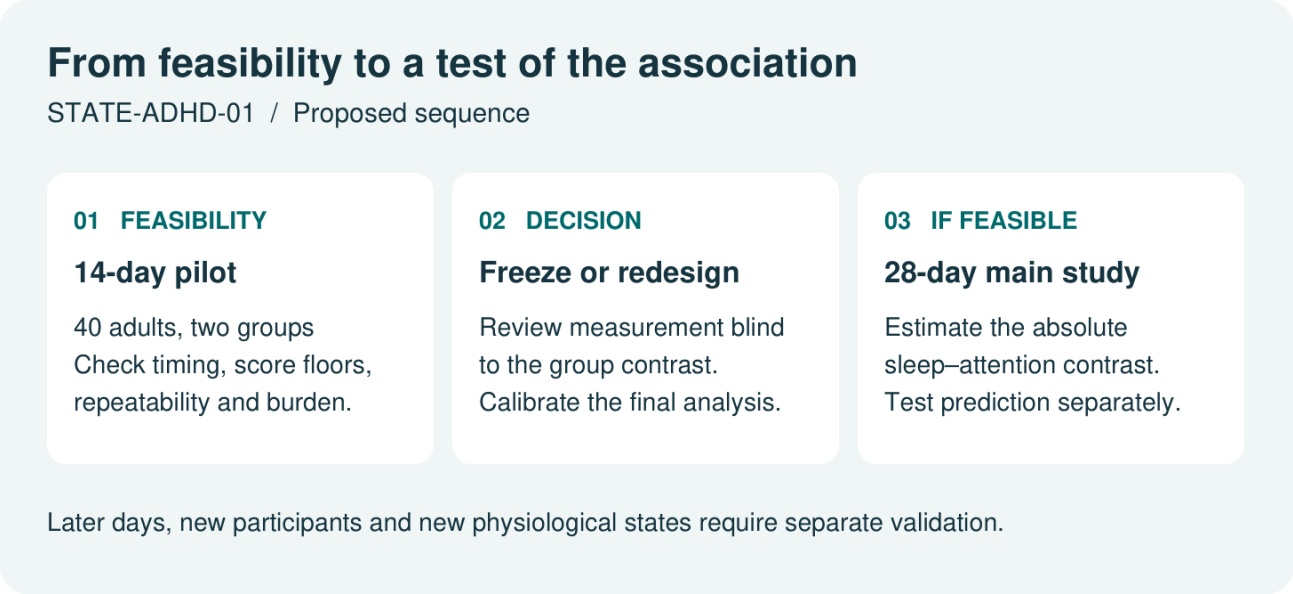


Figure 30. A proposed 14-day feasibility pilot precedes a conditional 28-day main study. The endpoint and analysis are frozen after a blinded measurement review and statistical calibration. Evaluation on later days, in new participants and in another physiological state answers different questions.

Who this comparison represents

Both groups continue their usual care. One group has clinician-confirmed ADHD. The comparison group has no known ADHD and clears screening. We will recruit through overlapping channels where possible. Each person serves as their own reference for sleep. Even so, the group comparison depends on who enrolls, including differences in treatment, sleep disorders and work schedules. Appendix L describes recruitment and the check among people with comparable sleep and testing schedules.

5.1 What the 14-day pilot must establish

The pilot would follow 40 adults for 14 days after two practice sessions. Half would have ADHD. We need to see whether the devices time the task accurately, whether enough sessions are valid, how often scores hit zero, and whether sleep varies enough to compare shorter and usual nights. We would also check whether participants can record the optional task. Appendix L gives the schedule and the rules for deciding whether to proceed.

Wrist actigraphy and a sleep diary would estimate sleep duration and timing. Morning tests would record attention alongside sleepiness, fog, fatigue and pain or headache. Record clock time, time since waking, caffeine, actual medication timing, missed doses and co-medications so changes in sleep can be interpreted in context. The optional handoff supplies an independent measure of daily function. Work with participants to choose a useful outcome and a manageable testing burden. Record sensory access needs and relevant health history at baseline, with clinical follow-up where indicated.

The feasibility pilot tests whether the task records enough valid opportunities and whether its scores capture day-to-day change. If it qualifies, the 60-target omission task supplies the primary outcome for the main study. Keep commission errors, response variability, fog and daily function separate. The pilot endpoint panel will review measurement quality without seeing the group-by-sleep contrast.

If the missed-target task fails the feasibility checks, the panel would review the brief vigilance test. Before switching, we would define its outcome and analysis for the main study.

5.2 The primary sleep–attention association

The feasibility pilot described in Section 5.1 would inform the proposed 28-day adult study. The main study's planning target of 200 remains provisional. Appendix L specifies the validation steps, endpoint decision and simulations needed before finalizing the design.

For each group, estimate how much the probability of a missed target changes after one hour less sleep than usual. Subtract the comparison group's change from the ADHD group's change. The proposed threshold for a meaningful difference is one additional miss per 60 valid targets. Appendix L specifies the model and how to interpret its 95% interval.

One miss is easy to picture on this task. If shorter sleep were associated with two additional misses per 60 in the ADHD group and one in the comparison group, the difference would be one. Participants and reviewers will judge whether that warrants a controlled sleep study. The pilot asks whether we can measure it reliably. The optional task asks a separate question about deadlines.

5.3 The secondary handoff prediction

In this optional part of the study, a participant chooses a recurring task they already submit online, such as a class or work check-in. We record when it becomes available, when it is due and when it is submitted. Those times show whether it was on time and help estimate how much time the participant had to spare. A confirmed missed submission is recorded separately from a missing record.

I want to know whether more morning misses predict more late or missed submissions when time is tight. Both models use the same information: morning attention, past task performance, time available, sleep, medication and context. One model represents time to spare explicitly and tests whether it changes the link between attention and deadlines. We'll compare the models on the same later days, check how closely predictions match outcomes and decide in advance how much improvement would be worth keeping.

If eligible participants rarely miss deadlines or have too little verified completion history, keep the handoff analysis exploratory. The core sleep–attention study can proceed if its own feasibility criteria are met.

Design for the days when participation is hardest. Reduce burden, record why assessments are missed and test whether monitoring itself changes reports. Recruit adults across ages, sexes, treatment regimens and levels of baseline impairment so the eventual result can be interpreted in context.

5.4 How to interpret the two tests

The primary analysis estimates a group difference in the sleep–attention association. The secondary handoff analysis tests prediction of an independently recorded deadline outcome. Each result answers its own question. Comparable physiological challenges and timed follow-up would be needed to distinguish reactivity, cognitive sensitivity and recovery.

Table 7A. How to interpret the sleep and handoff results.

Question	Would support	Would weaken or reverse	Would leave unresolved
Sleep and ADHD	The 95% interval lies wholly above the meaningful positive bound.	An interval wholly inside the negligible range weakens the predicted difference. A meaningful negative result reverses the prediction.	Wide intervals, omission floors, inadequate timing or poor overlap.
Functional margin	More misses predict a larger deadline cost at smaller margins. The margin model improves prediction by a worthwhile amount and is well calibrated.	A precise absence of the pattern weakens the prediction. Similar predictive performance limits the case for the margin model.	Few verified misses, poor pairing, inadequate history or imprecise results.

An imprecise result can leave room for both a negligible difference and one worth following up. We need a prespecified bound and enough precision to tell them apart. If we think a subgroup responds differently, that prediction needs a separate test. [60]

6. Interpretation and clinical boundaries

The two tests in Chapter 5 depend on whether the measures can track change within a person. Section 17.8 describes the separate work needed to validate the framework in new people or states.

If the relationship looks similar in adults with and without ADHD, the pattern could extend beyond ADHD. I would ask the same question in adults with relevant co-occurring conditions. I would also be specific about what changed. Someone might report worse concentration while making about as many errors as usual on the attention task.

A test that separates people can still miss changes within a person. In a cognitive ecological momentary assessment study, 198 adults with type 1 diabetes and 128 community participants completed brief tests three times daily. Between-person reliability was 0.95–0.99, while within-person reliability ranged from 0.20 to 0.80 across tests and samples. These samples overlap the research programme cited in Section 8.2. Read the results as related evidence. The proposed resilience score needs its own reliability study. [194]

6.1 Clinical boundary

This framework supplies no diagnostic criteria for autoimmune brain disease, MCAS, POTS, impaired cerebral perfusion or any other condition. It is not a reason to replace clinical assessment with a wearable score or symptom resemblance. It does not recommend antihistamines, GLP-1 drugs or improvised electrical stimulation for ADHD. An implanted vagal device, a research auricular protocol and a general TENS machine are not interchangeable interventions.

No personal medical records, medication response or wearable trace is analyzed as data. Prescribed treatment should not be withdrawn merely to make an experiment cleaner; any research change requires appropriate clinical and ethics oversight. The same applies to exposure challenges, particularly in participants vulnerable to delayed post-exertional worsening.

The broader review adds no indication for immune suppression, antimicrobial treatment or microbiome intervention for ADHD. It supports attention to diagnosed illness and treatable symptoms while keeping the proposed mechanisms open to disconfirmation. Mental health care, social support and ordinary environmental changes do not require a biomarker explanation to matter.

7. Conclusion: what the results would justify

Return to the made-up noon deadline. The attention lapse is the same for both people, but the consequences differ. The sleep study asks whether missed targets rise more after a shorter night in adults with ADHD than in the comparison group. The optional task asks whether time to spare helps predict a missed deadline.

If the measurements are reliable, the groups have enough sleep overlap and the 95% interval is entirely above the positive bound, I would test the sleep difference under controlled conditions. I would carry the optional task into a later study if it improved deadline prediction by a worthwhile amount and the predicted probabilities matched what happened on later days. Either result could hold on its own.

If the 95% interval falls entirely within the negligible range, I would narrow the claim about sleep and attention for this task. A meaningful result in the opposite direction would reverse the prediction. For deadlines, I would use the simpler model if the comparison rules out a worthwhile gain. Wide intervals, unreliable measurements or too few verified missed submissions leave those questions open and tell us what needs a better measurement.

Evidence modules

Supplement. Mechanisms and evidence beyond the first study

I turned to studies of other conditions to help decide what to measure and when. Some suggest ways to test a mechanism. The first sleep study can proceed while those mechanisms remain open.

Figure 2 maps the questions in this supplement. Chapter 8 collects design precedents and Chapter 9 compares the four intervention areas. Chapters 10–13 review immune, hormonal and recovery evidence. Chapters 14–17 cover treatment outcomes, context and possible follow-up studies.

Figure 2 starts with a bodily change and follows the routes it could take to affect three different outcomes. An immune, vascular or metabolic explanation needs to identify the link it predicts. Finding the same molecule in two conditions leaves open whether it plays the same role in both.

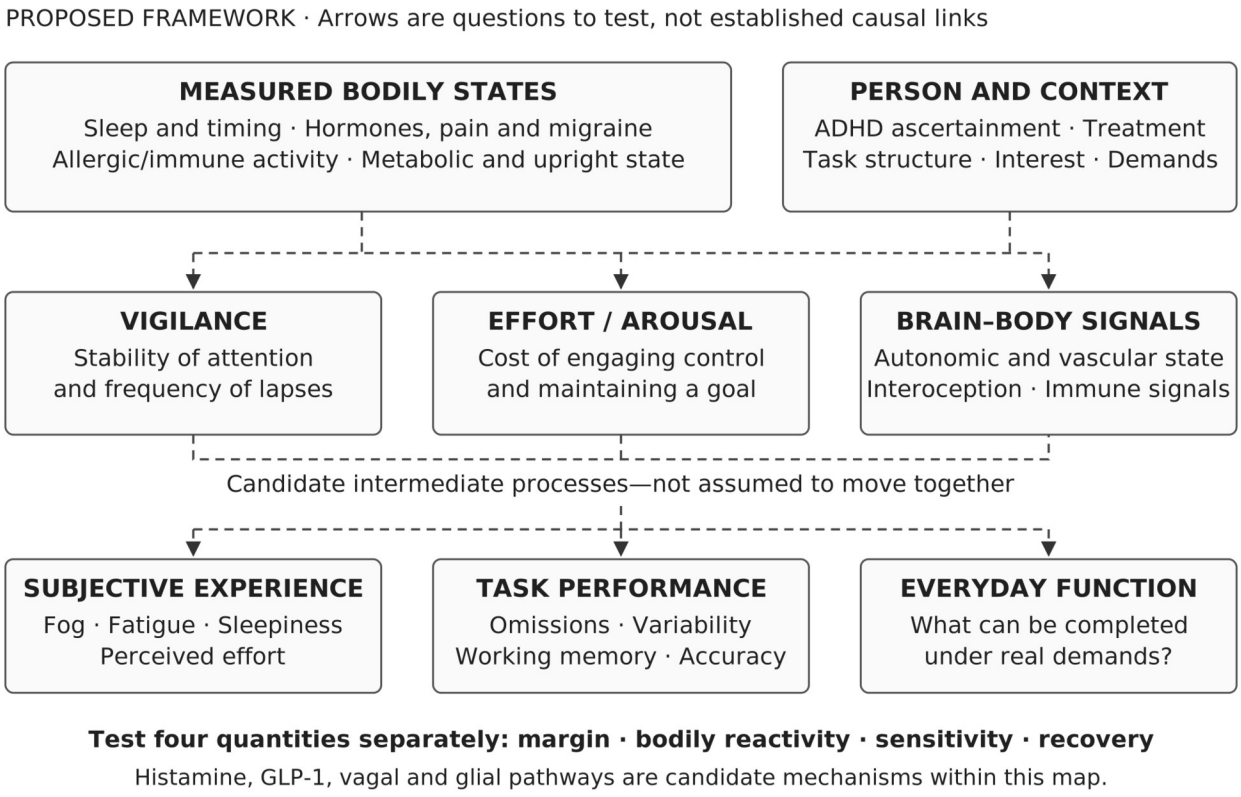


Figure 2. A map of the hypothesis, not a proven pathway. All connecting arrows are candidate relationships. "Person and context" may modify exposures, responses or consequences; the drawing is not a complete causal graph. Histamine, GLP-1, vagal and glial mechanisms belong within particular routes, not above them as universal explanations. The separate outcome boxes prevent subjective fog, test performance and everyday function from being treated as the same result.

The same symptom can follow different changes in the body. For each condition, I look for studies that measured more than one part of the proposed sequence.

For the sleep study, that means measuring attention and everyday performance separately. The work below also suggests how to design experiments that can distinguish specific mechanisms later.

Large datasets link ADHD with endometriosis, migraine and atopic dermatitis. Pain, sleep disruption, immune signalling, shared genetic liability and patterns of healthcare contact could all contribute to those associations. Diagnosis dates tell us when a condition was recorded. [34, 35, 67, 68, 69] In endometriosis, I would check fatigue and iron status before looking for a shared cognitive mechanism. [93, 94, 95]

Small studies report changes in ADHD symptoms across the menstrual cycle. A larger general-population synthesis found little evidence that objective cognition shifts consistently with menstrual phase. That difference makes it worth tracking symptoms and task scores separately in the same person. [32, 33, 66] Studies of postmenopausal estradiol also give us reason to test cognitive effects directly. [96]

Long COVID studies have measured cognitive differences. Smaller studies have examined brain barriers, autonomic responses and brain metabolites. These findings come from different samples. Testing the proposed sequence from immune activity to energy failure to brain fog would require measuring it in the same people. [44, 45, 70, 71]

Implanted vagal stimulation improved an arthritis outcome. Semaglutide reduced cardiovascular events in a selected population, and two large semaglutide trials found no slowing of symptomatic Alzheimer disease. Those outcomes set the limits of what each study tells us about ADHD. [56, 57, 73] Lifestyle and vascular trials in older or hypertensive populations found modest or outcome-specific cognitive effects. [97, 98]

Cognitive difficulty can be measured in PTSD and depression. Immune and infection histories are associated with some later diagnoses, and gut trials have produced a mix of promising and inconclusive results. Chapter 16 asks which of these findings suggest a testable question in ADHD. [99, 100, 104, 109, 110]

Chapter 14 starts with what treatment did for the illness it targeted, then asks what the same studies measured about cognition and how much they can tell us about the mechanism.

Some people have fatigue and cognitive complaints long after mono. We still need to find out whether ADHD is directly linked to mono or whether people with ADHD recover differently. Sections 13.4–13.9 take up those questions. Figures 37, 38, 39 follow fatigue-syndrome outcomes after infection.

The adult sleep, ELAD, HISTORI and Long COVID trials illustrate the range of outcomes at stake, from physical markers and ratings to objective performance and personal goals. Each trial can answer only the questions its outcomes were designed to test. [187, 188, 189, 190, 191, 192]

8. Comparisons that inform later study designs

A study can find a change in one kind of attention and very little in another. I use the comparisons in this chapter to choose the task and the time window for a follow-up.

Developmental history helps explain where a person starts. The comparisons below address a shorter timescale, from a few hours after a challenge to changes across ordinary days.

8.1 Which cognitive outcome changes?

In a counterbalanced exposure study, 26 adults were analyzed after particulate-pollution and clean-air sessions. Four hours after exposure, selective attention and emotion-expression discrimination differed, while working memory and psychomotor vigilance did not show the same effect. This was not an ADHD study; inflammatory mediation was not established, and co-exposures limit attribution. It argues for a prespecified domain and time window, not a general “cognitive health” score. It is not a reason to add a pollution challenge to the proposed ADHD study. [23]

A randomized endotoxin challenge offers a test of differential vulnerability. In a secondary spatial-testing sample of 84 older adults, 26 had insomnia. Endotoxin interacted with insomnia status on the room reconstruction measure of cognitive mapping. The corresponding interaction was not demonstrated for spatial memory, and within the endotoxin group, IL-6 and TNF-α responses showed no significant correlation with either spatial outcome. The small insomnia subgroup and secondary outcomes limit precision. The sample was not selected for ADHD. [G20]

A human immune-challenge experiment also linked inflammatory state to aspects of effort and reward allocation, while several proposed moderation effects were not supported. It did not compare people with and without ADHD. That comparison needs testing before similar motivational symptoms are assigned the same biological cause. [22]

8.2 How can we measure change in daily life?

A 28-day study followed 205 adolescents through two school weeks and two vacation weeks. It paired actigraphy and sleep diaries with morning and afternoon symptom ratings and a daily 3.2-minute vigilance task. Nights with more sleep or better sleep efficiency than a participant’s own average predicted better next-day task performance and less sleepiness and fatigue. The analysis separated these within-person associations from differences between adolescents. This was a convenience sample, not an ADHD study or a randomized sleep intervention; school always preceded vacation. Still, its design is much closer to the daily question in this paper than a single comparison between diagnostic groups. [173]

A separate ambulatory study asked 393 US adults to complete brief smartphone cognitive tests daily for ten days. Better performance was associated with higher reported energy and focus, fewer distractions and some positive mood states. Repeated objective testing in everyday settings helps separate context from test performance, although concurrent self-reports and observational associations leave direction and mechanism open. [G21]

The repeated measurements let us ask whether the day-to-day pattern differs by ADHD status. I would also ask whether a physiological measure adds anything to sleep, time of day and medication timing. To make that claim, the prediction has to work on later days and in people who were left out of model development.

A related design has already been used in type 1 diabetes. Researchers paired continuous glucose monitoring with cognitive tests over 15 days in 200 adults; 190 remained after data cleaning. Larger glucose excursions were associated with slower and less accurate processing speed, while sustained attention did not show the same association. Responses also varied between people. This is close to the kind of within-person test I have in mind. It remains observational evidence in diabetes, not proof of an ADHD mechanism or a reason to change glucose targets. The practical importance and stability of an individual response still need testing. [193]

An experimental glucose clamp gives a more direct, narrower result. In 94 adults with or without diabetes, researchers lowered glucose to about 2.8 mmol/L. Mean alertness reaction time increased by 32.1 milliseconds (SD 66.6), and working-memory errors and omissions increased by 2.0 (SD 5.5). These are standard deviations of the changes, not confidence intervals. The effects were largely similar across diabetes and hypoglycemia-awareness groups. Clinically low glucose can impair cognition; this fixed-sequence experiment does not show that ordinary meal-related fluctuations explain ADHD or that people with ADHD are more sensitive. [210]

Two comparisons guide measurement beyond sleep

A single appointment can miss how attention changes during and after a migraine attack. A later study should record assessment and medication timing and include symptom-free days. [37, G36] Repeated measurements after infection could likewise distinguish a brief drop from a prolonged change. Whether adults with ADHD follow the same course remains open. [70, 161, 172]

8.3 Preliminary findings for later studies

An orexin-2 agonist Phase 1b company announcement reported within-group clinical and translational signals after 14 days, without complete placebo-adjusted efficacy results. It remains preliminary [U05]. Gastrointestinal symptom burden also deserves measurement as a comorbidity, but the available synthesis does not identify a microbiome route to ADHD symptoms [U06].

A multicentre sham-controlled external trigeminal nerve stimulation trial in 150 young people with ADHD found no clear cognitive or autonomic benefit [U04]. A smaller single-blind photobiomodulation study reported acute performance and prefrontal oxygenation signals [U02]. The interventions differ, and neither establishes durable functional benefit or mitochondrial mediation.

8.4 Sleep exposures beyond nightly duration

Sleep opportunity, fragmentation, intermittent hypoxia and circadian mismatch are distinct exposures. A four-year cohort of 1,110 adults linked changes in apnea severity, perivascular diffusion and visual memory [U45]. Its statistical indirect path through DTI-ALPS is observational and depends on the validity of that surrogate. In a laboratory comparison of 84 people with apnea and 22 controls, sleep-stage shifts remained associated with slowed attention in the selected model, while apnea–hypopnea index and hypoxemia measures did not [U46]. This warrants measuring both fragmentation and hypoxic burden.

A secondary analysis of 833 participants in the sham-controlled APPLES trial found that estimated respiratory arousal threshold modified CPAP response on an executive-function outcome, but not attention or memory [U47]. It suggests a moderation question for prospective testing; it does not identify an ADHD-specific respiratory phenotype.

8.5 Longer-term and cardiovascular medication outcomes

Longer-term studies ask what happens outside the symptom scale. A Swedish target-trial emulation followed 148,581 newly diagnosed people under modeled treatment-initiation strategies for two years. Medication initiation was associated with lower rates of several first events, including suicidal behaviour and transport accidents; the first accidental-injury estimate was compatible with no difference. This was an observational comparison, not random treatment assignment. Dispensing does not prove adherence, and residual confounding remains possible. The study does not identify a cognitive or immune mediator. [81]

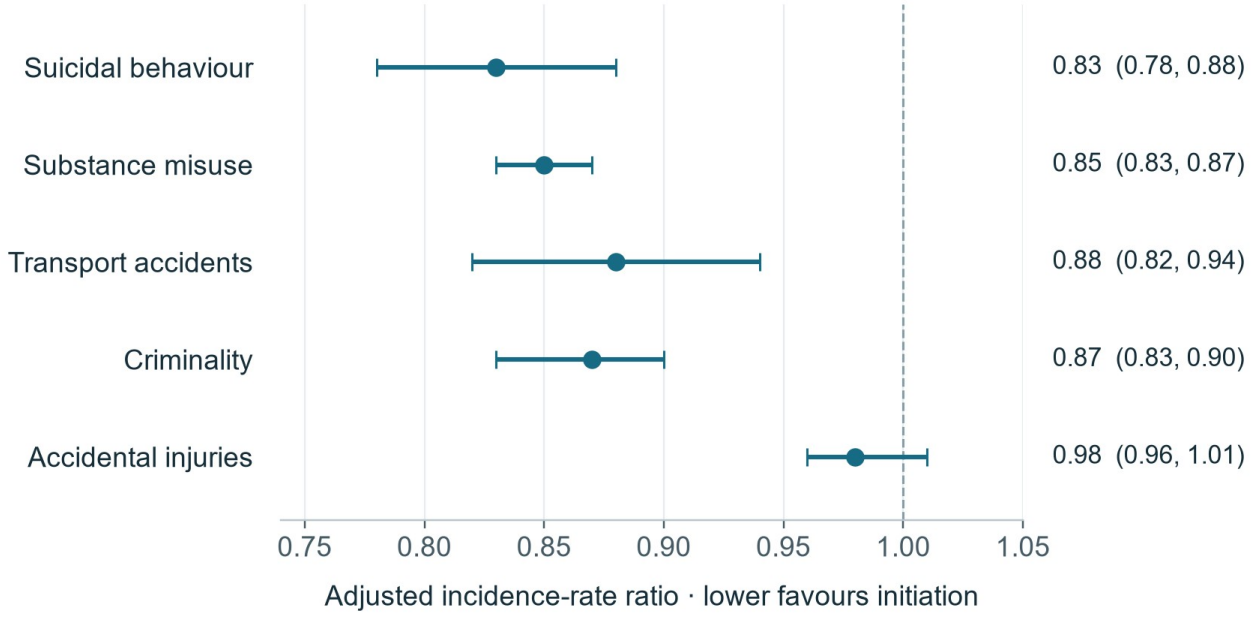


Figure 23. Longer-term events answer a different medication question. Adjusted first-event incidence-rate ratios with 95% CIs from a Swedish target-trial emulation in 148,581 newly diagnosed people. Follow-up was two years. This is observational evidence; residual confounding remains possible. Accidental-injury results are compatible with no difference. Ratios are not individual probabilities, and they do not identify a cognitive mediator. [81]

Cardiovascular effects depend on the drug. A 2025 synthesis of 102 randomized trials found small average increases in several blood-pressure or pulse measures with stimulant and some nonstimulant treatments, whereas guanfacine generally lowered them. Most trials were short: the median was seven weeks, and none supplied long-term randomized evidence. “Nonstimulant” therefore does not mean physiologically neutral. Heart rate, blood pressure, symptoms and clinical events are distinct outcomes. [83]

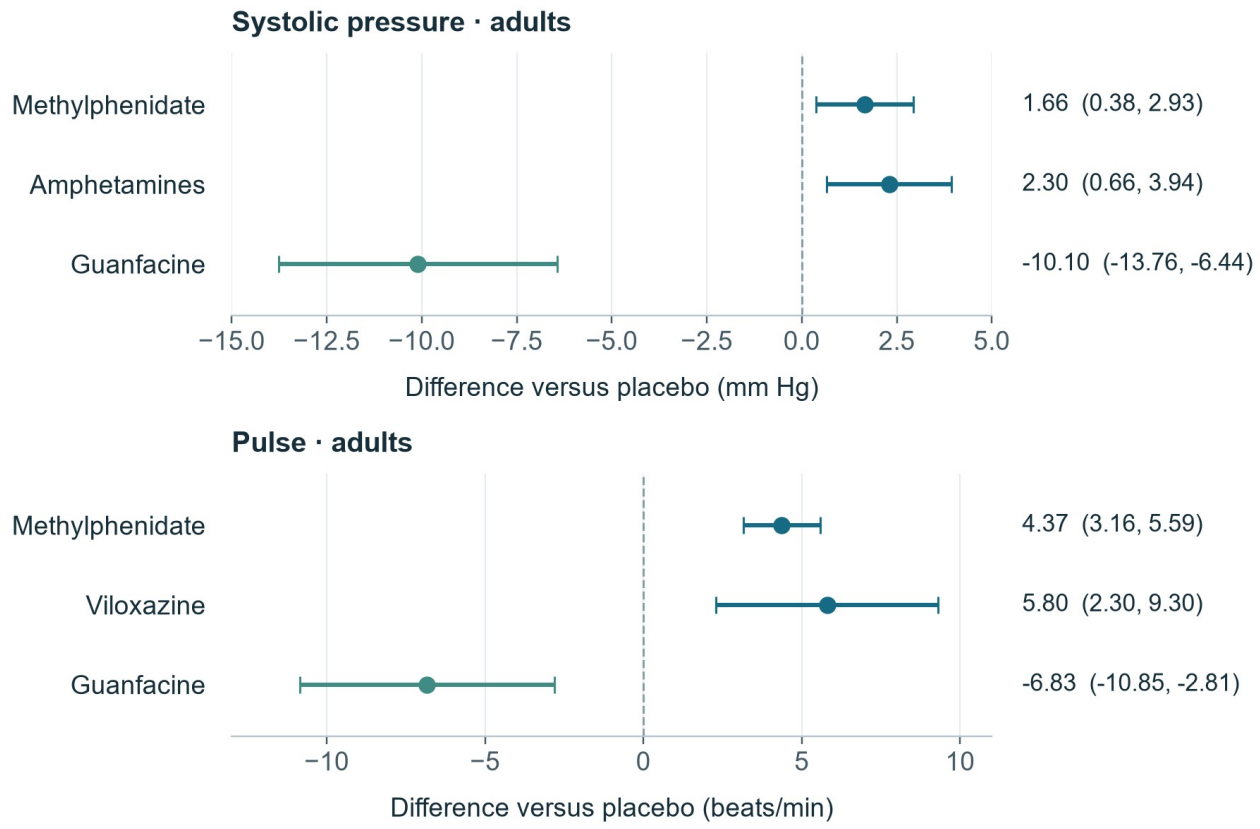


Figure 24. Medication can move physiological measures in different directions. Selected adult placebo comparisons from Farhat et al., with 95% CIs; all plotted estimates had very low evidence confidence. Median trial duration was seven weeks. Different drug selections illustrate the measures, not a complete ranking. Blood pressure, pulse and clinical cardiovascular events are separate outcomes; a lower value is not always better for a particular person. [83]

The person's underlying condition can change the response. In 27 people with POTS, an acute randomized crossover experiment found standing heart rate of 121 versus 105 beats/minute two hours after atomoxetine or placebo, respectively; symptoms also worsened on atomoxetine. This was a POTS experiment, not an ADHD efficacy trial. It supports checking autonomic context rather than assuming that the same noradrenergic effect is helpful in every physiological state. [84]

Longer follow-up adds a different caution. A Swedish nested case-control study associated more than five years of cumulative ADHD-medication use with higher odds of cardiovascular disease than nonuse: adjusted OR 1.23 (95% CI 1.12–1.36), with prominent hypertension and arterial-disease associations. This does not establish causation or a 23-percentage-point risk increase. Surveillance, severity and other confounding can affect the estimate. Benefits on one outcome and risks on another can coexist. [85]

8.6 Medicines for overlapping conditions

An allergy medicine can change attention through sedation as well as through symptom relief. In a randomized crossover study of 40 adults with seasonal allergic rhinitis, diphenhydramine impaired simulated driving, whereas fexofenadine performed similarly to placebo. Subjective drowsiness was a poor guide to impairment. This is a concrete reason to distinguish centrally sedating H1 drugs from other antihistamines. It does not show that allergy-related cognitive symptoms reveal a histamine cause of ADHD. [89]

Migraine treatment is another example. Topiramate's prescribing information reports cognitive adverse reactions in 22% of participants receiving 100 mg/day in six-month migraine-prevention trials, versus 10% on placebo. That is a trial exposure, not a dose recommendation. The endpoint was reported adverse reactions, not a dedicated ADHD test. A change in concentration during migraine prevention could reflect attacks, sleep, pain relief, treatment effects or several of these together. [88]

GLP-1 medicines can change appetite, gastrointestinal symptoms and gastric emptying. Semaglutide labeling flags possible effects on absorption of oral medicines, but does not establish a specific interaction with ADHD stimulants. A reported change in stimulant response after starting a GLP-1 drug deserves a medication and timing review; it cannot by itself demonstrate altered absorption or an immunometabolic mechanism. [90]

Tirzepatide has a specific hormonal-medication interaction worth distinguishing from a general absorption concern. Its US label advises switching to a non-oral contraceptive or adding a barrier method for four weeks after starting and after each dose escalation because oral hormonal contraception may be less effective. This is a prescribing boundary relevant to the endometriosis and hormonal-state discussion, not evidence of an established stimulant interaction. [143]

Mental-health safety and mental-health efficacy also answer different questions. In January 2026, the FDA requested removal of the suicidal-behaviour/ideation warning after its review found no increased risk with GLP-1 medicines. That finding does not establish an antidepressant effect or replace assessment of an individual's symptoms. [142]

8.7 A prescribed dose is not the whole exposure

Some interactions are well defined. In normal CYP2D6 metabolizers, fluoxetine or paroxetine can increase atomoxetine exposure about six- to eightfold. That is an interaction with specified enzyme inhibitors, not an effect of all antidepressants. Guanfacine exposure changes with CYP3A4 inhibitors or inducers; sedation can add to other sedating medicines. Abrupt guanfacine withdrawal can cause rebound hypertension. These label findings explain why co-treatment and medication changes belong in the record. [86, 87]

Amphetamine-product labeling also identifies serotonin-syndrome risk with serotonergic comedications, including some antidepressants and triptans, and contraindicates concurrent or recent MAOI use. This is a specific prescribing interaction, not evidence that all migraine or mood treatments are incompatible with ADHD medication. The examples here are not an exhaustive interaction check or a self-adjustment guide. Prescribing and changes remain an individual clinical decision. [91]

9. Measuring change across four intervention areas

9.1 The same outcomes across studies

When a treatment helps, I want to know what changed for the person taking it. They may feel better without finding a demanding task any easier. I would keep the same four outcomes across intervention studies, measure each treatment's own target and schedule follow-up when a change could plausibly begin or wear off.

For attention, count misses among valid targets and record reaction-time lapses where the task supports that measure. Ask how much effort the task took. Use timestamps to check whether an everyday task was finished on time and how much time was available. Record adverse effects, missed sessions and treatment burden. Each outcome keeps its own unit. Appendix L explains how past completion times and the time available give an estimate of functional margin. The table separates outcomes the studies measured from ones I would add. A better score in the lab leaves the question about everyday tasks open.

Comparing the four intervention areas

Area	Population and comparison	Measured outcome	What a future study would add
Sleep	Sleep treatment added to usual adult ADHD care. Positive airway pressure vs sham in sleep apnea.	Reported sleep and fatigue improved in the ADHD study. Its primary ADHD comparison was inconclusive. APPLES found no primary cognitive difference at six months.	Measure duration, continuity or circadian phase and record other treatment timing.
GLP-1 drugs	Major depression, semaglutide vs placebo. Obesity with sleep apnea, tirzepatide vs placebo.	Depression trial missed its executive endpoint. Effort analysis uses the same trial. Tirzepatide reduced apnea severity.	Measure the clinical target, sleepiness and gastrointestinal effects alongside the shared outcomes.
Noninvasive vagus nerve stimulation	Ear devices, a research TENS setup and a cervical device.	Results differ by device, population and outcome. Adult ADHD efficacy remains unestablished.	Specify model, target, waveform and sham. Measure attention and task completion separately.
Histamine pathways	H3 agents vs placebo in adult ADHD. H1 drugs in allergic rhinitis. Famotidine regimens after COVID-19.	H3 efficacy was unestablished. Rhinitis studies measured vigilance and symptoms. Post-COVID studies measured global cognition or fatigue.	Measure allergy relief, sedation and sleep continuity separately.

9.2 What the four intervention areas show

Sleep interventions and treatment timing

Direct adult ADHD and sleep apnea. Adding sleep treatment to usual ADHD care improved self-reported sleep quality and fatigue. The primary self-reported ADHD outcome was inconclusive. Melatonin plus bright light advanced circadian phase without a clear ADHD symptom benefit in that arm. A one-week crossover comparison in 15 adults selected for a good stimulant response found better self-reported sleep quality and less fatigue on medication. In sleep apnea, APPLES found a difference in one primary cognitive measure at two months and none at six months. Measure sleep duration, continuity or circadian phase as appropriate, then assess the four shared outcomes separately and record when other treatments were taken. [187, 19, 20, N10, N11]

GLP-1 drugs and related agonists

Other conditions. In major depression, semaglutide did not improve the prespecified executive-function composite versus placebo. A planned secondary global cognition measure favored semaglutide. Separate analyses of the same trial found changes in reward choices and willingness to choose physical effort at higher expected reward. In adults with obesity and obstructive sleep apnea, tirzepatide reduced the apnea–hypopnea index. None of these trials enrolled an ADHD treatment population. Alongside the four shared outcomes, measure the indication-specific response and record gastrointestinal effects, sleepiness and treatment timing. [U54, G22, N01, 113]

Vagus nerve stimulation

Studies of noninvasive stimulation report different results across devices, populations and tasks. Adult ADHD efficacy remains unestablished. The pediatric symptom and cognition reports tested the trigeminal nerve through the forehead and found no superiority to sham on the reported outcomes. Section 15.4 compares the devices and evidence. Section 15.5 sets out a separate trial proposal for adults with ADHD, including how to verify the matched sham. [N12, U04]

Antihistamines and histamine pathways

Direct adult ADHD and other conditions. Trials of the experimental H3 agents bavisant and MK-0249 did not establish efficacy in adult ADHD. In a single-dose ragweed challenge, diphenhydramine impaired vigilance, while desloratadine relieved symptoms without the same impairment on that battery. A small famotidine trial in early COVID-19 recovery reported better global screening scores among those with follow-up. In an open-label Long COVID trial, famotidine plus loratadine modestly improved self-reported fatigue at 12 weeks. The difference was inconclusive 12 weeks after treatment ended. Alongside the four shared outcomes, record the drug and receptor target, allergy symptoms, sleep continuity and daytime sedation. [27, U55, N09, 49, 50]

Timing matters when interpreting these outcomes. Pain can disrupt sleep, and a difficult day can change when someone takes medication or stops a task. Record the order of events and compare competing explanations before attributing a change to treatment.

Using the same measures across studies makes it possible to test whether a relationship found in one state predicts outcomes in another. That test needs independent observations and adequate measurement in both settings.

A relationship may transfer for vigilance and still fail for effort or task completion. Evaluate each outcome separately and check whether the relationship differs by ADHD status.

9.3 From observation to intervention

Chapter 5 estimates how attention varies after naturally shorter nights. A randomized add-on or factorial trial would ask what changes when we assign an intervention. The measures can carry over once their suitability has been checked for the new setting and time window.

10. Histamine and immunity: distinct compartments, possible links

Immune signalling can affect sleep and alertness, so an immune study needs separate measures of symptoms, effort and performance. I would specify which signal to measure, where to measure it and which outcome I expect to change.

10.1 There are at least two distinct histamine questions

Central neuronal histamine participates in arousal and cognitive regulation. Peripheral mast-cell histamine is part of immune signalling alongside other mediators. Receptor subtype and anatomical compartment matter. A proposed connection between them is not evidence that circulating histamine directly determines brain histamine or attention. [24, 25, 26]

In a circuit experiment, reducing H2-receptor signalling in specific parvalbumin-positive neurons produced ADHD-like behaviours in mice. The accompanying human tissue observations concerned patients described as having ADHD symptoms. The causal manipulation was in mice. It does not predict that blocking peripheral H1 receptors with an ordinary allergy medication treats ADHD. [24]

A pitolisant experiment tested central histamine signalling in humans. Sixty healthy participants were randomized; 58 formed the final sample and 52 contributed to key imaging/task analyses. An H3 inverse agonist altered learning-related network activity and working-memory computations. This is evidence that the intervention can change human cognitive processing, not an ADHD treatment trial or proof that its effects are exclusively histaminergic. [25]

The earlier ADHD trial was negative. In 72 adults, the H3 inverse agonist MK-0249 did not improve the primary ADHD outcome relative to placebo, whereas the active methylphenidate comparator did. This constrains that compound and protocol, not every histamine strategy. A molecule’s involvement in cognition is not sufficient evidence of treatment efficacy. [27]

10.2 Allergy overlap does not identify the route

An umbrella review supports ADHD associations with asthma and rhinitis, but diagnoses cannot separate immune effects from sleep disruption, symptoms, shared liability or ascertainment. In a daily time-series study, 21 children with ADHD and atopic disease were enrolled and 16 were analyzed; relationships between atopic and ADHD symptoms varied in timing and direction. These findings justify investigating individual relationships, not declaring stable allergy-sensitive ADHD subtypes. [28, 29]

A 2026 cytokine synthesis covered 74 studies across autism and ADHD together, including 990 ADHD participants, 4,331 autistic participants and 5,037 controls. Network analyses found lower TNF-α, IL-2 and MCP-1 in ADHD, rather than a broad elevated profile. These are pooled peripheral measurements, not a diagnostic signature or evidence that lowering inflammation would improve ADHD. They constrain a generic systemic-inflammation account without settling local brain immune activity. [30]

The question is whether allergic or immune changes affect cognition in some people through symptoms, sleep, nerve signalling or another measured route. A study must specify the route it is testing. Relief after an

antihistamine could come from less discomfort or better sleep; it does not diagnose mast-cell activation syndrome or identify the cause of ADHD.

Atopic dermatitis studies put numbers on the overlap. A US survey analysis found adjusted ADHD odds ratios of 1.14 (95% CI 1.03–1.26) in children and 1.61 (1.25–2.06) in adults. A later synthesis estimated 1.28 (1.18–1.40). These observational estimates do not isolate histamine. Itch, disrupted sleep, treatment and diagnostic contact remain plausible contributors, and atopic dermatitis is not equivalent to mast-cell activation syndrome. [68, 69]

10.3 Immune communication is not the same as an autoimmune attack

The proposed immune model concerns communication and regulation. It does not require immune attack on brain tissue, and it should not use “neuroinflammation” as a catch-all explanation for pressure, dizziness or fog. In mice, Jin and colleagues identified vagal–brainstem circuitry that senses inflammatory signals and modifies peripheral responses. That establishes a specific experimental feedback route, not a diagnosis from a symptom cluster. [31]

Suppressing inflammation can also come at a cost. In the same circuit research, strong artificial activation that suppressed inflammation impaired protection in a bacterial-infection model. Appropriate regulation is a more defensible target than “the lowest possible inflammation.” Those mouse manipulations are not equivalent to ordinary vagal stimulation in people. [31]

10.4 What controlled human inflammation experiments show

Human immune challenges offer a more direct test than diagnostic overlap. In a randomized crossover study, 22 healthy volunteers received low-dose bacterial endotoxin or saline. After endotoxin, they rated their performance more poorly, but the researchers did not detect a change in average reaction time or errors on two brief tasks. The small sample and three-minute tasks leave uncertainty. Still, the result is useful: feeling cognitively worse and showing a lower mean task score are outcomes we should measure separately. [153]

A newer experiment gets close to another distinction in this paper. Among 160 adults aged 60 or older, researchers randomized endotoxin or placebo within groups with and without insomnia. Of the endotoxin recipients, 26 had insomnia and 53 did not. The insomnia group had a larger self-rated depressed-mood response across the observation period, with a standardized difference of d = 0.64 (95% CI 0.16–1.12), despite similar measured rises in two inflammatory signals, IL-6 and TNF. Insomnia itself was not randomized. This was an acute mood outcome, not a cognitive test, a diagnosis of depression or an ADHD study. [154]

Two people can have similar measured changes in cytokines and feel quite different afterwards. That is why I would measure both the bodily response and the person’s sensitivity to it. The experiments leave the roles of histamine, barrier leakage and other pathways open. I use them here to choose measurements for an observational study.

10.5 Inflammation needs more than a positive association

A 2025 analysis combined ALSPAC observations with genetic approaches. ADHD and Creactive protein (CRP) had a genetic correlation of 0.27 (95% CI 0.19–0.35). That is evidence of shared common-variant influences, not proof that inflammation causes ADHD. The apparent CRP-to-ADHD effect was inconsistent across Mendelian-randomization sensitivity analyses, with evidence that variants could act through other pathways. Associations between childhood ADHD traits and later measured CRP were also not consistent across ages. Attrition, ancestry restriction and uncorrected multiple testing limit interpretation. [185]

Timing changes the question further. In Generation R, 6,555 mother–child pairs were followed with ADHD symptom ratings at ages 6, 10 and 14. Each additional reported prenatal fever exposure was associated with a 0.06-SD increase in symptoms (95% CI 0.01–0.11). Adjusted associations with the measured cytokine composite and CRP were not significant. The blood samples were taken in mid-pregnancy and could miss brief inflammatory spikes; fever was self-reported. The study neither proves an inflammatory mechanism nor rules out every prenatal immune effect. It also concerns developmental exposure, not the cause of an adult's difficult afternoon. [186]

10.6 Mast-cell disease and cognitive symptoms

Mastocytosis involves abnormal mast-cell accumulation. In one clinical sample of 57 patients, 42 reported cognitive complaints and 22 met the study's memory or attention impairment threshold. The study used standardized test norms, without a separately recruited healthy comparison group. Medication, referral selection and other symptoms complicate interpretation. It shows that cognitive difficulties deserve assessment in a defined mast-cell disease; it cannot establish their mediator or be treated as an MCAS or ADHD prevalence estimate. [198]

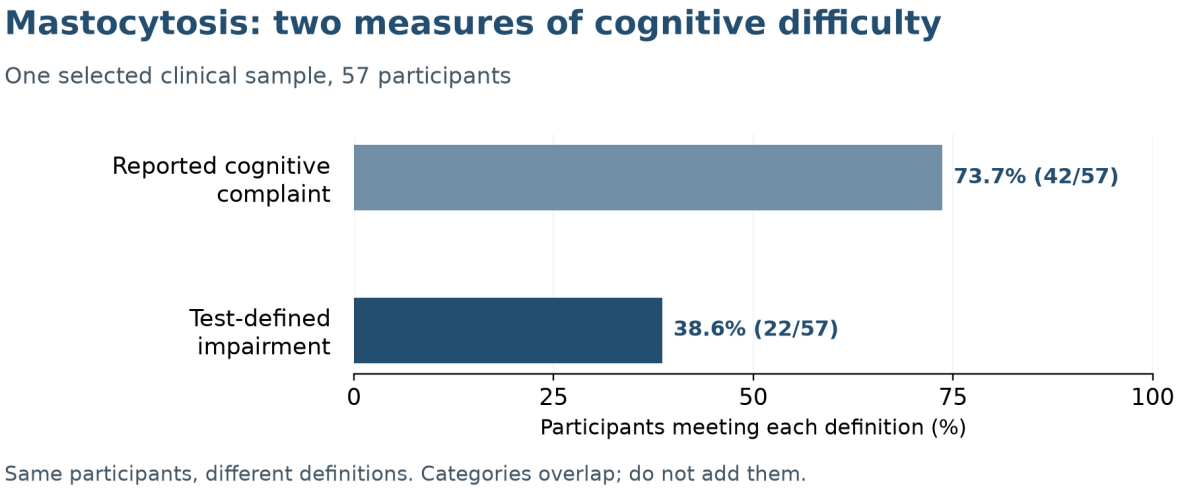


Figure 5. Two definitions in the same 57 patients. Percentages are calculated from reported counts and rounded. Symptoms and test thresholds are not interchangeable; the bars are neither treatment effects nor independent groups. [198]

Mast-cell activation and mast-cell activation syndrome (MCAS) also need separating. Clinical frameworks combine recurrent systemic symptoms, objective mediator evidence and response to directed treatment; tests and definitions still have limitations. A 2024 expert research report specifically identifies weak evidence for several neurological and psychiatric attributions. For our study, the diagnostic criteria, event

timing and laboratory method must be recorded. Improvement after an antihistamine alone cannot identify the cause of brain fog. A negative test does not make the symptoms unreal, and a symptom cluster does not identify mast cells as its source. [199]

10.7 Brain barriers and fluid drainage

The blood–brain barrier regulates exchange between blood and brain tissue. Blood flow describes delivery through vessels; cerebrospinal fluid (CSF) movement is another process. A finding in one cannot substitute for a measurement of the others. The small Long COVID imaging comparison in Section 13.1 is human evidence about barrier permeability. It does not establish mast-cell causation or an ADHD abnormality. [44]

A 2025 Cell study adds a specific mechanism at the brain's outer coverings. In mice, dural mast-cell activation released histamine, widened bridging veins and restricted CSF passage at arachnoid cuff exit points. BBB permeability was unchanged at the measured one-hour time point in the activation experiment. Mast-cell depletion increased brain bacterial burden in meningitis models, showing a protective role in that setting. Human tissue observations established the presence of meningeal mast cells; the functional experiments were in mice. Fluid regulation and barrier leakage therefore need separate tests. Neither result demonstrates an explanation of human brain fog. [200]

A different study exposed cultured human brain endothelial and microglial cell lines to products from activated mast cells and found inflammatory responses and reduced tight-junction proteins. It also examined infected mice. Cell lines derived from humans are not a clinical human experiment, and an acute infection model does not establish persistence in Long COVID. Together, these studies identify routes worth testing while leaving their relevance to ADHD unresolved. [201]

Table 2. Different brain-related measurements and their limits

Measurement	What it addresses	What it does not establish
Barrier permeability	Movement across a specified barrier, using an appropriate imaging or laboratory method.	The responsible cell, a cognitive mediator, or the cause of ADHD.
Cerebral blood flow	Blood delivery or a measured velocity proxy under specified conditions.	Barrier leakage or impaired CSF drainage.
CSF dynamics	Fluid movement around and through brain-associated spaces.	A general brain "detox" deficit or a proven human treatment target.
Cognitive outcome	A defined symptom report, task score or functional measure.	Which physiological route produced the difficulty.

10.8 Attention, histamine compartments and immune boundaries

Three preregistered within-person experiments involving 57 healthy adults found that attention allocation altered an exogenous histamine skin-prick response. This demonstrates a short-timescale attention–periphery interaction, but does not measure endogenous mast-cell release, chronic allergy, ADHD cognition or a treatment effect [U08]. Mouse H2-receptor experiments in substantia nigra circuitry and separate NSF/D2-receptor-lineage manipulations further illustrate receptor and circuit specificity [U09, U10].

The brain–dura mechanism in Section 10.7 [200] depends on the tissue and disease context. Whether peripheral H1 treatment reverses the local mouse response remains unknown. In adult ADHD, the highest bavisant dose (10 mg) failed its primary comparison with placebo ($p = .161$). The planned testing sequence stopped there, so the lower doses were not formally tested. Both active comparators separated from placebo. [U55]

The childhood asthma–ADHD association also requires restraint. In ALSPAC, the crude association among 7,165 children was OR 1.35 (95% CI 1.10–1.65), attenuating to 1.14 (0.92–1.41) after adjustment for 15 early-life factors. Non-atopic asthma accounted for the stronger crude association, and Mendelian randomization offered little support for causality [G01]. This developmental question is different from within-person symptom variation during an allergy flare.

Human PET and pharmacological work supports distinguishing peripheral allergy relief from central H1 occupancy [G02]. The cited PET study examined levocetirizine with fexofenadine as an active control; it was not an ADHD efficacy study. The peripheral cytokine findings in Section 10.2 provide a further constraint on a generalized inflammation account [30]. Systemic markers, neuroinflammation and day-to-day cognitive effects require separate evidence.

Astrocyte-derived fibronectin in APOE4-related experimental models and complement-factor-H studies identify candidate barrier and neuronal-stress mechanisms [U24, U25]. Neither demonstrates an allergy-to-ADHD pathway in living humans. The next test must specify the compartment and cell type before selecting a mediator.

11. Hormones, PMDD and endometriosis

Recurring hormonal and endometriosis symptoms let us compare the same person across different conditions. Tracking pain and sleep alongside cognitive performance would help distinguish routes that a comparison of diagnoses leaves open.

11.1 The endocrine signal should be measured, not assumed

Roberts and colleagues collected saliva and symptom reports for 35 days in 32 regularly cycling women. Hormonal combinations predicted next-day ADHD symptoms, with trait impulsivity moderating relationships. This was a community sample, not proof of a diagnosed-ADHD interaction or objective cognitive decline. A later 35-day pilot in 30 stimulant-treated females found phase-related symptom variation without establishing hormonal or immune mediation. [32, 33]

A counterweight is a 2025 meta-analysis of 102 articles, 3,943 participants and 730 comparisons: it found no systematic, robust menstrual-phase differences in objective cognitive performance. Its mainly nonclinical scope does not exclude symptomatic ADHD or PMDD subgroups, or increased effort to maintain a score. It argues against treating small symptom studies as proof of universal cycle-related cognitive decline. [66]

For the proposed study, I would measure endocrine state repeatedly where it is relevant to the question. Calendar phase alone is not a mechanism. Hormones, mood, pain, migraine, sleep and treatment can change together; measuring them together would help us distinguish their roles.

11.2 PMDD and the direct ADHD overlap

PMDD involves impairing symptoms that follow a recurring premenstrual pattern and ease after menstruation begins. In a UK survey of 715 young adults, provisional PMDD was more frequent among those reporting a clinical ADHD diagnosis. Figure 6 shows the comparison. Retrospective screening cannot confirm PMDD, which needs prospective daily symptom tracking across at least two cycles. ADHD was oversampled, and depression and anxiety were common. Selection, recall and symptom overlap limit interpretation; neither hormones nor a cognitive mechanism were measured. [204]

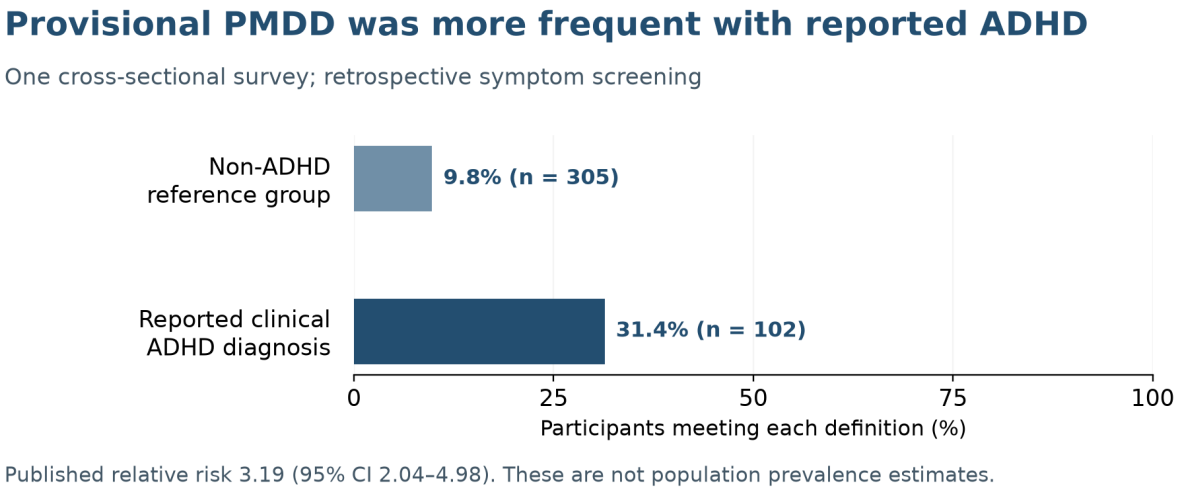


Figure 6. Reported proportions from Broughton et al. Bars show provisional screening results, not confirmed PMDD diagnoses. The ADHD group and reference group are selected subsets of the survey. The relative risk is an association, not evidence of shared causation. [204]

Someone with ADHD or depression may have symptoms throughout the month that worsen before menstruation. PMDD can follow a more restricted cyclical pattern. Daily observations would help separate a heavier usual symptom burden from a larger change around that part of the cycle.

Hormonal manipulation provides a more direct test. In a 2017 study, 22 women with PMDD were studied and 12 selected responders to ovarian suppression entered the relevant hormone add-back comparison. Overall symptom ratings rose during the first month of combined estradiol and progesterone exposure, then subsided during continued exposure in months two and three; daily irritability ratings remained elevated in month two. This supports sensitivity to a change in hormones in that subgroup. The small, selected sample and combined treatment do not isolate estrogen, and cognition in ADHD was not tested. [205]

A 2025 extension compared 34 women with PMDD and 76 healthy participants during suppression and separate estradiol and progesterone add-back. Affective symptoms returned during add-back in the PMDD group but not the comparison group. Ten PMDD participants and 15 comparators came from the original cohort, so this is not a wholly independent replication. The result strengthens the case for differential sensitivity while leaving the downstream mechanism open. [206]

11.3 Estrogen progesterone and metabolic signalling

Estrogen is a family of hormones; estradiol is the form studied in these experiments. Progesterone and its metabolites deserve their own place in the model. Allopregnanolone, a progesterone-derived neuroactive steroid, modulates GABA-A signalling. In two small dose studies, each with eight PMDD participants and eight controls, the higher-dose dutasteride experiment blocked the luteal allopregnanolone rise and reduced several symptoms; the lower-dose experiment did not. Several steroid pathways are affected, and these were mood outcomes. The experiment is a mechanistic clue, not an established PMDD treatment or an ADHD cognitive result. [208]

There is also a laboratory estrogen–mast-cell connection. Estradiol increased mediator release in cell lines and mouse-derived mast cells through estrogen-receptor-alpha-dependent mechanisms. The work included

a human mast-cell line, not a trial in women with PMDD. It supports testing hormone–immune interactions; it does not establish an estrogen–histamine feedback loop as the cause of PMDD, BBB disruption or ADHD. Our model should leave those connections open until measurements in the same people connect them. [202]

The insulin connection has a small human experiment behind it. Eleven healthy, naturally cycling women completed insulin-versus-placebo nasal-spray comparisons during glucose clamps in follicular and luteal phases. Insulin increased the glucose infusion requirement in the follicular phase; the adjusted luteal result was not significant. A separate 15-person imaging study examined hypothalamic responses but lacked a contemporaneous placebospray comparison. Phase was verified with hormone measurements, yet estrogen and progesterone were not independently manipulated. This is a physiological interaction, not proof of cognitive impairment, PMDD or a pathological luteal state. [212]

11.4 Testosterone and estrogen treatment need cognitive outcomes

Menopause needs its own evidence. In ELITE-Cog, 567 healthy postmenopausal women contributed to a randomized estradiol comparison. The verbal-memory difference was −0.06 SD (95% CI −0.22 to 0.09) after a mean 57 months, with no clear advantage from starting nearer menopause. This does not test symptomatic perimenopause or ADHD, and does not negate relief of other menopausal symptoms. A hormonal hypothesis should not assume that replacement improves cognition. [96]

Testosterone belongs in the discussion on the same terms. In the Testosterone Trials, 493 men aged 65 or older with low testosterone and age-associated memory impairment formed the primary cognitive subgroup. One year of testosterone gel did not improve delayed paragraph recall versus placebo: the adjusted difference was −0.07 points (95% CI −0.92 to 0.79). This does not settle treatment for established endocrine indications, younger people or ADHD. It does show why a low hormone level and a plausible brain mechanism cannot promise a cognitive benefit. [207]

I would ask whether a measured endocrine change predicts a specific difficulty and whether treating an established condition improves that difficulty. The studies do not establish an optimal estrogen or testosterone level for attention or resilience. Reproductive stage, prescribed hormones, sleep, pain and mood all shape what the comparison can tell us.

11.5 Why the ADHD–endometriosis overlap stays in the paper

In a Swedish register study of 854,361 women, 14,144 had an endometriosis diagnosis. The sibling comparison found a higher rate of subsequent recorded ADHD diagnosis (HR 1.98, 95% CI 1.34–2.93). Comparing sisters reduces some shared familial confounding; it does not remove differences in pain, sleep, medical contact or diagnostic delay. Later ADHD recognition cannot establish that endometriosis caused a developmental disorder. No cognitive task was measured. [34]

Cross-trait genetics provides another kind of evidence. One analysis estimated endometriosis –ADHD genetic correlations of 0.21 and 0.17 using two methods. These are population-level correlations in common-variant effects, not a 21% individual risk or a percentage of shared cause. The methods reuse source data, and the result does not identify an inflammatory pathway or establish causal ADHD-to-endometriosis effects. [35]

A human endometriosis study found higher lesion histidine decarboxylase expression and serum histamine, but no difference in peritoneal histamine or urinary methylhistamine. Histamine-related receptor transcripts were not uniformly increased either. Those null findings belong in the account: local tissue biology, circulating markers and excretion are not interchangeable compartments. [36]

A more specific 2025 study combined human lesion observations with cell and mouse experiments. It implicated the mast-cell receptor MRGPRX2 and a histamine–H1–TRPV1 route in sensory-neuron sensitization and pain. This is a candidate local immune-to-nerve mechanism. It is not a trial showing that an antihistamine treats endometriosis, and it did not measure ADHD cognition. [26]

My proposed connection is that recurring endocrine or local immune activity may increase pain, alter sleep or migraine burden, and thereby change cognitive experience or performance. A direct immune-to-cognition route is an additional hypothesis, not a necessary assumption. If pain and sleep explain the association, we should accept that answer rather than keep searching for a more exotic mechanism.

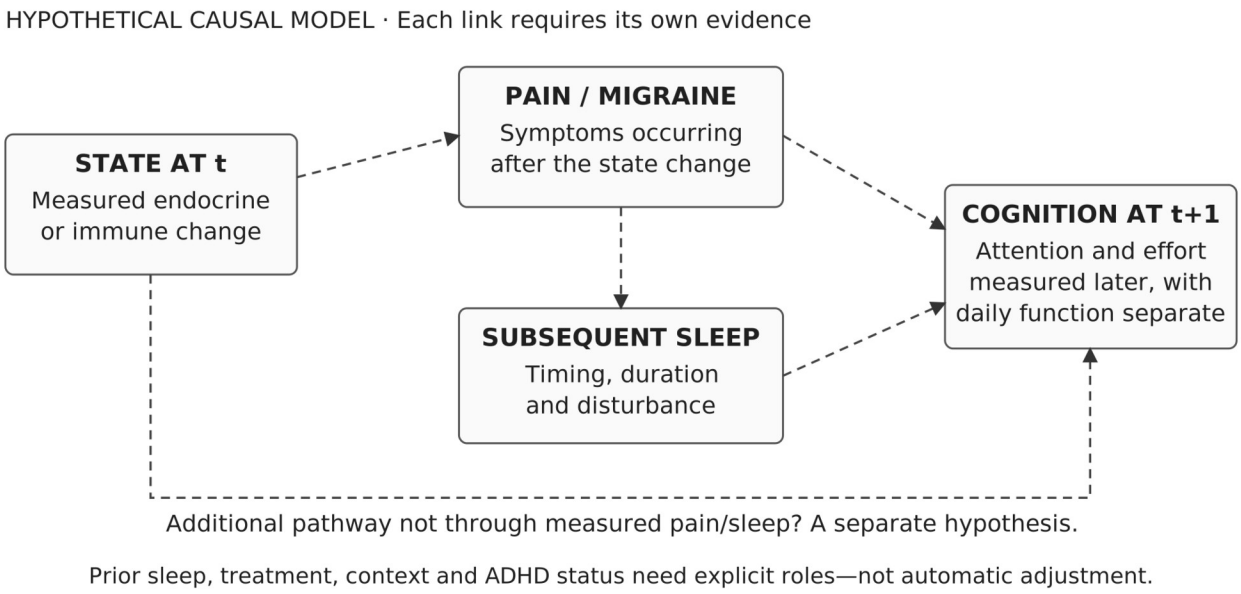


Figure 7. A plausible connection still contains competing explanations. Hypothetical time-ordered diagram for the endometriosis module; arrows do not assert demonstrated mediation. A total-effect question includes routes through subsequent pain and sleep. A question about a remaining association after those routes requires different assumptions. This is not a complete causal graph: prior exposures, treatment and confounders need explicit specification.

Worked example 3 · Do not adjust away the question

Imagine that attention worsens during a pain flare that has already disrupted the preceding night's sleep. A model including sleep might reduce the estimated flare–attention association. That could be consistent with sleep carrying part of the relationship, not proof that the original experience was unrelated to the flare. The study must state whether it is asking about the total relationship or a pathway not operating through sleep. This is an illustration, not a patient case or a causal conclusion.

11.6 Fatigue and iron status need their own measurements

In 560 women with confirmed endometriosis and 560 matched controls, frequent fatigue was reported by 50.7% versus 22.4%. Within the endometriosis group, fatigue was associated with insomnia, pain, depression and occupational stress. These are reasons to measure competing routes, not proof that they mediate ADHD cognition. [93]

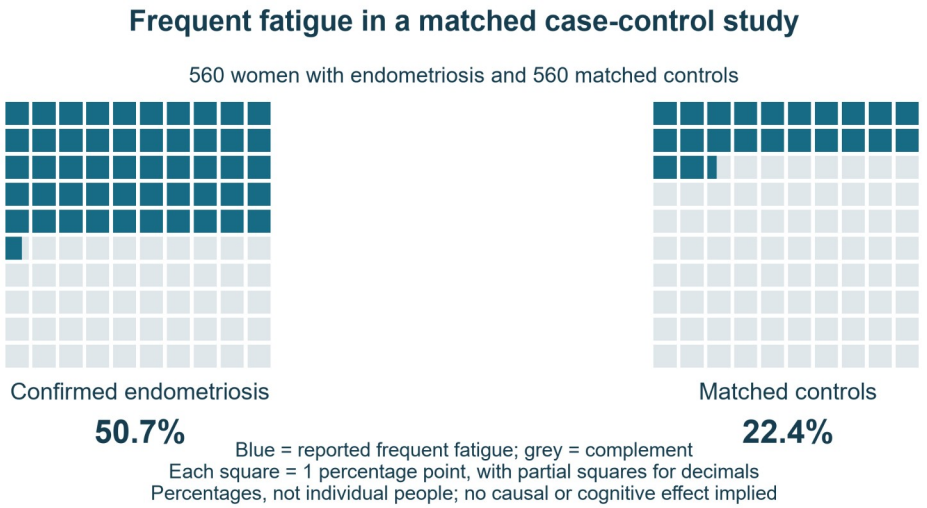


Figure 8. Fatigue is common in this endometriosis sample, but fatigue is not a cognitive test. Reported proportions from a matched case-control study; no confidence intervals are reconstructed. Selection, symptom reporting and unrecognized endometriosis in controls may affect the comparison. It cannot identify a shared cause with ADHD. [93]

Fatigue is more common in endometriosis

In this matched study, about half of women with endometriosis reported frequent fatigue, compared with about one in five controls. That makes fatigue worth measuring alongside cognition. It does not identify the cause of either symptom.

The observed difference is 28.3 percentage points. It is an association in this sample, not a treatment effect. Pain, insomnia, depression and occupational stress were also associated with fatigue within the endometriosis group. The study does not demonstrate that an immune pathway causes ADHD or that fatigue equals objective cognitive impairment. [93]

Iron status also needs checking. In a selected clinic cohort of 251 people with symptomatic endometriosis, 134 (53.4%) had iron deficiency and 34 (13.5%) had iron-deficiency anemia. Anemia is included within iron deficiency; the percentages must not be added. This was a referral sample, not a prevalence estimate for everyone with endometriosis, and it did not measure objective cognition. [94]

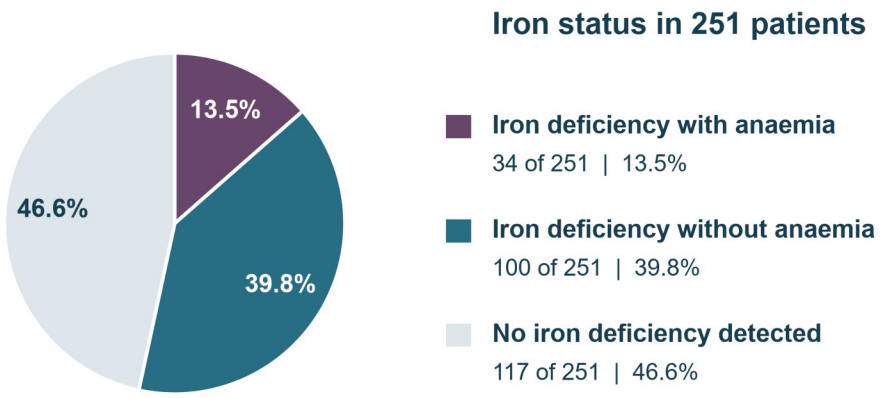


Figure 9. Iron-status categories in a selected endometriosis clinic cohort. Counts partition the same 251 patients: 34 with iron-deficiency anaemia, 100 with iron deficiency without anaemia (134 minus 34), and 117 without detected iron deficiency (251 minus 134). Percentages are calculated from these counts and rounded. This is not a general-population prevalence estimate or a cognitive outcome. [94]

A separate randomized trial enrolled 198 fatigued, nonanemic menstruating women with low ferritin. At 12 weeks, fatigue scores fell by 47.7% with iron and 28.8% with placebo; the reported contrast was −18.9 percentage points (95% CI −34.5 to −3.2). Those are mean score reductions, not the proportions cured. The trial did not test ADHD or cognition. Assessment guided by clinical history can identify a contributor without treating fatigue or fog as proof of an immune–brain mechanism. These findings do not support routine supplementation without an indication. [95]

11.7 Recurring pain, lesion biology and objective performance

Spatial transcriptomics of endometriosis lesions found anatomically distinct nerve, fibroblast and immune-cell interactions, supported by patient-derived models. This specifies local pain-related candidate inputs without demonstrating systemic histamine elevation or impaired ADHD attention [U11]. A sponsor-reported Phase 3 menstrual-migraine trial in 468 participants found fewer perimenstrual migraine days with cycle-timed atogepant than placebo. Cognitive secondary outcomes were mentioned without full numerical reporting [U12].

In a small neuropsychology/EEG comparison, endometriosis with chronic pelvic pain (n=22) showed broader cognitive differences and reduced P300 amplitudes relative to controls (n=42), while endometriosis without chronic pain (n=13) showed narrower differences. Pain and affective symptoms moderated the findings [G05]. Medication, mood, selection and multiple comparisons limit causal interpretation. A longitudinal dysmenorrhea pilot in 32 participants found poorer attention, control and processing-speed performance on the most painful day than on a pain-free day [G06]; order, hormone phase, expectations and analgesics may contribute. This pilot concerns primary dysmenorrhea, not endometriosis or ADHD.

These designs motivate within-person attack, interictal and recovery comparisons. Measure pain intensity, fatigue, treatment and hormonal timing separately, and test cognition independently of reported symptom relief. A pain-related functional cost need not operate through an assumed inflammatory mediator.

12. Migraine, circulation and body-signal processing

With migraine, we can compare the same person during an attack and between attacks. A posture study asks a related question on a shorter timescale: how does someone's performance change when they take the same task in a different position?

12.1 Migraine is more than a convenient analogy

In a Taiwan cohort, 81,441 people with ADHD were matched to 81,441 controls without a prior migraine diagnosis. During follow-up, 462 versus 212 received a migraine diagnosis; the adjusted hazard ratio was 1.92 (95% CI 1.64–2.34). The young-adult subgroup estimate was imprecise and not significant: 1.31 (0.58–2.98). These are newly coded diagnoses, not lifetime migraine prevalence. Figure 10 shows the observed counts on an absolute scale. [67]

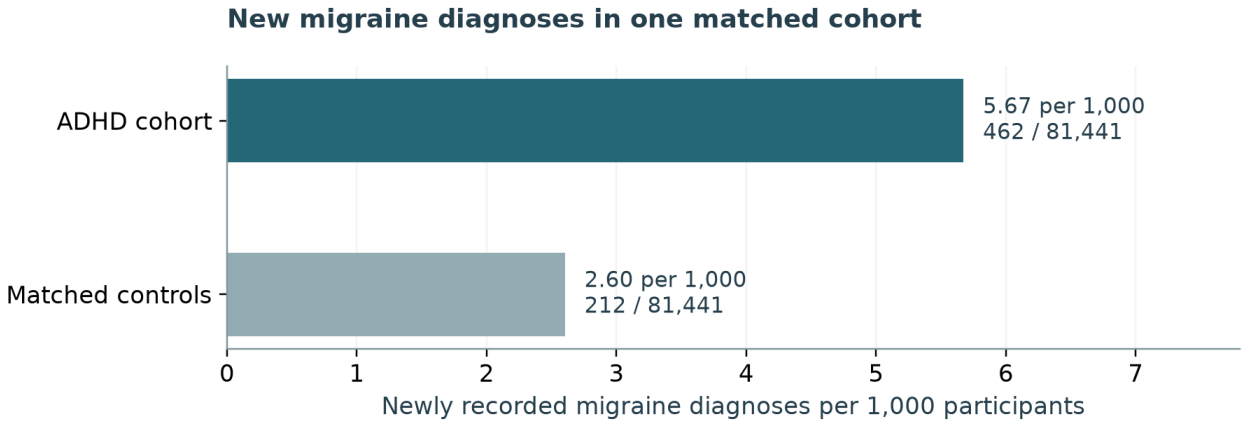


Figure 10. One cohort, two ways to read the result. Counts from Hsu et al.; bars are calculated as count / 81,441 × 1,000. Adjusted HR 1.92 (95% CI 1.64–2.34). Enrollment 2001–2009, follow-up through 2011: these descriptive proportions have variable follow-up and are not annual incidence rates or migraine prevalence. They cannot identify a shared cause. [67]

A migraine study randomized 58 people to assessment order; 39 completed at least one assessment and 24 completed both an untreated-attack and a headache-free assessment. Processing speed and verbal-memory performance were worse during attacks. This supports within-person neurological-state variation; it does not establish a larger effect in ADHD or identify the entire mechanism. [37]

Migraine also supplies an example of a defined, clinically testable pathway. In the 955-participant STRIVE trial, blocking the calcitonin gene-related peptide (CGRP) receptor with erenumab reduced migraine frequency and interference with activities compared with placebo. That supports a specific migraine target, not a histamine explanation or evidence that ADHD improves when that target is treated. [38]

A randomized provocation study found that PACAP38 could still trigger migraine after CGRP blockade with eptinezumab. This supports testing PACAP as a distinct target under those experimental conditions. In June 2026, the sponsor reported a positive phase 2b intravenous bocunebart prevention comparison with placebo. The announcement supplies preliminary migraine-frequency evidence. [G34, G35]

A prospective study enrolled 100 patients treated with an anti-CGRP antibody or botulinum toxin. Among 90 completers, 47 met the study's headache-day response criteria. Responders improved more on an executive-function composite than nonresponders. The comparison lacked an untreated randomized control, and retesting, treatment selection, pain and sleep could contribute. It motivates measuring cognitive change alongside migraine relief. [G36]

The relevant ADHD study would assess attack, interictal and recovery states, alongside pain, sensory symptoms, treatment and independent cognition. A treatment-related reduction in headache days and a reduction in cognitive fog are related but separate outcomes. Migraine medication itself is another possible contributor to cognitive change.

12.2 Orthostatic physiology offers a measurable challenge

Autonomic studies in adult ADHD are heterogeneous, and differences are more evident in some task/stress settings than as one resting marker. Interoception research—the detection and interpretation of bodily signals—is similarly limited and mixed. Neither literature justifies diagnosing dysautonomia from ADHD or a wearable HRV value. [39, 40]

POTSKog tested the relationship in POTS: 30 participants with postural orthostatic tachycardia syndrome (POTS) and 30 controls completed cognitive assessment across body positions in counterbalanced order. Attention and executive differences were found, with posture-related Stroop effects in the POTS group; leg crossing did not restore executive performance. It was not an ADHD study and did not establish that cerebral blood flow alone explained the results. [41]

I would record posture, cardiovascular response, respiratory carbon dioxide and cognitive performance during the same test. Transcranial Doppler measures blood-flow velocity, not absolute flow; vessel diameter and carbon dioxide affect interpretation. Cognitive effort can itself alter haemodynamic measurements, so a repeated-task control condition is essential. [42, 43]

Feeling light-headed or pressured does not establish impaired cerebral perfusion. Conversely, a symptom report should not be dismissed merely because one cardiovascular measurement is normal. Physiology, perception, symptoms and performance are different measurements whose relationships remain to be tested.

12.3 A normal pulse response does not answer every circulation question

In a specialist-clinic study, 429 people with ME/CFS and 44 healthy controls underwent head-up tilt with extracranial Doppler measurements of cerebral blood flow. Among the 247 patients whose heart rate and blood pressure responses were classified as normal, the mean end-tilt flow reduction was 24%, compared with 7% in controls. Referral bias may have enriched the patient sample for orthostatic problems. This was not an ADHD study and did not demonstrate that reduced flow caused cognitive symptoms. It gives a concrete reason to avoid treating pulse and blood pressure as complete substitutes for cerebral circulation. [177]

I would repeat the circulation measurement outside the original clinic and record respiratory carbon dioxide and cognitive performance at the same time. The question is whether the measures change together in a way that supports the proposed mechanism. The same safeguards for symptoms and exertion would apply.

12.4 Perfusion and transport as separate outcomes

Across CREST-2 cognitive analyses involving 2,165 older adults with severe asymptomatic carotid stenosis, adding stenting or endarterectomy to medical management did not improve cognitive outcomes at four years. A smaller hemodynamic substudy did not identify a clear responder subgroup [U28, U29]. This constrains those interventions in that population without resolving rapidly varying perfusion during a task.

A randomized crossover physiology study enrolled 29 healthy volunteers: 27 underwent vibration alone and 26 the combined nitric-oxide-mediated vascular change and vibration condition; an exploratory nitric-oxide-only session included 12. Combined treatment altered brain-fluid imaging patterns and circulating brain-derived proteins. Neither reduced brain amyloid burden nor cognitive improvement was demonstrated [U30]. The protocol involved nitroglycerin and monitored blood-pressure changes; individual imaging and blood analyses used smaller available samples.

A post-COVID case-control study of 22 patients and 19 controls reported lower global cerebral blood flow (49.2 versus 51.7 mL/100 g/min), slower reaction time over a 20-minute vigilance task and greater subjective fatigue. The slope of within-task slowing did not significantly differ [G03]. In a clinically selected POTS imaging series (N=56), 61% had abnormal perfusion relative to a reference distribution. Selection for cognitive complaints and absence of an equivalent control group prevent prevalence or causal inference [G04].

A discriminating postural challenge would pair posture, heart rate, blood pressure, end-tidal CO2 where feasible, a validated perfusion measure, objective vigilance and symptoms. Test whether perfusion explains the cognitive response beyond anxiety, sleep, medication and demand.

12.5 ADHD neurovascular imaging and measurement validity

Adult (41 ADHD/108 controls) and pediatric (80/110, 66/41 and 159/163) cohorts report differences in DTI-ALPS or related neurovascular measures, with selected associations with memory, executive tests and sleep [U39, U40, U41, U42]. In one pediatric cohort, lower ALPS coexisted with no significant difference in global BOLD–CSF coupling [U40]. These are convergent diffusion findings, not independent demonstrations of deficient solute clearance.

A methods study of 850 younger and 150 older healthy participants found substantial ALPS dependence on region placement, ventricular morphology and periventricular anatomy, with median hemispheric asymmetry of 12.5% [U43]. A separate acute stimulant experiment in 36 adults with ADHD and 42 controls found apparent structural/diffusion changes within one to two hours, but did not measure ALPS [U44]. Record medication timing, anatomy, motion, scanner conditions and physiology before interpreting a disease-specific transport signal.

12.6 Sleep, neurofluid dynamics and recovery

A randomized crossover experiment in 39 people found plasma amyloid/tau patterns consistent with a model of sleep-dependent transport [U48]. Phase-timed auditory stimulation during human EEG/fMRI increased slow-wave and CSF flow-wave responses [U49]. Plasma appearance, CSF displacement and net tissue solute removal are distinct endpoints, and neither result establishes cognitive benefit.

Mouse experiments link norepinephrine oscillations with vasomotion and CSF rhythms during NREM sleep [U50]. A male-mouse intermittent-hypoxia experiment implicates adenosine transport, vascular/AQP4 organization and tracer dynamics [U51]. These mechanisms concern particular states and compartments. They do not imply that reducing waking norepinephrine or manipulating adenosine would improve ADHD.

Counterevidence is material. Miao and colleagues reported reduced fluorescent-tracer clearance during sleep/anesthesia in mice [U52]. Differences in tracer, compartment, influx, efflux and analysis may explain disagreement. A human scoping review found direct or near-direct transport endpoints in only five of 105 studies [U56]. A critical human intervention review likewise distinguishes fluid movement, diffusion, transported proteins and clinical benefit [G08]. ACX-02 remains a small sponsor-linked unreviewed preprint [U57]. The cited version could not be reverified in the present source audit; its detailed findings remain provisional.

Neurofluid measurements remain an exploratory module. They would need to add prospective information about cognitive recovery beyond sleep duration, fragmentation, circadian timing, medication, pain and baseline ability. A mediation claim would additionally require valid transport measurement, temporal order and an intervention design capable of excluding causal alternatives.

13. Long COVID and ME/CFS: acquired disruption and delayed effects

An infection gives us a date to work from, but recovery can be uneven for months. In Long COVID and after mono, repeated testing can show how long difficulties last and whether performance returns to its earlier level. The mechanism is a separate question in each condition.

A large English community study measured cognition after COVID. Of 141,583 people who began an online battery, 112,964 completed it. Unresolved persistent symptoms were associated with a −0.42 SD adjusted global-cognition difference from the no-COVID group (95% CI −0.53 to −0.31). Smaller differences remained after reported symptom resolution. The study did not measure each person before infection; selection and confounding remain. These are group task-score differences, not observed individual IQ losses or evidence of irreversible decline. [70]

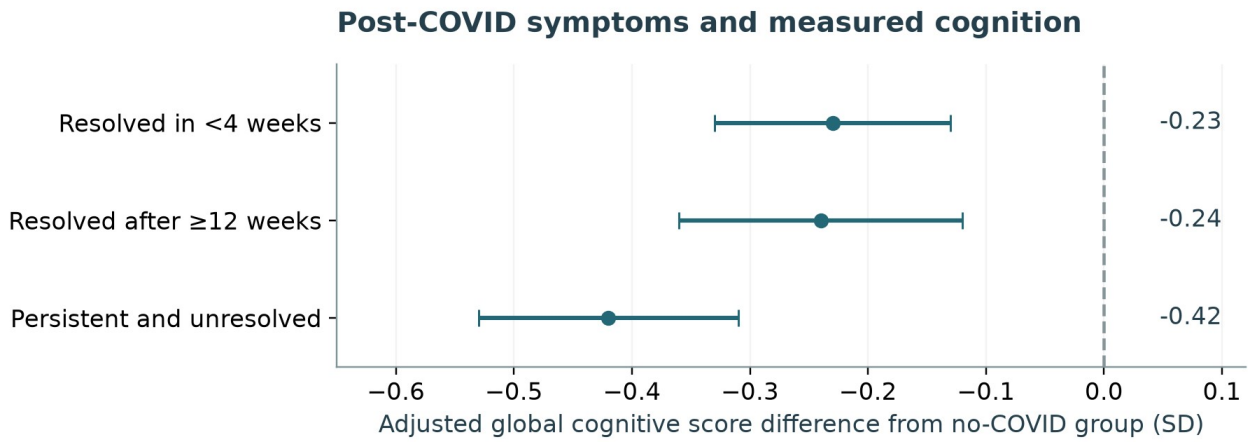


Figure 11. Published adjusted group differences with 95% confidence intervals, Hampshire et al. The reference group included people with no confirmed infection. Selected symptom groups from one community study are shown; they are not a pooled analysis or repeated stages in the same people. SD means standard-deviation units. No personal pre-infection cognitive baseline was available. [70]

13.1 Brain barriers and energy measures are distinct pieces of evidence

A Long COVID study associated cognitive symptoms with blood–brain barrier abnormalities and immune findings. Its dynamic-contrast MRI comparison included 10 recovered participants, 11 with Long COVID without brain fog and 11 with brain fog. This is a small mechanistic comparison, not evidence that ADHD shares its cause. [44]

A separate phosphorus spectroscopy study compared 27 post-COVID participants with 23 recovered controls. The authors reported lower ATP-to-phosphocreatine ratios and associations between anterior cingulate measures and cognition. A ratio is not a direct ATPproduction measurement. The barrier and spectroscopy findings came from different studies; combining them does not establish one causal chain in the same participants. [45]

Long COVID and myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) also require a different sampling timescale. Post-exertional malaise can involve delayed worsening after physical or mental exertion; CDC guidance describes typical onset 12–48 hours later and persistence for days or longer. That is not adequately captured by an immediate “before and after” test. [46]

For this programme, post-infectious cohorts require symptom-informed limits, low-burden measures and delayed follow-up. Maximal exercise should not be introduced merely to create a strong signal. The module is intended to study naturally occurring change or a clinically justified protocol, not provoke avoidable deterioration.

Autonomic and muscle findings need equally careful denominators. In 24 selected patients referred for post-COVID symptoms, 23 had orthostatic intolerance on tilt, but only four showed POTS; 15 developed intolerance after nitroglycerin provocation. That cannot estimate population prevalence. A separate exercise study compared 25 people with Long COVID and post-exertional malaise with 21 recovered controls, finding skeletal-muscle abnormalities. Muscle biopsies are not brain-energy measurements, and neither study demonstrates mediation of ADHD cognition. [71, 74]

13.2 The mast-cell findings conflict—and that is part of the evidence

An exploratory report in 13 symptomatic post-acute COVID participants found elevations in selected mast-cell-derived proteases. A later case-control study of 24 Long COVID participants and 24 recovered controls found no significant differences in two mast-cell proteases and no corrected associations with the physical-function measures tested. [47, 48]

Timing and assays differed; the later Long COVID group was sampled a median of 60 weeks after diagnosis. These differences are reasons for prospective replication, not proof that mast-cell activity starts early and then disappears. Nor do two null blood markers rule out every local or episodic process. The defensible conclusion is narrower: persistent systemic mast-cell activation is not established as a general explanation, and a subgroup claim needs prespecified, reproducible evidence. [48]

13.3 There are two different antihistamine trial questions

The blinded famotidine trial enrolled previously hospitalized adults with low cognitive screening scores about 29 days after symptom onset. The full-text flow reports 60 randomized, 50 analyzed after a first follow-up and 45 completing 12 weeks. The abstract instead calls 50 randomized. At week 12, endpoint contrasts favored famotidine by 2.44 MMSE points (95% CI 1.47–3.41) and 2.56 MoCA points (1.69–3.43). Excluding early withdrawals and carrying observations forward limit interpretation. The report labels MMSE primary and MoCA secondary, but says the registry did not distinguish them explicitly. Independent registry verification remains open. [49]

STIMULATE-ICP randomized 778 participants across four groups, with 193 assigned famotidine–loratadine. Against no study drug, the adjusted difference on the self-reported Fatigue Assessment Scale was −1.48 points at 12 weeks (95% CI −2.88 to −0.08). At 24 weeks, 12 weeks after treatment ended, the between-group difference was inconclusive. This was an open-label trial with fatigue as its primary outcome. [50]

These trials are not replications: early recovery after hospitalization and established post-COVID condition differ, as do single-drug and combination treatment, masking, and cognitive screen versus fatigue

endpoints. Neither establishes a mast-cell mechanism. A positive screen result or a small fatigue benefit warrants a better-targeted trial; it does not justify a general antihistamine claim.

The vagal research offers another possible explanation for a drug effect. In a mouse endotoxin model, famotidine reduced selected cytokines even in mast-cell-deficient animals, while vagal interruption or deletion of an acetylcholine receptor abolished the effect. This is a possible neural anti-inflammatory route, not a demonstrated human Long COVID mechanism. The model and administration routes differ from oral clinical use. It illustrates why benefit from an H2 drug need not identify mast cells as the cause. [51]

13.4 Mono gives the recovery question a starting point

The overlap with this hypothesis is prolonged fatigue, cognitive complaints and recovery after infection. A direct causal overlap with ADHD has not been established, and the studies below do not show that pre-existing ADHD changes recovery after mono. Section 13.6 separates that unanswered question from the broader postinfectious evidence.

Mono is relevant because it gives us a recognizable illness to follow forward. Infectious mononucleosis is most often caused by Epstein–Barr virus, although other infections can produce the syndrome. Having had EBV is not the same as having had mono. Most people with mono improve within two to four weeks, but fatigue can last longer. The important question here is what happens when recovery takes months rather than weeks. [158]

In a prospective study of 301 adolescents with mono, 39 met the study's chronic fatigue syndrome criteria at six months, 22 at 12 months and 13 at 24 months. The published rounded proportions were 13%, 7% and 4%. Figure 37 shows these counts against the original cohort. Some participants were lost to follow-up, so everyone outside those counts cannot simply be labelled recovered. These were clinical fatigue-syndrome classifications, not ADHD diagnoses or cognitive-test scores. [160]

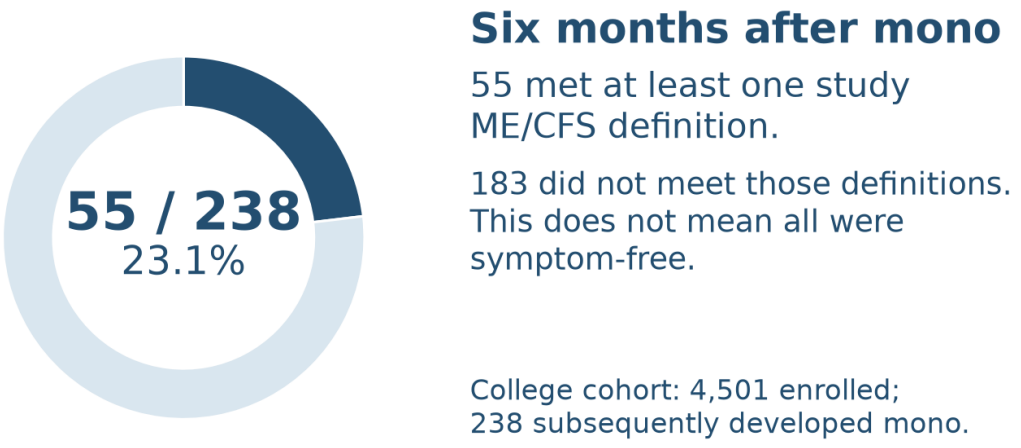


Figure 12. The denominator is the 238 students who developed mono, not all 4,501 enrolled students. Percentages are calculated from reported counts. This is a fatigue-syndrome classification, not a cognitive score or ADHD outcome. [161]

A college study adds something unusual: measurements from before the infection. Researchers enrolled 4,501 students; 238 subsequently developed mono, and 55 of those students—23%—met at least one of the study's ME/CFS definitions six months later. That percentage should not be averaged with the adolescent

estimate. The samples and diagnostic definitions differ. What matters for this hypothesis is the design: a person's condition before infection can be compared with their later course, rather than reconstructed entirely from memory. [161]

A 2026 follow-up of that college cohort assessed 104 selected participants about seven years later. Among those who returned, 11 of 14 from the earlier "severe" group and 7 of 23 from the "moderate" group met ME/CFS criteria. Those labels originally meant meeting multiple case definitions versus one. The results and figure report 11/14, approximately 79%; the abstract's 81% is inconsistent with that count. Small groups, attrition and selected follow-up prevent treating these figures as the long-term risk for everyone with mono. They do show that persistent illness can extend well beyond two years. [172]

EBV is not the only infection that can precede prolonged illness. In the Dubbo cohort, which combined EBV, Ross River virus and Q fever, 28 of 253 participants met chronic fatigue syndrome criteria at six months. Acute illness severity predicted persistence. This supports studying recovery across infections, while leaving open whether the processes that maintain symptoms are shared, pathogen-specific, or different between people. [162]

13.5 Brain fog, test performance and autonomic symptoms

The cognitive evidence is more mixed than the phrase "brain fog" suggests. A Dubbo study assessed 107 people during acute infection and retested a subset after recovery. Some working-memory, planning and attention measures were poorer during illness. This was a mixed-infection study from the same research programme, not an independent EBV replication. Recovery-selected retesting and practice effects also limit what the comparison can establish. [163]

At six months after EBV infection, another study compared 195 adolescents, grouped by fatigue status, with 70 healthy controls. The fatigued participants reported more cognitive difficulty, and the fatigued group scored lower on backward digit span before adjustment ($p = .037$). The difference was no longer detected after adjustment ($p = .445$). The other objective tests showed no differences in the main fatigue-group comparison. That does not establish equivalent ability in every setting, or make the symptoms imaginary. It tells us that a brief test and the experience of getting through a day may capture different things. [164]

A related study from the same Norwegian programme enrolled 200 adolescents after EBV infection and analyzed 195 at six months (91 fatigued and 104 non-fatigued), alongside 70 healthy controls. Fatigued and non-fatigued participants had similar infection markers and mostly similar immune, hormonal and autonomic measurements. There were some small differences, but no strong relationship between symptoms and the measured biological changes. Persistent fatigue did not correspond to a higher measured viral load. These overlapping samples should not be counted as independent confirmations. [165]

For our hypothesis, this is a reason to measure symptoms, sustained performance, standingrelated symptoms and recovery separately. A report of dizziness is not a diagnosis of POTS. A normal short task does not tell us how much effort it took, how long performance can be maintained, or what happens afterward. Those are questions for the proposed measurements, not conclusions supplied by the mono studies.

13.6 Is there a direct ADHD link?

The direct evidence located in this pass does not establish one. A US NHANES analysis had ADHD status available for 2,844 children and adolescents. EBV antibody positivity was not significantly associated with reported ADHD after adjustment: odds ratio 0.77, with a 95% confidence interval of 0.42–1.38. The same report found an association with reported learning disability, OR 2.76 (1.53–4.96). This was a cross-sectional survey using antibodies to past infection and reported diagnoses, with different available samples across outcomes. It did not measure mono severity, establish which came first, or test cognitive recovery in people with ADHD. [166]

The null ADHD result offers no basis for claiming protection, and the learning disability association does not establish that EBV causes ADHD. I would test a narrower question: does a substantial infection add fatigue or cognitive difficulty in people who already have ADHD? A prospective comparison by ADHD and infection status could test whether decline or recovery differs. I did not locate that comparison in this targeted search.

13.7 EBV and Long COVID: an unresolved connection

A study of 280 adults after COVID associated particular EBV antibody patterns with fatigue or subjective cognitive symptoms. It lacked pre-COVID samples, and antibody findings were indirect evidence about viral activity. The result generates a hypothesis; it does not show that EBV reactivation caused those symptoms or that treating EBV would improve them. [167]

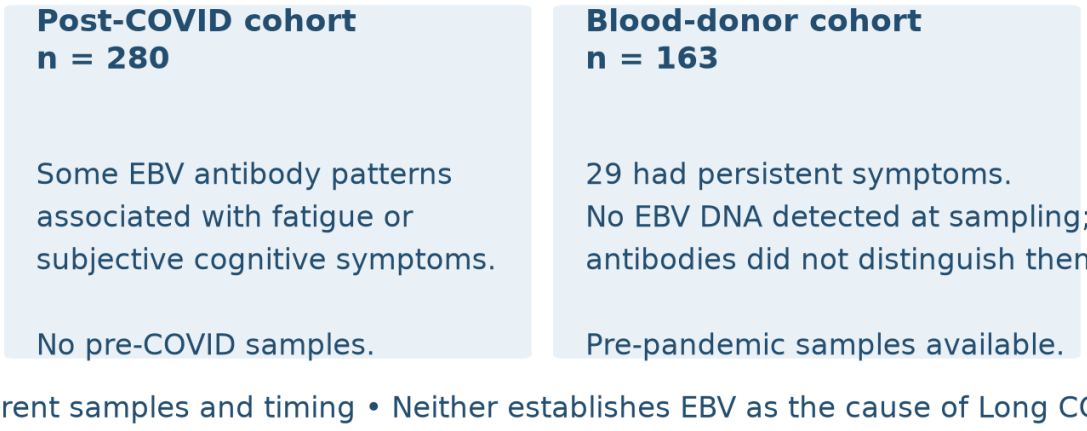


Figure 13. Evidence comparison, not a pooled estimate or demonstrated mechanism. Antibodies are indirect evidence of viral activity; the blood-donor study could miss transient or tissue-localized activity. The findings do not establish an ADHD link. [167, 168]

A different study could compare stored prepandemic samples in 163 previously EBV-infected blood donors. Among 29 with persistent post-COVID symptoms, EBV DNA was not detected at the sampled times, and antibody patterns did not distinguish the symptomatic group. Its selected, generally mild-infection population and sampling schedule leave room for transient or tissue-localized activity that blood testing could miss. Taken together, these studies do not support using EBV reactivation as a general explanation for Long COVID. [168]

The laboratory distinction matters. More than 90% of adults have evidence of past EBV infection. VCA IgG together with EBNA antibodies usually indicates past infection; high antibody levels can persist for years. Even early-antigen IgG can remain detectable in healthy people. A positive antibody result on its own cannot explain a person's current cognitive symptoms. [159]

13.8 What mono changes in the hypothesis

Mono strengthens the case for treating recovery as something to measure. It also forces us to separate an initiating event from whatever maintains a later problem. An infection can precede persistent illness without a high blood viral load continuing to drive it. Immune changes, sleep disruption, pain, autonomic symptoms and altered activity may be candidates, but the evidence above does not identify one shared pathway or establish a histamine mechanism.

I would record when the infection occurred, how the person functioned before it and whether they had ADHD. Where feasible, I would recruit before infection and follow people who recover as well as those who remain unwell. Repeated measures would track cognition, symptoms and everyday activity separately. A prespecified interaction would test whether ADHD predicts a larger change or slower recovery after accounting for baseline function and illness severity. Similar changes across groups would weaken the case for an ADHD-specific difference.

This remains an observational proposal, with the existing safeguards around post-exertional symptom worsening. It does not justify provoking a crash or replacing assessment for other causes of persistent symptoms. Nothing in these studies establishes antiviral, GLP-1 or experimental-peptide treatment for ADHD or post-mono brain fog.

"Chronic active EBV disease" also needs its proper meaning. It is a distinct systemic inflammatory illness involving EBV-infected T or NK cells, with specific diagnostic requirements. It is not a synonym for prolonged fatigue after mono or a high IgG result. [171]

13.9 What a before-infection baseline and a comparison group add

UK Biobank provides an important addition to the post-COVID evidence. Researchers compared repeat brain measurements in 785 adults aged 51–81: 401 infected between visits and 384 controls. The second scan was, on average, 141 days after diagnosis in the infected group. Infection was associated with greater change in some brain measures and slower performance on the Trail Making Test between visits; findings remained after excluding 15 hospitalized cases. Pre-infection measurements reduce the chance that a pre-existing difference is mistaken for an illness effect. They do not eliminate selection or confounding, identify a viral mechanism, establish permanence, or make the findings applicable to ADHD. Imaging and cognitive analyses also have their own available samples. [174]

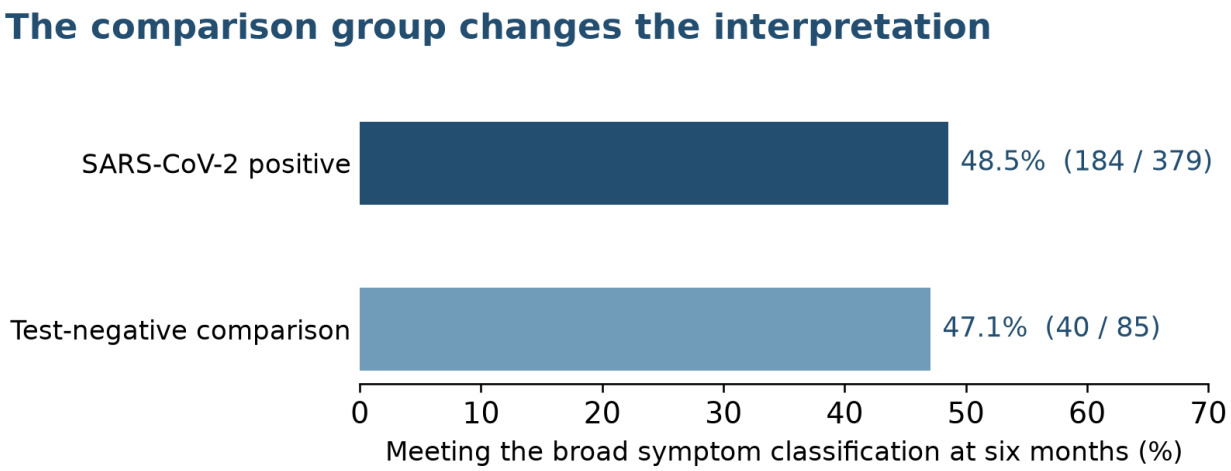


Figure 14. A broad persistent-symptom classification in nonhospitalized adolescents and young adults. The published difference was 1.5 percentage points (95% CI −10.2 to 13.1). Meeting symptom criteria does not give test-negative participants a confirmed Long COVID diagnosis. Selection, age and infection period limit generalization. [175]

A younger Norwegian cohort illustrates why comparison groups can change the answer. At six months, a broad persistent-symptom classification applied to 184 of 379 SARS-CoV-2positive participants and 40 of 85 test-negative participants: 48.5% versus 47.1%. The published difference was 1.5 percentage points (95% CI −10.2 to 13.1). Applying symptom criteria to test-negative participants does not mean they had confirmed Long COVID. The selected controls may have had more symptoms than the general population; participants were young, nonhospitalized and mostly infected in the Alpha period. This result cannot erase findings in other populations or prove a psychological cause. It shows how much an estimate can depend on the outcome definition and comparator. [175]

For a mono–ADHD study, I would therefore want both the person's previous function and a comparison group followed over the same period. The central question remains whether existing ADHD changes the response to infection. Similar symptoms after different illnesses make that question worth asking, but do not answer it.

13.10 Antiviral treatment is a test of a specific proposal

RECOVER-VITAL, reported in August 2026, randomized 964 adults with Long COVID; 959 entered the modified intention-to-treat analysis. Neither 15 nor 25 days of nirmatrelvir–ritonavir showed a significant primary benefit in the cognitive, autonomic or exercise symptom phenotypes. The cognitive analysis included 332 participants and used a patient-reported improvement threshold at day 90, not normalization of an objective cognitive test. Figure 42 shows its estimates and uncertainty. The comparator included ritonavir with nirmatrelvirmatching placebo. These results constrain the tested regimens; they do not disprove every viral-persistence mechanism or test EBV-directed treatment. [190]

RECOVER-VITAL is listed under both [190] and [U19]; these identify one trial. Its publisher correction added Marta Cerda as an author without changing the estimates shown in Figure 42 [N02]. A separate small placebo-controlled monoclonal-antibody trial failed its major efficacy comparisons [U20]. These results constrain the particular antiviral hypotheses and regimens.

13.11 Useful adaptation and cognitive restoration are different outcomes

RECOVER-NEURO randomized 328 adults with cognitive symptoms to five conditions over ten weeks. Computerized cognitive training, additional structured rehabilitation and active brain stimulation did not outperform their respective comparators on the primary self-reported cognitive-difficulty measure. For training versus its active comparator, the adjusted difference was 0.0 points (95% CI −0.2 to 0.2). Improvement within every group does not establish an intervention effect. More than half lacked performance-based impairment at entry, and an active comparator may itself help. [191]

CICERO asked a different question in 78 adults with objective cognitive impairment. Individualized rehabilitation improved attainment of trained personal goals versus usual care: an adjusted 2.88-point difference (95% CI 2.03–3.73) at three months and 1.72 (0.86–2.57) at six months. These were participant-reported goal ratings. Participants were unblinded, contact was not matched, and more than 40% of six-month cognitive tests were missing. Most cognitive and symptom outcomes did not differ. This supports a defined functional benefit, with limits; it does not establish general cognitive restoration. Different participants, comparators and endpoints prevent a direct ranking against RECOVER-NEURO. [192]

13.12 Cognitive, autonomic and recovery heterogeneity

A 109-person quantitative MRI/cognition comparison identified distributed diffusion and free-water-related features, without directly assaying inflammation or barrier leakage [U13]. A separate PET study in 24 cases associated lower striatal dopamine-VMAT2 binding with selected neuropsychiatric symptoms; cross-sectional binding cannot distinguish terminal loss from reversible expression or functional change [U14].

A large observational cognitive analysis indicated contributions from fatigue and objective attention/memory measures [U15]. In a separate 44-versus-44 comparison, greater subjective fatigue and lower resting HRV did not imply proportional immediate objective fatigability, and HRV did not predict fatigue [U16]. Measure capacity, effort, subjective symptoms and delayed post-exertional outcomes independently.

A retrospective German cohort associated incomplete recovery from a previous post-acute infection syndrome with worse subsequent symptoms and recovery [U17]. Prospective RECOVER follow-up also found heterogeneous remission trajectories [U18]. These designs motivate measuring recovery over time while accounting for baseline burden, self-report and selection.

CSF/serum studies in two neurological Long-COVID cohorts did not identify a reproducible shared brain-directed autoantibody pattern [U21]. A sham-controlled magnetic-treatment pilot in 30 people reported selected exploratory cognitive differences [U22]. EBV-linked immune shifts preceding MS relapse illustrate time-resolved immune sampling in another disease [U23]. None establishes a universal Long-COVID neural signature, routine magnetic-treatment efficacy or an EBV mechanism of ADHD.

14. GLP-1 and other peptides in integrated health

For each GLP-1 or related peptide trial, I start with the condition the drug was meant to treat and the outcome the trial was designed to change. I take the same approach to vagal stimulation in Chapter 15.

Trials in dementia and trials that found little cognitive benefit help set the limits of the proposed mechanisms. Chapter 4 covers medications studied as ADHD treatments.

Blood glucose, insulin signalling and HbA1c answer different questions. Glucose tracks circulating fuel availability; insulin helps regulate its use, while HbA1c summarizes glycemic exposure over preceding months. Insulin sensitivity describes responsiveness to that signal. Peripheral measures do not directly establish how the brain responds to insulin. Daily glucose excursions, long-term diabetes complications and a drug's receptor effects should therefore be tested as separate proposals. [209, 212]

14.1 How GLP-1 signalling could connect brain and immune state

GLP-1 is relevant because it allows a test of communication between metabolic signalling and other systems. Wong and colleagues found that anti-inflammatory effects in mouse Toll-like-receptor and sepsis models required central neuronal GLP-1 receptors rather than hematopoietic or endothelial receptors. Adrenergic and opioid signalling contributed to those effects. The experiment demonstrates a specific brain-to-immune route in those models; it does not establish that semaglutide improves human ADHD through that route. [52]

There is a shortcut to avoid here. An immune effect does not require assuming that every affected immune cell expresses the relevant receptor. Direct action on mast cells or microglia, indirect neural regulation, changes in body weight, and changes in other physiology are competing explanations that need different experiments.

Brain access is another distinction. In rodent tracing experiments, semaglutide reached selected brain regions through circumventricular and ventricular-adjacent sites, without broad passage across the BBB. Neural activity also changed in regions without direct drug access. A drug can therefore influence brain circuits through selective access and relays. This does not mean every GLP-1 drug has the same distribution, or that it repairs barrier leakage in humans. [211]

Immune effects can also involve cells other than mast cells. In an aspirin-exacerbated respiratory disease study, liraglutide reduced selected platelet activation responses outside the body. The Results section reports 33 affected participants recruited and 30 with usable samples, alongside 11 controls; the abstract and Figure 3 instead report 31 affected participants. That denominator discrepancy remains unresolved. Airway experiments were in mice. These findings suggest a peripheral route, not a clinical asthma, MCAS or cognitive treatment result. [203]

14.2 Human studies measure different outcomes

A 47-person uncontrolled MCAS case series reported clinical benefit in 89% after GLP-1 receptor agonist treatment. That is the proportion reported to benefit in this selected series, not an 89% reduction in symptoms or a controlled response rate. Variable treatments, cointerventions, expectation and the absence of a comparison group prevent causal inference. The series is a lead for a controlled trial; it cannot demonstrate direct mast-cell stabilization, BBB repair or an ADHD benefit. [53]

Two patients with treatment-resistant chronic spontaneous urticaria improved shortly after starting semaglutide or tirzepatide. In a separate report, clinicians considered tirzepatide a possible cofactor in exercise-induced anaphylaxis. Urinary leukotriene E4 fell after withdrawal in that case. These reports describe different clinical events in different patients and cannot establish a common mast-cell mechanism or its frequency. [G31, G32]

A separate placebo-controlled study of 48 adults with alcohol use disorder found that semaglutide changed laboratory drinking, craving and some—but not all—drinking outcomes over nine weeks. This supplies human evidence of behavioural effects relevant to reward, but it did not directly measure dopamine or test executive function in ADHD. Reduced craving and improved attention are not interchangeable outcomes. [54] Larger and newer alcohol-use trials are discussed in Section 14.6.

The proposed ADHD link remains conditional: a metabolic intervention might affect cognition directly, through symptoms and sleep, through food intake or another physiological route, or not at all. Body-weight change should be measured rather than dismissed. A nonsignificant weight change or a statistical adjustment for weight alone does not establish a weightindependent neural mechanism.

The stronger human GLP-1 evidence is outcome-specific. SELECT randomized 17,604 adults with overweight or obesity and established cardiovascular disease, without diabetes. Major cardiovascular events occurred in 6.5% on semaglutide and 8.0% on placebo (HR 0.80, 95% CI 0.72–0.90). That is vascular benefit in a high-risk population; attention and immune mediation were not established. Adverse-event discontinuation was more frequent, 16.6% versus 8.2%. [73]

A prespecified SELECT secondary analysis estimated a placebo-corrected change in geometric mean hsCRP of −37.8% at week 104 (95% CI −39.7% to −35.9%). Differences appeared by weeks four to eight and among participants without weight loss. The timing and subgroup pattern motivate mediation studies. Whether weight change explains part of the effect or peripheral hsCRP tracks inflammation in the brain remains open. [G23]

REWIND tested a closer cognitive question in type 2 diabetes. Among 8,828 participants with baseline and follow-up testing, the exploratory cognitive-impairment outcome gave HR 0.93 (95% CI 0.85–1.02). A post-hoc adjustment for baseline cognitive scores gave 0.86 (0.79– 0.95). Reporting only “14% lower risk” would hide that analytic distinction. The endpoint was a low MoCA or processing-speed score, not an incident dementia diagnosis or ADHD improvement. [72]

GRADE provides a larger cognitive comparison in type 2 diabetes. Of 5,047 randomized participants, 3,721 had baseline and year-four cognitive assessments. Adding liraglutide, insulin glargine, glimepiride or sitagliptin to metformin produced no significant between-group cognitive differences. Higher average HbA1c was associated with poorer performance, but that association was observational within the trial;

glucose targets were not randomized. Open-label treatment, follow-up selection, rescue insulin and modest glycemic separation limit the inference. The study did not demonstrate cognitive superiority for liraglutide in this population. [209]

The clinical benefits in the following sections remain valuable on their own endpoints. For this paper, a GLP-1 mechanism becomes more convincing if changes in the proposed mediator precede a cognitive change, the result survives appropriate comparisons, and simpler explanations are measured. Weight loss, improved sleep-disordered breathing, reduced symptoms and direct neural signalling are possibilities to distinguish, not competing stories we can settle from an individual treatment response.

14.3 Metabolic benefit during psychiatric treatment

HISTORI studied 154 antipsychotic-treated adults with schizophrenia, prediabetes and overweight or obesity. At 30 weeks, semaglutide reduced HbA1c by 0.46 percentage points (95% CI −0.56 to −0.36) and weight by 9.21 kg (−11.68 to −6.75) relative to placebo. Physical quality of life improved by 3.75 SF-36v2 points (1.52–5.98). Average schizophrenia symptom ratings and mental quality of life did not show significant benefit; cognition was not an efficacy endpoint. Gastrointestinal symptoms were more frequent, and repeated hospitalizations occurred in a few treated participants. Metabolic care can matter during psychiatric treatment without proving a cognitive or immune mechanism. [189]

14.4 Where treatment of overlapping illness has stronger evidence

For each treatment, we need to ask which illness improved, in whom and on what measure. GLP-1 receptor agonists such as semaglutide and the dual GIP/GLP-1 agonist tirzepatide now have substantial human evidence outside glucose control. These results are relevant because sleep, pain, circulation and illness burden could affect a person's available capacity. That last connection remains a hypothesis unless cognition and daily function are measured.

Sleep apnea. SURMOUNT-OSA comprised two placebo-controlled trials totaling 469 adults with obesity and moderate-to-severe obstructive sleep apnea. At 52 weeks, tirzepatide reduced breathing interruptions substantially more than placebo (Figure 15). One trial enrolled people without positive airway pressure treatment; the other enrolled people receiving it. For the latter, pressure treatment was withheld before endpoint assessment under the study protocol. This does not justify stopping prescribed treatment. Gastrointestinal adverse effects were common. Improved sleep-related reports are relevant, but the trials do not establish an ADHD cognitive effect. [113]

A later randomized-trial synthesis likewise found fewer breathing events with GLP-1-based therapies, with substantial variation across studies. Its pooled breathing endpoint adds context to SURMOUNT-OSA. Attention and executive function would need separate testing. [G24]

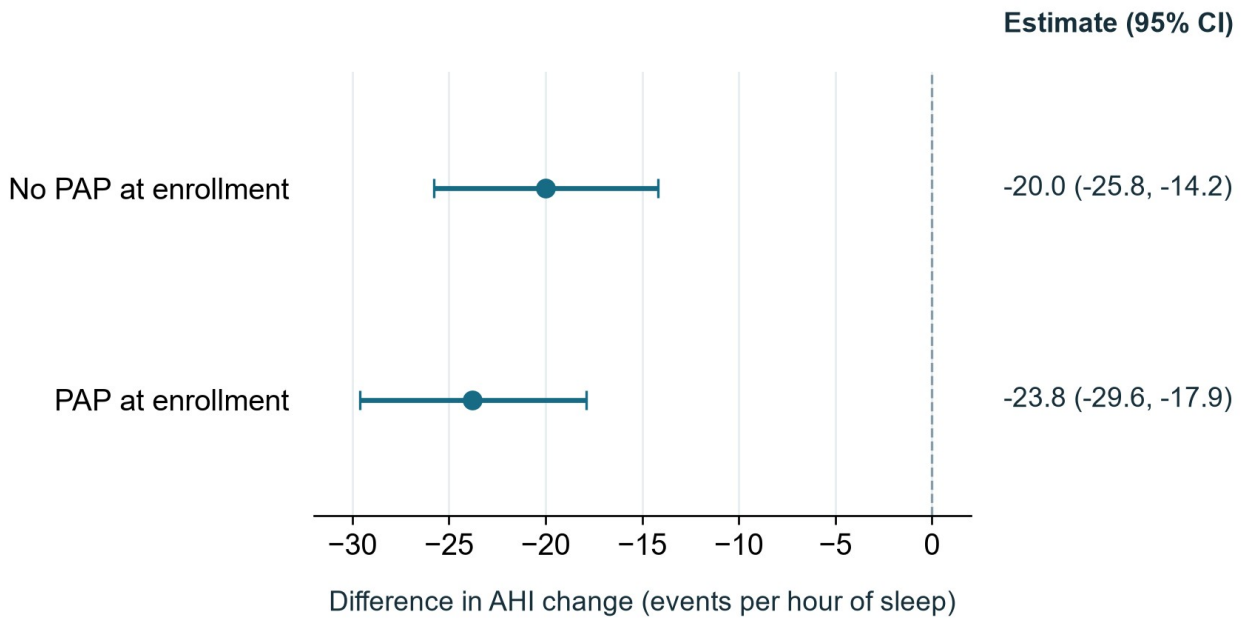
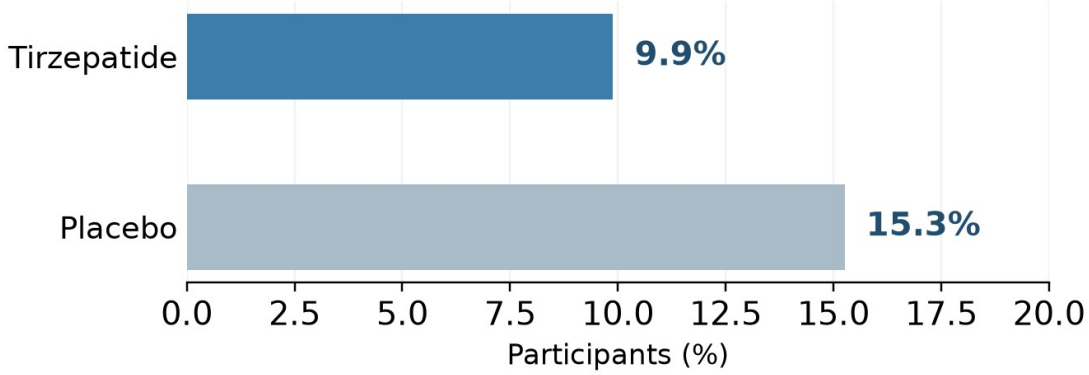


Figure 15. Tirzepatide minus placebo at 52 weeks, with 95% CIs. Lower favours tirzepatide. AHI counts breathing interruptions; PAP means positive airway pressure. No pooling. [113]

Heart failure. In STEP-HFpEF, 529 people with obesity and heart failure with preserved ejection fraction were randomized for 52 weeks. Semaglutide improved the Kansas City Cardiomyopathy Questionnaire clinical summary score by an additional 7.8 points (95% CI 4.8–10.9) on its 0–100 scale, and six-minute walking distance by 20.3 metres (8.6–32.1). These are symptoms, physical limitations and exercise capacity, not attention tests. [114]

SUMMIT tested a harder clinical outcome in 731 people with the same broad heart-failure phenotype and obesity. Over a median 104 weeks, cardiovascular death or worsening heart failure occurred in 36/364 participants on tirzepatide versus 56/367 on placebo: 9.9% versus 15.3%, HR 0.62 (95% CI 0.41–0.95). The advantage was driven by fewer worsening heart failure events; cardiovascular deaths were 8 versus 5, with wide uncertainty. Calling this a demonstrated mortality benefit would be wrong. [115]

SUMMIT heart-failure outcomes



Heart-failure event comparison. SUMMIT: cardiovascular death or worsening heart failure in 731 people with preserved-ejection-fraction heart failure and obesity. Events: 36/364 on tirzepatide and 56/367 on placebo over a median 104 weeks. Percentages calculated from these counts. [115]

The hazard ratio was 0.62 (95% CI 0.41–0.95). Fewer worsening-heart-failure events drove the benefit; this does not establish a mortality benefit or ADHD efficacy. Bars show observed proportions, not annual risks or hazard ratios. See Section 14.4.

Pain and mobility. STEP 9 randomized 407 people with obesity and knee osteoarthritis for 68 weeks. WOMAC pain scores, on a 0–100 scale, fell by 41.7 points with semaglutide versus 27.5 with placebo; weight fell by 13.7% versus 3.2%. A meaningful pain result can coexist with uncertainty about mechanism. These data do not demonstrate cartilage regeneration, an effect independent of weight loss, or improved executive function. [116]

Kidney disease. FLOW randomized 3,533 people with type 2 diabetes and chronic kidney disease. Over a median 3.4 years, its primary composite occurred in 331/1,767 on semaglutide and 410/1,766 on placebo, HR 0.76 (95% CI 0.66–0.88). The endpoint combined kidney failure, sustained major filtration loss, and kidney or cardiovascular death. It is not a 24-percentage-point reduction in kidney failure alone. This is strong organ-outcome evidence in a defined population, without a demonstrated cognitive mediator. [117]

14.5 Liver and immune disease sharpen the mechanism question

ESSENCE provides tissue-level evidence in metabolic dysfunction-associated steatohepatitis, or MASH: liver inflammation and injury associated with metabolic dysfunction. Its planned 72-week interim analysis covered the first 800 of 1,197 randomized participants with moderate or advanced fibrosis. Figure 16 shows two biopsy outcomes favouring semaglutide. These are histological responses, not yet proof of preventing cirrhosis-related death or improving cognition. Gastrointestinal events were more frequent, and the bodily-pain outcome did not show a significant between-group difference. A positive liver result is not a general symptom result. [118]

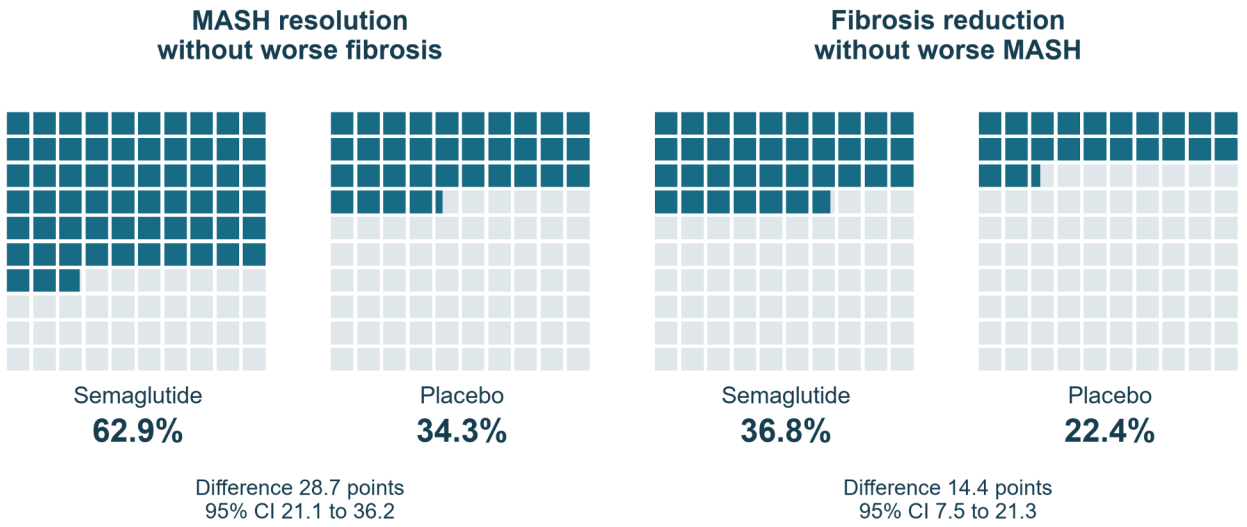


Figure 16. ESSENCE liver-biopsy responses at 72 weeks. Each grid represents 100%; one square is one percentage point, with partial squares for decimals. Blue shows the reported response proportion; grey is its complement. Squares are not individual participants. The two outcomes overlap and must not be added. First 800 randomized: semaglutide n=534, placebo n=266. Published differences and 95% CIs are retained. [118]

The smaller SYNERGY-NASH trial gives a related signal with tirzepatide. Among 190 randomized participants, 157 had evaluable week-52 biopsies; missing outcomes were handled by imputation. MASH resolution without worsening fibrosis occurred in 44–62% across the three tirzepatide groups versus 10% on placebo. This phase 2 study supports further testing. Its numbers cannot be used as a head-to-head comparison with ESSENCE because the trials differ. [119]

Psoriasis offers a direct overlap experiment. TOGETHER-PsO randomized 274 adults with psoriasis and overweight or obesity to the immune-targeted antibody ixekizumab, with or without tirzepatide. Treatment was open-label, but skin assessors were masked. At week 36, adding tirzepatide improved the joint skin-clearance/weight-loss primary endpoint and the key secondary skin-clearance endpoint (Figure 17). Nausea occurred in 24.6% versus 2.2%. This demonstrates added benefit in that treatment setting. It cannot separate immune signalling from weight change, drug exposure or other routes, and it did not test cognition. [141]

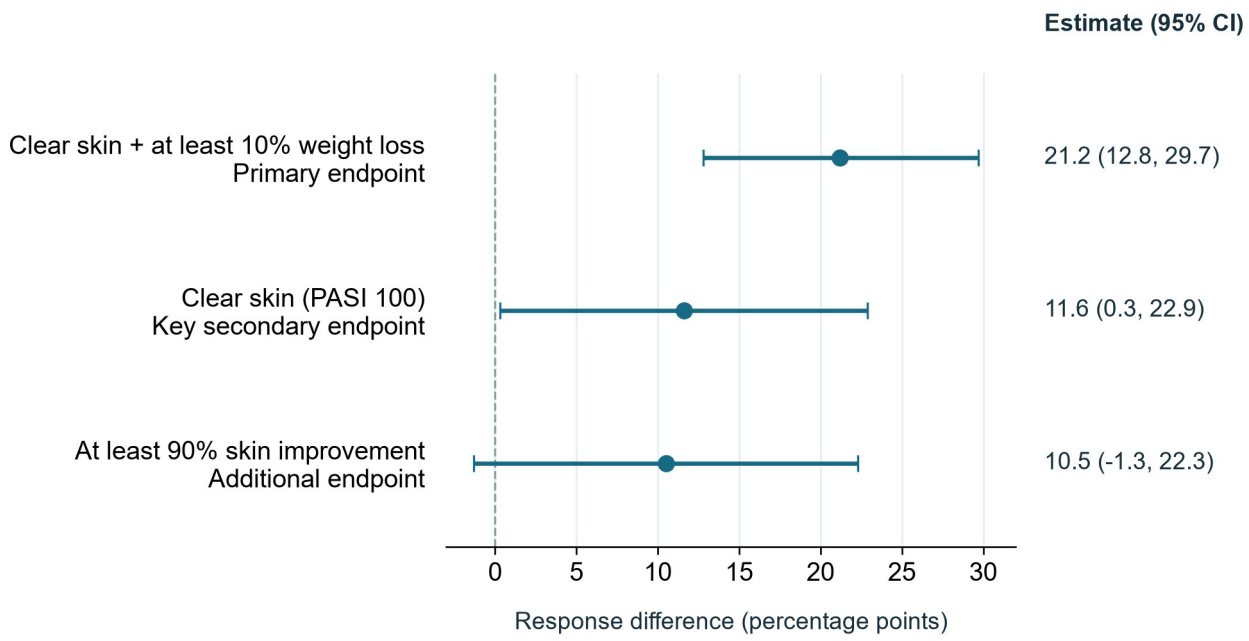


Figure 17. Combination minus ixekizumab alone; higher favours combination. Published 36-week contrasts and 95% CIs. The model estimates outcomes without discontinuation or prohibited treatment, imputing missing data. The additional endpoint lacks multiplicity control. [141]

Rheumatoid arthritis evidence is less secure. A retrospective study compared 173 people who took a GLP-1 drug with 42 who were prescribed one but did not start. Pain changed by −0.6 versus +1.3 cm on a visual scale. Access, adherence, other treatments and baseline differences can affect that comparison; roughly one-third of treated patients discontinued. A randomized trial would be needed to test whether the drug modifies the disease. [127]

The inflammatory bowel disease findings are also mixed. One electronic-record study reported lower five-year odds of surgery among GLP-1 users, OR 0.36 (95% CI 0.29–0.46). [128] A separate 2026 target-trial emulation in stable IBD found relapse rates of 7.9% in both groups in its first matched cohort, RR 1.01 (0.82–1.24). Its advanced-therapy cohort was also inconclusive, RR 0.80 (0.55–1.15). [129] The populations, outcomes and follow-up differ. An odds ratio for surgery and a risk ratio for relapse should not be pooled or treated as direct contradictions. Neither study randomized treatment; “emulation” describes the design, not removal of confounding.

14.6 Reward and headache outcomes need their own tests

The alcohol-use evidence has progressed beyond the small trial in Section 14.2. A 2026 trial randomized 108 adults with alcohol use disorder and obesity to semaglutide or placebo, with cognitive behavioural therapy in both groups. Over 26 weeks, the reported treatment difference in percentage of heavy-drinking days was −13.7 percentage points (95% CI −22.0 to −5.4); 88 participants completed treatment. Gastrointestinal effects were more common. This is a clinically relevant behavioural result in a specific population. It does not measure dopamine restoration, executive function or ADHD treatment response. [120]

A separate eight-week oral-semaglutide trial randomized 50 adults. Its primary laboratory cue-induced craving outcome was not significantly different, while the key secondary heavy drinking-days outcome

favoured treatment; drinks per day did not clearly differ. Reporting only the positive measure would hide the outcome hierarchy. Together, the studies support further alcohol-use treatment research without establishing one uniform effect across formulations, populations and outcomes. [121]

Headache evidence is earlier. In an uncontrolled 31-person pilot involving obesity and frequent or chronic migraine, headache days fell from 19.8 to 10.7 per month after 12 weeks of added liraglutide. Without a control group, expectation, other treatment and regression to the mean remain plausible contributors. A nonsignificant weight change does not establish a weight-independent mechanism. [122]

Idiopathic intracranial hypertension is a distinct condition. A small exenatide trial randomized 15 women and found lower intracranial pressure at 24 hours: difference −4.7 mm Hg (95% CI −8.8 to −0.5). At 12 weeks the interval crossed zero, −4.1 (−8.4 to 0.1); the study had prespecified a more permissive significance threshold of 0.10. This is an early pressure result, not a general migraine trial. [123] The related exploratory cognitive analysis found no significant between-group cognitive advantage. Improvements within the treated group alone cannot supply that missing comparison. [124]

The pressure trial also reported an effect at 2.5 hours. A later systematic review reported generally favorable headache and papilledema findings, with variable visual and pressure results and substantial reliance on observational evidence. Intracranial pressure, perfusion and fluid clearance require distinct measurements. [123, G38]

14.7 Endometriosis and Long COVID remain open questions

There is a recent endometriosis signal, but its design matters. A February 2026 preprint analyzed 161 survey participants with confirmed or suspected endometriosis who had used GLP-1 therapy. Retrospective reports of symptom improvement are vulnerable to selective participation, recall, co-treatment and changes in weight or menstrual state. They do not establish lesion regression, a shared cause with ADHD, or cognitive benefit. A prospective controlled study should separate pelvic pain, bleeding, sleep, fatigue and attention rather than combine them into “feeling better.” [130]

For Long COVID, a tirzepatide trial had enrolled 1,058 participants in an August 2026 preprint. The report describes recruitment, baseline characteristics and the operation of a remote trial; it does not report treatment efficacy. Forecasts of improvement are not observed treatment effects. As of this review, that report cannot support a claim that tirzepatide treats Long COVID fatigue, autonomic dysfunction, blood–brain barrier disruption or mitochondrial impairment. Those are separate outcomes needing results. [131]

In a separate POTS study, fasting insulin and C-peptide correlated with blood pressure in participants with POTS. Other gut hormone comparisons were mixed. The associations motivate measuring metabolic and autonomic responses together. The study did not assign an incretin treatment. [G33]

14.8 Other peptides are different treatments

Metabolic trials now offer comparisons across hormone targets. The phase 3 TRIUMPH-1 obesity trial tested retatrutide, which activates GIP, GLP-1 and glucagon receptors. The 12 mg arm showed greater weight loss than placebo at 80 weeks. In REIMAGINE 2, 2,713 adults with type 2 diabetes were randomized. Higher-dose cagrilintide plus semaglutide improved HbA1c and weight more than either component alone. These trials tested metabolic outcomes. [G26, G27]

The phase 2 ZUPREME 1 study randomized 493 adults to petrelintide or placebo and found greater weight loss at week 28 with the amylin analogue. Phase 2 work on selective amylin agonist eloralintide and on zenagamtide, which acts at GLP-1, amylin and calcitonin receptors, also tested metabolic endpoints. A prospective comparison would need to measure weight trajectory, sleep, fatigue and cognition in parallel to assess whether outcomes differ by pathway. [G28, G29, G30]

“Peptide” describes how a molecule is built. It does not identify a shared therapeutic target, benefit or safety profile. An oral IL-23 receptor antagonist, a GLP-2 analogue and a GnRH agonist ask different clinical questions. Table 3 includes positive disease-specific examples alongside direct brain-related trials that did not confirm benefit. Antibodies such as ixekizumab are identified separately as protein biologics; sharing an immune target does not make treatments interchangeable.

Table 3. Different peptide targets and their measured outcomes

Treatment and target	Population and comparison	Measured result and boundary
Icotrokinra IL-23 receptor peptide [132]	Plaque psoriasis; 684 randomized; placebo comparison at 16 weeks.	Clear/almost-clear skin with ≥2-grade improvement: 65% vs 8%. ≥90% severity improvement: 50% vs 4%. Skin endpoints; no cognitive test.
Teduglutide GLP-2 analogue [134]	Short bowel with intestinal failure; 86 randomized; 24 weeks.	Weekly intravenous support volume reduced by 20–100% at weeks 20 and 24: 27/43 (63%) vs 13/43 (30%). Not general gut repair.
Leuprolide GnRH agonist [140]	Endometriosis; 201 randomized to add-back regimens; everyone received leuprolide.	Pain improved from baseline. Randomization tests add-back, not leuprolide versus placebo. Bone-density trade-off discussed below.
Intranasal insulin [135]	Mild cognitive impairment/Alzheimer dementia; 289 randomized; primary device cohort 240; 12 months.	ADAS-cog12 cognitive score difference 0.026 points (95% CI −1.771 to 1.822): no primary advantage. Delivery-device problems limit generalization.
Intranasal oxytocin [136]	Autism; 290 enrolled, ages 3– 17; placebo comparison; 24 weeks.	Modified social-withdrawal score difference −0.2 (95% CI −1.5 to 1.0). No primary benefit; this was not an ADHD trial.

Table sources (in row order): [132], [134], [140], [135], [136].

Icotrokinra is particularly useful here because it is a targeted oral peptide with a positive immune-disease trial, rather than a general “immune support” claim. The FDA approved it in March 2026 for eligible adults and adolescents with moderate-to-severe plaque psoriasis. That is a US regulatory indication, not approval for ADHD or cognitive symptoms. [133]

The endometriosis example also makes the trade-off concrete. In the leuprolide study, everyone received the GnRH agonist and randomization concerned hormonal add-back. Pain improved from baseline across groups, but this was not a placebo-controlled estimate of leuprolide's pain effect. After a year, the group without add-back lost 6.3% of bone density; add-back groups preserved it. The hormone state created by treatment belongs in the model alongside any symptom relief. Cognition was not measured. [140]

14.9 Negative trials and research peptides set limits

Neurological plausibility has not produced a consistent GLP-1 treatment effect. A 156-person phase 2 Parkinson's trial of lixisenatide found a favourable motor difference at 12 months, 3.08 points (95% CI 0.86–5.30). [125] A separate 194-person phase 3 exenatide trial did not confirm benefit on its 96-week motor endpoint: adjusted difference +0.92 points (−1.56 to 3.39). [126] That exenatide report is subject to a Lancet expression of concern and should not be treated as secure evidence for the absence of benefit [N03]. Different drugs and on/off-medication assessments prevent a direct numerical comparison. Neither motor result is a cognitive outcome. The negative Alzheimer trials in Section 14.11 remain a further boundary.

The FDA granted accelerated approval of elamipretide for a defined Barth syndrome population based on knee extensor strength. A confirmatory trial is required. This is a mitochondrial-targeted treatment example with a specific muscle outcome and a rare genetic indication. [G37]

The same standard applies to products sold as research peptides. FDA staff's 2026 BPC-157 review identified a small ulcerative-colitis trial reported only as a meeting abstract and several limited human reports. Human reports exist, but their designs and reporting do not establish the broad claims made for systemic treatment. [137]

For the nominated TB-500 fragment and MOTS-c products, the FDA reviews identified no clinical studies involving administration to humans. Evidence about full-length thymosin beta-4 cannot simply be assigned to the shorter TB-500 fragment. Measurements of naturally occurring MOTS-c, animal experiments and studies of modified analogues do not establish benefit from injected MOTS-c. These were staff evidence reviews for a compounding advisory process, not efficacy trials or final approval decisions. [138, 139]

14.10 Where mitochondria, mTOR, AMPK and glia belong

Immunometabolism means that cellular energy management and immune behaviour interact. A TREM2 study combined human Alzheimer tissue observations with cellular and mouse experiments, linking impaired microglial metabolic function to defective mTOR signalling and altered responses to pathology. It demonstrates a cell-specific immune–metabolic relationship; it does not establish the same defect in ADHD. [55]

The earlier mTOR/AMPK proposals fit here as candidate regulatory mechanisms, not as a master switch that should always be turned up or down. The study above is a reminder that reducing one signalling pathway is not inherently beneficial. Cell type, timing and the process being measured matter. Peripheral mitochondrial markers or muscle findings should not be relabelled as direct measurements of neuronal energy supply.

A metabolic measure needs to do more than look abnormal alongside the symptoms. It should predict the relevant cognitive or functional outcome and add something beyond simpler explanations.

14.11 Dementia is a different timescale, not the endpoint of ADHD fog

The TREM2 work makes glial and metabolic biology relevant to neurodegeneration, but that is not a disease progression from ADHD through Long COVID to Alzheimer’s. Mechanistic overlap does not establish that a fluctuating symptom is an early form of a progressive disease. [55]

In 3,808 participants across evoke and evoke+, oral semaglutide did not slow early symptomatic Alzheimer disease. At 104 weeks, differences in change on the Clinical Dementia Rating–Sum of Boxes were −0.08

points (95% CI −0.35 to 0.20) and +0.10 (−0.17 to 0.38), respectively. Lower values favor semaglutide; both intervals span zero. Biomarker improvement cannot replace this clinical outcome. These trials do not settle prevention in other populations, but their null results are not positive evidence for this hypothesis. [56]

A separate systematic review of GLP-1 agonists and SGLT2 inhibitors found exploratory cognitive, imaging and biomarker signals, with inconsistent clinical cognitive benefit and low certainty. The review adds context to the trial results and may include overlapping evidence. [G39]

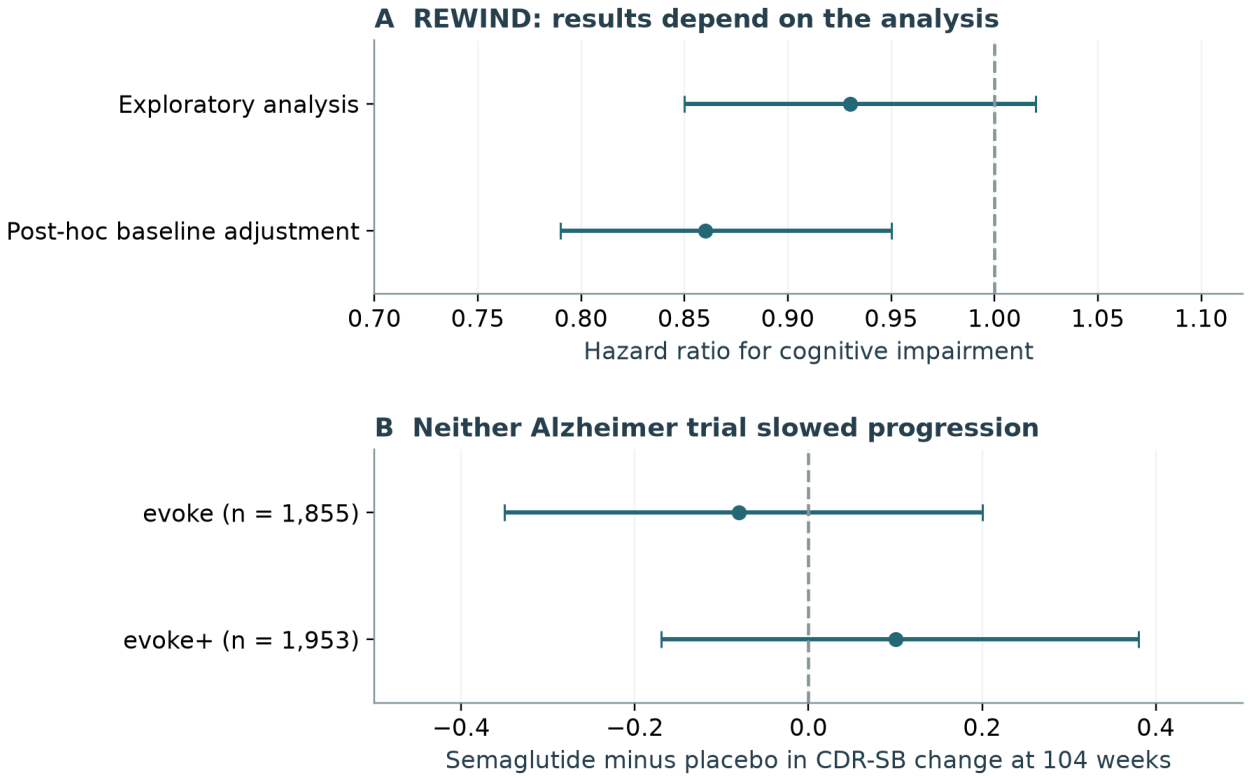


Figure 18. Cognitive outcomes do not follow automatically from a plausible mechanism. Panel A: two analyses of the same exploratory REWIND outcome, not independent trials. Panel B: two separate Alzheimer trials. Bars are 95% confidence intervals; lower values favor active treatment. The panels use different populations, drugs, endpoints and scales and must not be pooled. CDR-SB measures dementia severity through cognition and function. [56, 72]

The ELAD trial gives a mixed liraglutide result. Among 204 randomized participants with mildto-moderate Alzheimer disease and no diabetes, the 52-week primary brain glucosemetabolism endpoint did not differ significantly from placebo: −0.17 in PET standardized uptake value (95% CI −0.39 to 0.06). A secondary cognitive composite favored liraglutide by 0.15 z-score units (0.03–0.28), but the secondary analyses were hypothesis-generating, without confirmatory multiplicity control. Daily-living and global clinical ratings did not show significant differences. Only 154 participants had the primary follow-up scans. The cognitive signal deserves further testing; it is not a 15% improvement, an established diseasemodifying effect or ADHD evidence. [188]

14.12 Integrated health interventions need a measured cognitive endpoint

US POINTER provides a direct test of a broad lifestyle package in a different population: 2,111 adults aged 60–79 at risk of cognitive decline. Both groups received lifestyle guidance; the structured programme added support and intensity across exercise, diet, cognitive and social activity, and cardiovascular monitoring. Global cognition improved by an additional 0.029 SD per year (95% CI 0.008–0.050) over two years. Figure 19 shows that the domain estimates were not uniform. The gain is modest, and the package cannot identify which component caused it. It does not establish ADHD benefit or dementia prevention. [97]

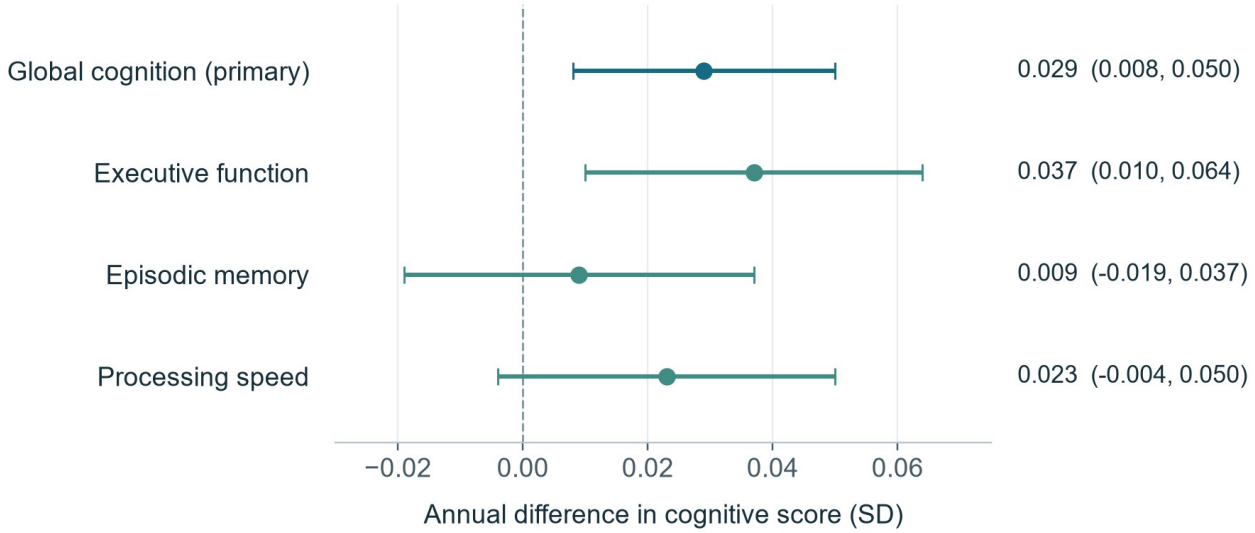


Figure 19. A small average cognitive gain from a structured lifestyle programme. US POINTER Table 2: annual differences versus self-guided intervention, with 95% CIs. Global cognition was primary; the domains are related secondary outcomes from the same participants. Higher values favour the structured programme. Memory and processing-speed intervals include zero. Everyday clinical meaning and durability remain separate questions. [97]

Vascular treatment supplies another boundary. SPRINT randomized 9,361 adults aged 50 or older with hypertension and elevated cardiovascular risk, excluding diabetes and prior stroke. Assigned blood-pressure treatment lasted 3.3 years. After a median seven years of cognitive follow-up, intensive versus standard treatment gave HR 0.89 (95% CI 0.79–0.99) for mild cognitive impairment or probable dementia, but 0.86 (0.72–1.02) for probable dementia alone. Extended ascertainment reached 4,232 of 7,221 eligible participants. This supports a specified vascular intervention and outcome, while leaving attrition and dementia-specific uncertainty visible. It cannot be carried over to normal blood pressure, orthostatic intolerance or an ADHD treatment target. [98]

14.13 A corrected health measure may not produce a cognitive benefit

Vitamin B12 provides a useful example. A trial's eligible cohort included 201 adults aged 75 or older with moderate biochemical deficiency, but without anemia or neurological or cognitive symptoms. Daily B12 for 12 months improved vitamin status. It did not improve the main cognitive measure: the adjusted difference on the first three learning trials of the California Verbal Learning Test was −1.4 words (95% CI −2.9 to 0.1), analyzed in 91 treated participants and 93 controls. Cognitive outcomes were secondary to the trial's neurological outcome. This does not argue against treating severe or symptomatic deficiency. It shows why a corrected blood result cannot stand in for a demonstrated cognitive benefit. [156]

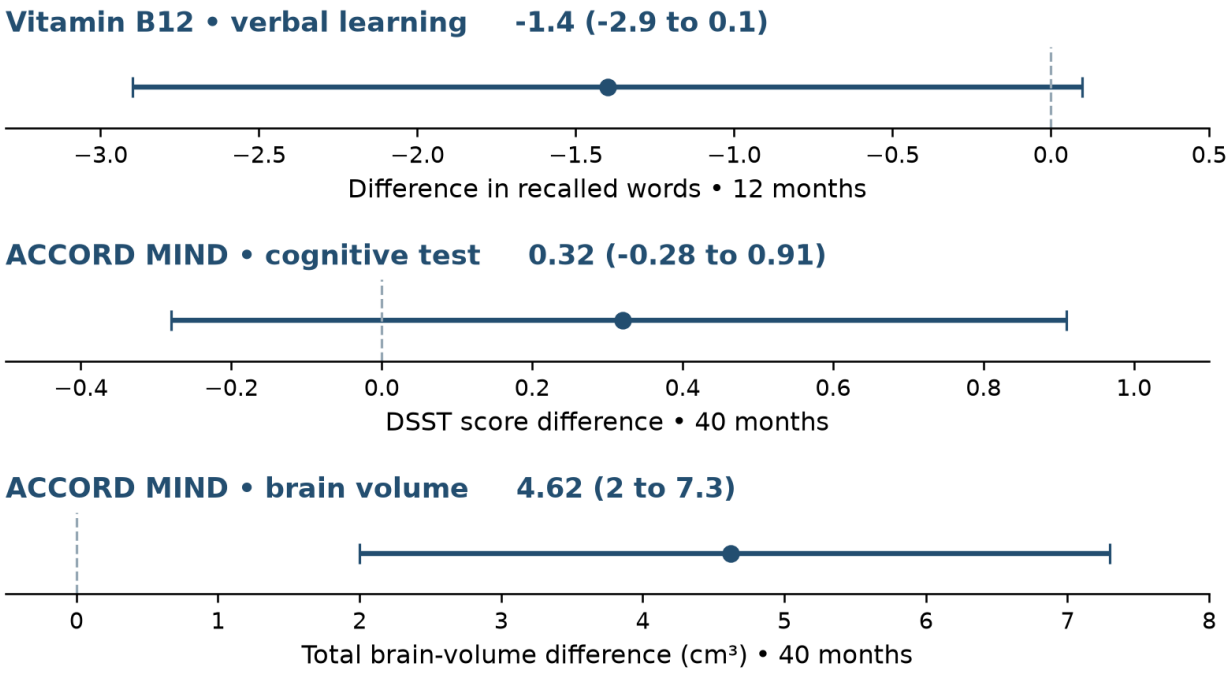


Figure 20. Published between-group estimates with 95% CIs, on separate axes and in different units. B12 verbal learning: 184 analyzed. ACCORD cognitive follow-up: 2,794; MRI subset: 503. The two cognitive intervals cross zero; the brain-volume interval does not. Do not pool these outcomes. The intensive glucose strategy was stopped for increased mortality in the parent trial. [156, 157]

ACCORD MIND makes a related point about glucose and brain imaging. It enrolled 2,977 adults with type 2 diabetes and high cardiovascular risk. Targeting HbA1c below 6%, versus 7.0–7.9%, did not improve the principal cognitive outcome, the Digit Symbol Substitution Test, at 40 months: the adjusted difference was 0.32 points (95% CI −0.28 to 0.91). The cognitive analysis included 2,794 people with follow-up. An MRI subset of 503 nevertheless showed 4.62 cm³ more total brain volume with intensive treatment (95% CI 2.00–7.30). The intensive strategy was stopped because of increased mortality in the main trial. These results concern that treatment strategy and population; they do not test modern GLP-1 therapy or ADHD. [157]

For the proposed studies, I would record whether a change on a scan or in a laboratory measure is accompanied by a change on a cognitive task, in symptoms or in everyday function. Report each result separately so a positive marker cannot obscure a null cognitive finding.

14.14 Hearing and access to the task

The ACHIEVE trial randomized 977 adults aged 70–84 with untreated hearing loss to hearing intervention or health education. The overall three-year cognitive result was null. A prespecified analysis found less decline in the 238 participants recruited from the Atherosclerosis Risk in Communities (ARIC) cohort, but no demonstrated benefit in the other 739 volunteers. The groups differed in baseline risk, and the interaction was statistically significant. Figure 36 shows the estimates together. The often-quoted 48% slowing refers to cognitive-score decline in the ARIC subgroup, not a reduction in dementia risk across the trial. [155]

Before interpreting an attention score, I would check whether the participant could comfortably see and hear the instructions. The study would record any access difficulty alongside the score.

14.15 Metabolic and cognitive endpoints

A review of 37 human neuroimaging studies found GLP-1 drug effects that varied with population, metabolic phenotype and exposure duration [U26]. A seven-trial synthesis of 4,132 older participants with Alzheimer disease or MCI found no convincing average cognitive or progression benefit and more adverse events [U27]. This substantially overlaps the EVOKE/EVOKE+ trials in [56], so the synthesis is not an independent set of participants.

In a metformin/placebo study of 40 older adults with insulin resistance, selected brain glucose-uptake, structural, connectivity, processing-speed and working-memory outcomes were preserved or improved [U53]. The proposed cellular mechanism comes from separate animal work. A 72-person semaglutide trial in major depression missed its prespecified executive-function endpoint. A planned secondary global cognition measure favored semaglutide [U54]. Two further analyses of the same trial reported more choices of known high-value options, directed exploration and choice repetition, and greater willingness to choose physical effort at higher expected reward. Modeling suggested reduced effort discounting. These tasks measured choice behavior in depression. Independent replication and effects on daily functioning remain open. [N01, G22]

An electronic-record study of 13,007 adults with psychiatric diagnoses associated semaglutide initiation with fewer clinician-recorded cognitive symptoms than several comparators. The difference was unclear against sitagliptin, and attention and orientation showed no significant advantage in domain analyses. Treatment selection and clinical documentation limit interpretation. These records measure recorded symptoms rather than performance on an objective executive-function test. [G25]

A 15-study case-control meta-analysis reported modest ADHD-related differences in malondialdehyde (g=0.32), 8-OHdG (g=0.27) and total oxidant status (g=0.35), with heterogeneous antioxidant findings [G07]. Peripheral redox markers are nonspecific and do not measure mitochondrial flux or brain ATP. The mitochondrial review cited in the evidence-gap work also describes limited direct ADHD evidence [G09].

Metabolic state, effort, adaptation to sleep loss and neurofluid physiology remain separate propositions. AMPK, mTOR, glia, immunometabolism and incretin signalling identify candidate levels of explanation. A convincing mediator test would show a specified physiological change preceding objective cognition while excluding weight change, sleep, sedation and motivational alternatives.

14.16 Late-life risk and cognitive reserve

In an EHR/genomic cohort of 187,341 older adults, ADHD diagnosis and polygenic liability were associated with later recorded mild cognitive impairment or dementia. Diagnosis timing, residual confounding and reverse causation remain important; this study does not test an acute state-change or reserve mechanism [U07]. A separate longitudinal cohort of 3,119 older adults found that a residual-defined cognitive-resilience score modified dementia risk alongside pathological burden [U31]. A statistical residual does not identify a molecular pathway. Both studies concern a different timescale from the daily regulation proposed here, and neither establishes one common resilience scale.

14.17 What the treatment results mean

The trials report benefits on specific sleep, cardiac, renal, liver, pain and skin outcomes in the populations studied. Some behavioural outcomes are promising too. The route from those changes to attention remains largely unmeasured in these studies. A person can benefit from treating an overlapping illness even if it does not explain their ADHD.

I would start measuring before people begin clinically indicated treatment. Then I would follow the targeted illness, medication exposure, sleep, food intake, hydration, pain and objective cognition over time. Reports of brain fog and everyday functioning would have their own measures. Where feasible, random assignment or an active treatment that produces similar weight loss could help separate a drug effect from the effects of weight change and other circumstances. Statistical adjustment can narrow the possibilities. Testing mediation would require further evidence.

An improved bodily endpoint with no meaningful cognitive change would weaken the proposed route for that population, outcome and time window, given adequate precision. If cognition tracks pain or sleep more closely than the proposed immune marker, test those routes first. Track adverse effects alongside benefits. These studies motivate focused tests, while clinical use for ADHD would require its own evidence.

15. Vagal signalling: what human interventions establish

An implanted vagal device changed an arthritis outcome in a human trial. The ear and neck studies ask a different question about stimulation and cognition. Their results help decide what to measure in a direct ADHD study.

15.1 Neural control can change immune disease in humans

In RESET-RA, 242 people with inadequately controlled rheumatoid arthritis received an implanted cervical vagus-targeted system and were randomized to active or sham stimulation for three months. ACR20 arthritis response occurred in 35.2% versus 24.2%: a raw difference of 11.0 percentage points. The stratification-adjusted difference was 11.8 points (95% CI 0.6– 23.1), with a prespecified one-sided primary test. Device-related serious events occurred in 1.6% and were perioperative. Later follow-up was open-label. This is human immune-disease intervention evidence, not cognitive benefit. [57]

The arthritis result does not demonstrate cognitive benefit or ADHD treatment. It also cannot establish equivalence with ear stimulation or the safety of a repurposed TENS unit.

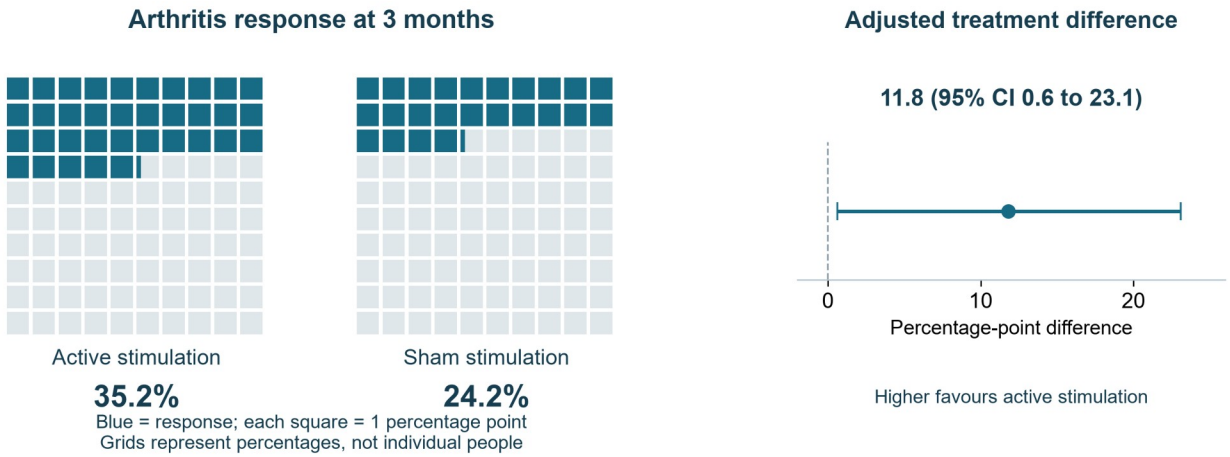


Figure 21. An implanted vagal intervention changed an arthritis outcome. RESET-RA, 242 participants: ACR20 response at three months. The observed proportions differ by 11.0 percentage points; the published stratification-adjusted contrast is 11.8 points (95% CI 0.6–23.1). The primary test was one-sided. This is immune-disease intervention evidence, not cognitive benefit or equivalence with ear stimulation. [57]

Implanted vagal stimulation and arthritis

The raw difference is 11.0 percentage points. The published adjusted difference was 11.8 points (95% CI 0.6–23.1), using a prespecified one-sided primary test. This is evidence for an arthritis outcome, not cognitive improvement or equivalence with ear stimulation. See Section 15.1.

Compare the populations and outcomes of the treatment studies before drawing a conclusion about cognition.

15.2 The cognition and HRV findings answer different questions

A 2026 review included 53 auricular vagus-stimulation studies, 30 in quantitative synthesis. The pooled cognitive effect was Hedges g = 0.41 (95% CI 0.30–0.53); working-memory/attention studies gave a smaller estimate, 0.185 (0.043–0.327). This follows the abstract and main results; the figure description gives a lower limit of 0.29. The populations, protocols and outcomes varied. These are standardized differences, not percentages improved, and neither estimate is ADHD-specific efficacy or proof of an anti-inflammatory route. [58]

A separate Bayesian synthesis of 16 sham-controlled studies in healthy participants found essentially no acute effect on cardiac vagally mediated HRV (g = 0.014; shortest credible interval −0.103 to 0.132). This is a Bayesian credible interval, not a conventional confidence interval. The result limits HRV as a check that auricular stimulation engaged its intended target; it does not show that every neural effect is absent. [59]

Separate evidence of target engagement from evidence of cognitive benefit. Ear-site fMRI responses can differ across sites. In the 33-person ear–neck study, an exploratory salivary alpha-amylase rise followed neck stimulation even though modeling suggested poor vagal engagement. A 94-person crossover found no clear pupil dilation advantage over a sensation-matched sham. These findings make HRV, pupil size and salivary biomarkers insufficient on their own to verify which nerve was engaged. Pair a credible target-engagement assessment with an appropriate control, a masking check and direct measures of cognition and adverse effects. [N23, N24, N30]

The Vita retail manual gives no research sham specification, and I found no peer-reviewed trial of that model in this review. Research devices from BrainCLOS appear in a pediatric headache protocol, a 26-person sham-controlled conference abstract reporting less menstrual pain, and a manufacturer report on heart-rate variability. [N35, N36, N37, N40]

Researchers used the pain-relief TENS 7000 with specialized ear electrodes. Knee osteoarthritis pain improved versus sham. A low-back-pain trial compared two electrically active ear sites and found no clear clinical difference. ADHD outcomes remain untested. [N34, N38, N39]

15.3 Network state, effort and regulatory capacity

A preregistered consortium spanning 40 datasets found poor convergence among proposed behavioural and neural indicators of emotion-regulation capacity [U32]. Human intracranial insula recordings/stimulation identify organized sensory and salience territories, but do not establish successful top-down control of immune state in ADHD [U34].

A small rTMS experiment found that cognitive-control responses depended on whether the targeted network was engaged during stimulation [U35]. Exploratory depression MEG phenotypes and an EEG/pharmacology study of language similarly show state, task and neurochemistry interactions [U33, U37]. None identifies an immune route or an established ADHD intervention.

Across six intensive longitudinal cohorts (n=1,088), fatigue and unusually fast performance were associated with subsequent fatigue/performance dynamics. The small observational effects do not establish cellular-energy depletion [U36]. FKBP5 identifies a stress-feedback candidate, but the cited review is from 2015 and is not an ADHD state-cognition experiment [U38]. Measure effort and fatigue alongside task results, adjusting prospective models for prior performance, context and medication.

15.4 External devices and evidence for cognition

Ear and neck stimulators deliver current through the skin at different sites. Ear devices aim at sensory fibers of the auricular vagus branch. Neck devices aim at the deeper vagal trunk. Transcutaneous electrical nerve stimulation (TENS) describes a way to deliver current. The site and waveform shape which nerves it may recruit. In a study of 33 adults, modeling suggested that the matched neck settings would miss the vagal target, leaving the ear and neck effects unresolved. [N16, N23, N30]

External VNS devices and the evidence for cognition

Device	Intended target	Attention and executive-function findings	Untested in ADHD
Nuropod	Auricular vagus branch.	Flanker scores improved in an uncontrolled Parasym Long COVID pilot. PK01 showed an immediate-recall signal. [N13, N14, N22]	Retail-model effects on attention, executive control and daily functioning.
Nurosym	Auricular vagus branch via tragus.	One model suggested less vigilance slowing in 48 sleep-deprived conscripts. Inhibition showed no clear benefit. [N21]	Vigilance on ordinary days and effects of repeated use.
Research TENS setup	Varies by site. Ear setups aim at the vagus branch.	FlexiStim tested fatigue. A separate stroke trial found no clear advantage on Trail Making attention and switching measures. [N16, N25]	Attention and executive function with a specified setup.
gammaCore	Cervical vagal trunk.	Mixed vigilance and multitasking results after sleep loss. Faster vocabulary learning in one study. A visual-learning replication was inconclusive. [N26, N27, N31, N32]	Attention and executive control outside sleep loss.
Pulsetto	Cervical vagal trunk.	The acute study measured EEG and blood pressure. Cognitive performance was untested. [N28]	Objective attention and executive function versus sham.
Truvaga	Cervical vagal trunk.	The manufacturer manual describes the device. No controlled cognitive result was identified in this review. [N20]	Objective cognition and daily functioning by model.

Parasym describes shared technology in Nuropod and Nurosym. The PK01 trial enrolled 76 nonclinical adults and found inconclusive results for the broader memory measures. In other ear-device studies, NEMOS did not reduce vigilance loss. The tVNS501 study found immediate improvements in conflict and task switching after cognitive load. A separate experiment changed autonomic measures without improving stress-task performance. [N07, N08, N13, N14, N15]

TENS and condition-specific outcomes. An ear-electrode FlexiStim protocol missed its primary post-COVID fatigue endpoint at eight weeks. In a 90-person chronic stroke trial, auricular stimulation added to exercise improved global cognitive screening and episodic memory compared with sham plus exercise. Trail Making measures of attention and switching showed no clear advantage. A separate insomnia trial found a 4.2-point greater improvement in self-reported sleep quality at eight weeks (95% CI 2.6 to 5.9). [N16, N18, N25]

GammaCore has U.S. clearances for specified headache indications. PRESTO missed its prespecified two-hour pain-free endpoint, while earlier time points favored active treatment. The vocabulary study enrolled military linguists. Pulsetto and Truvaga are consumer neck devices with their own protocols. [N17, N19, N20, N26, N29]

Application to ADHD. Adult ADHD efficacy remains unestablished. The pediatric symptom and cognition reports tested the trigeminal nerve through the forehead and found no superiority to sham on the reported outcomes. RESET-RA used an implanted vagal device and measured arthritis response. For a future noninvasive study, specify the model, stimulation site, waveform, dose and sham. Match the sensation and assess masking. Track attention, effort, real-task completion and adverse effects separately from autonomic or sleep measures. [57, N12, U04]

15.5 Proposed Nurosym trial in adults with ADHD

The proposed trial begins with a one-week baseline and a randomized feasibility pilot. If masking, tolerability and data quality are adequate, adults with confirmed ADHD would be assigned 1:1 to active or sham stimulation for a provisional six weeks alongside stable usual care. In the Nurosym field study, both versions delivered current at first. The sham current faded to zero in the first minute while active stimulation continued. The study did not formally assess whether participants could tell them apart. This trial would verify output from matched Nurosym units, use identical placement and routines, ask for assignment guesses and mask assessors. Vita could enter feasibility testing with a verified matched sham. See Section 15.2. [N21, N24]

If the pilot shows that the 60-target task is repeatable and leaves room for improvement, missed targets per 60 valid presentations would be the primary outcome. Repeat testing in the final two weeks before that day's stimulation, at a consistent time and interval from the preceding session. Record false alarms and response speed separately. For executive function, a prespecified cued task-switching test would measure the extra time needed to switch rules alongside accuracy. [N08]

Daily functioning would be measured with an existing digital handoff identified before randomization. Use verified submission times to determine whether each eligible handoff met its deadline. Record minutes to spare separately and distinguish confirmed misses from unavailable records. Collect the Weiss Functional Impairment Rating Scale at baseline and study end. Appendix L defines the handoff. [N33]

Compare participants as randomized, accounting for baseline performance and repeated observations. Report absolute differences with 95% confidence intervals. Prespecify the handling of missing data and the two key secondary comparisons. Record sleep, medication timing, adherence and adverse effects. Use pilot data to size the trial for a meaningful difference and revisit the provisional threshold of one miss per 60. An inconclusive handoff result would leave the effect on everyday function uncertain.

16. Mental health, trauma and the wider environment

A difficult day may involve poor sleep, pain, treatment effects or demands that changed. The studies in this chapter help decide which of those to measure in ADHD and which warrant a separate study.

Someone can have ADHD, depression, painful bowel disease and a difficult history at the same time. The task is to find out what each contributes. Counting overlaps does not tell us which condition came first, which route matters today, or whether treating one will change the others.

16.1 Mental health belongs inside the model

A meta-analysis of PTSD included 60 studies and 4,108 participants. The largest average differences included verbal learning (d = −0.62), processing speed (−0.59) and attention/working memory (−0.50). These are differences between groups, not percentages of ability lost. Comparison groups included people with and without trauma exposure; study selection, coexisting ADHD and other clinical differences affected estimates. The findings establish cognitive relevance, not whether the difference began before the trauma or resulted from a particular immune pathway. [99]

Depression also needs more than a mood score on test day. A synthesis of 252 studies compared 11,882 people in remission after depression with 8,533 controls. Differences remained on 55 of 75 cognitive variables. That is 73% of the variables examined, not 73% of people, and it is not proof of permanent damage. Recovery of mood and recovery of cognition should be measured separately. [100]

For this hypothesis, depression, anxiety, intrusive memories and dissociation are candidate contributors, outcomes or confounders depending on the question. Their role cannot be decided after seeing which adjustment gives the preferred result. Bipolar disorder, psychotic disorders and substance-related conditions also require their own clinical histories; a shared complaint of poor concentration does not make their mechanisms interchangeable. This edition does not attempt a separate evidence review of each.

16.2 Trauma is relevant; ADHD is not a synonym for trauma

The temporal evidence is mixed. In the E-Risk cohort of 2,232 British twins, childhood abuse/neglect did not predict young-adult ADHD after accounting for childhood ADHD. Childhood ADHD predicted later maltreatment particularly with coexisting conduct disorder. This identifies a need for protection, never responsibility on the child's part. [101] An analysis of 2,480 children and adolescents from the Boricua Youth Study found some prospective evidence involving physical abuse and ADHD, as well as concurrent associations. Different exposure windows, informants and adjustments can change the answer. Neither cohort settles every form of adversity or trauma-related inattention. [102]

Trauma may affect someone's sleep, arousal, avoidance or experience of a setting today. The origin of their ADHD is a separate question. With consent, I would ask about timing and current symptoms and give participants a choice about how much to disclose. An adversity score describes reported exposures and needs separate evidence to support a biological explanation.

16.3 Stress, autoimmunity and infection: association has a direction

In Swedish registers, 106,464 people diagnosed with stress-related disorders were compared with 1,064,640 matched people without that diagnosis. Subsequent autoimmune disease rates were 9.1 versus 6.0 per 1,000 person-years; the adjusted hazard ratio was 1.36 (95% CI 1.33–1.40). These were diagnosed stress-related disorders, not an experiment in everyday stress. The study also used siblings, but residual differences in health, exposure and care-seeking remain possible. It measured autoimmune diagnoses, not cognition. [103]

Other studies start with immune disease or infection and follow later mental health diagnoses. A Danish cohort of 3.56 million people recorded 91,637 first mood-disorder diagnoses through hospital, outpatient or emergency contacts. Figure 25 shows higher subsequent diagnosis rates after autoimmune disease, hospitalized infection, or both. These histories may mark several pathways, including illness burden and treatment. They do not show that infection persists or that mood symptoms are caused by brain inflammation. [104]

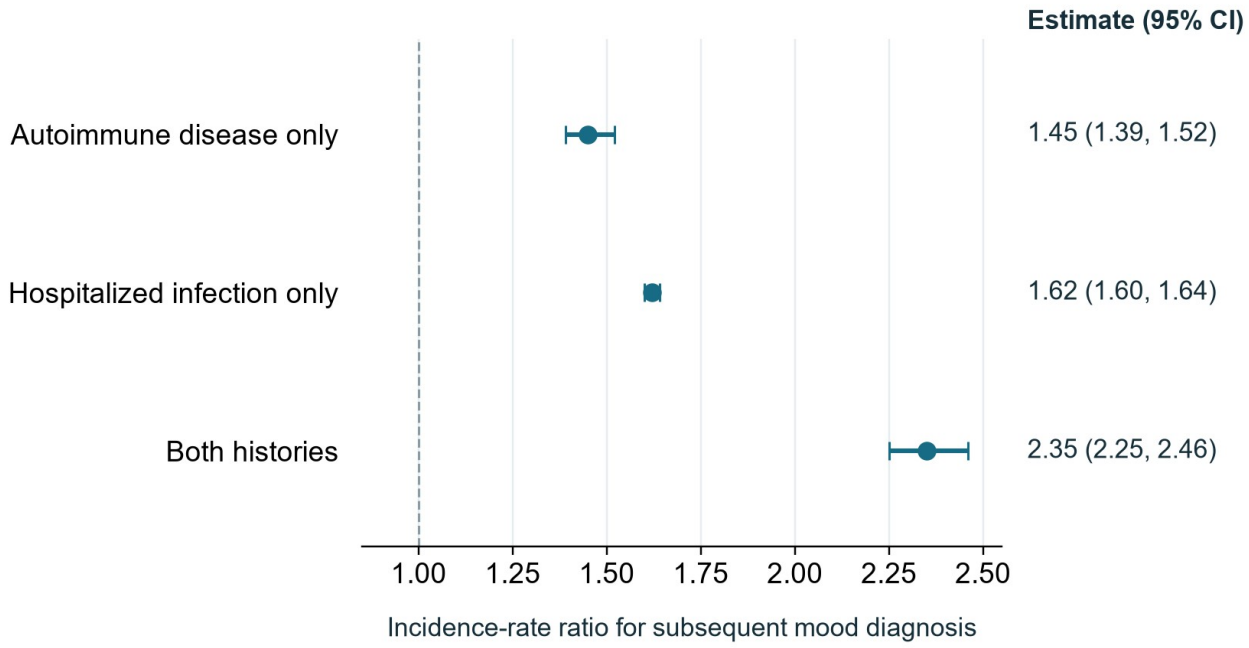


Figure 25. Immune and infection histories predict later mood diagnoses in a register study. Table 2 estimates, with 95% CIs, adjusted for age, sex and calendar year; the reference is neither history. "Only" excludes the other exposure category. A ratio of 2.35 does not mean a 235% probability. Statistical interaction depends on the scale; the joint estimate is not proof of biological synergy. [104]

An immune-mediated condition can have cognitive relevance without establishing a microbiome cause. In an inflammatory bowel disease synthesis, all 12 included studies were cross-sectional (687 participants; 11 studies contributed to meta-analyses), generally involving younger adults with inactive disease. Attention and executive measures differed from controls, while learning/recall and broad screening estimates were inconclusive. Pain, fatigue, nutrition, mood and medication remain competing routes. This is not evidence that cognition tracks a flare within one person. [105]

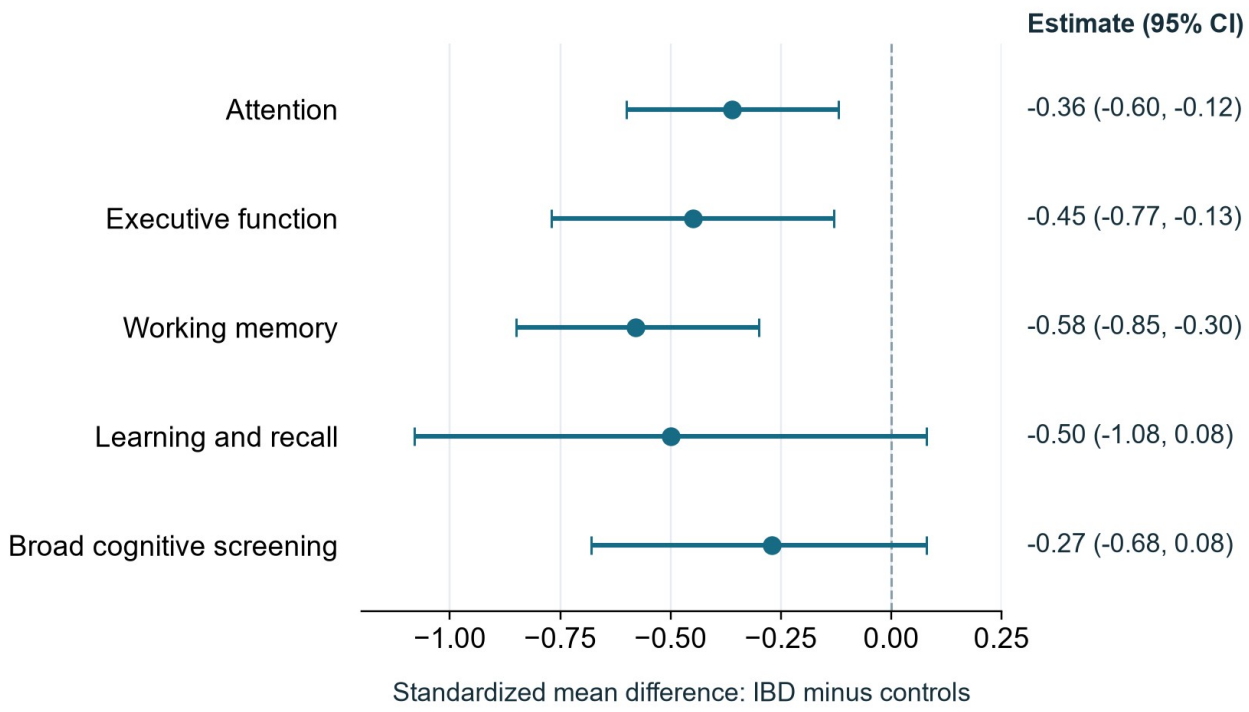


Figure 26. Cognitive domains do not all give the same answer in inflammatory bowel disease. Published standardized differences with 95% CIs; negative values indicate poorer average performance. Working memory overlaps with executive function, so these are not independent replications. The broad-screening interval is reproduced as published. The studies cannot identify a cause or measure decline over time. [105]

An autoimmune label alone tells us too little. We need the diagnosis, treatment, timing and organs involved before proposing a route to cognitive change.

16.4 Infections can be causal in a specific disease without explaining ADHD

Epstein–Barr virus and multiple sclerosis provide a stronger example of temporal evidence. The underlying military cohort included more than 10 million people and 955 MS diagnoses; the analysis with available serum samples included 801 cases and 1,566 matched controls. Among those initially EBV-negative, becoming antibody-positive rather than remaining negative was associated with an MS hazard ratio of 32.4 (95% CI 4.3–245.3). The large estimate has wide uncertainty. It concerns EBV seroconversion, not a mono diagnosis or a 32% absolute risk. The absence of a similar cytomegalovirus association strengthens the specific etiological case. It does not establish ongoing infection as the cause of ordinary fog or an antiviral treatment for ADHD. [106]

Long COVID already supplies a distinct postinfectious setting in Section 13. Acute infection, postinfectious symptoms, latent infection and active infection are different exposures. Combining them into “infection load” would lose the timing and biological distinctions the hypothesis needs. Prospective work should record documented episodes, severity, recovery, sleep and treatment, rather than infer an infectious cause from a symptom checklist.

Autoimmune encephalitis is a separate clinical boundary. The diagnostic approach concerns a subacute syndrome—typically progression within three months—with memory, mentalstatus or psychiatric changes

and supporting neurological, laboratory or imaging features after alternatives are considered. Chronic concentration difficulty alone is insufficient; normal routine tests do not by themselves exclude every case. Rapid new confusion, seizures or neurological decline warrant prompt assessment rather than interpretation through this framework. [107]

There is also a specific molecular lead in MS. Lanz and colleagues identified antibodies that cross-reacted with an EBV protein, EBNA1, and a nervous-system protein, GlialCAM. This is an example of molecular mimicry: an immune response can recognize both a viral target and part of the body. The finding concerns a subset of MS biology. It is not evidence that the same mechanism explains ADHD or ordinary post-mono fog. [169]

Lupus offers another, still limited, connection. A 2025 study examined B cells from 11 people with systemic lupus erythematosus and 10 controls. Mean EBV-infected cell frequencies were about 25 versus 1 per 10,000 sequenced B cells. The researchers also found programmes that could help infected, self-reactive B cells present targets to other immune cells. That is a cellular frequency difference in established disease, not a 25-fold lupus risk after mono. It supports mechanistic work, not a treatment conclusion or a common explanation for cognitive symptoms. [170]

16.5 Gut bacteria: a plausible route with several missing steps

A microbiome study analyzed 1,054 Rotterdam participants and replicated associations in 1,539 Dutch HELIUS participants after exclusions. Thirteen microbial taxa were associated with depressive symptoms. The association replicated. Whether those organisms can guide a diagnostic test or improve mood when altered remains unproven. The genetic analyses were exploratory and did not remove uncertainty about direction. [108]

The microbiome argument requires several steps. Bacteria must differ in a relevant way, their activity must change a signal, and that signal must affect a meaningful outcome. Finding a microbial difference does not complete that sequence. A bacterium’s ability to produce a chemical says little about how much reaches the human brain. Diet, bowel transit, antibiotics, other medication and sample handling need to be recorded before assigning a causal role.

The intervention evidence deserves the same restraint. A depression pilot included 49 treated participants in its analysis, testing a multistrain probiotic added to ongoing antidepressants for eight weeks. Its purpose included feasibility and estimating effects for a later trial. Figure 27 shows all four week-eight symptom measures: two intervals cross zero. These correlated outcomes are not four replications, and none was an objective cognitive endpoint. [109]

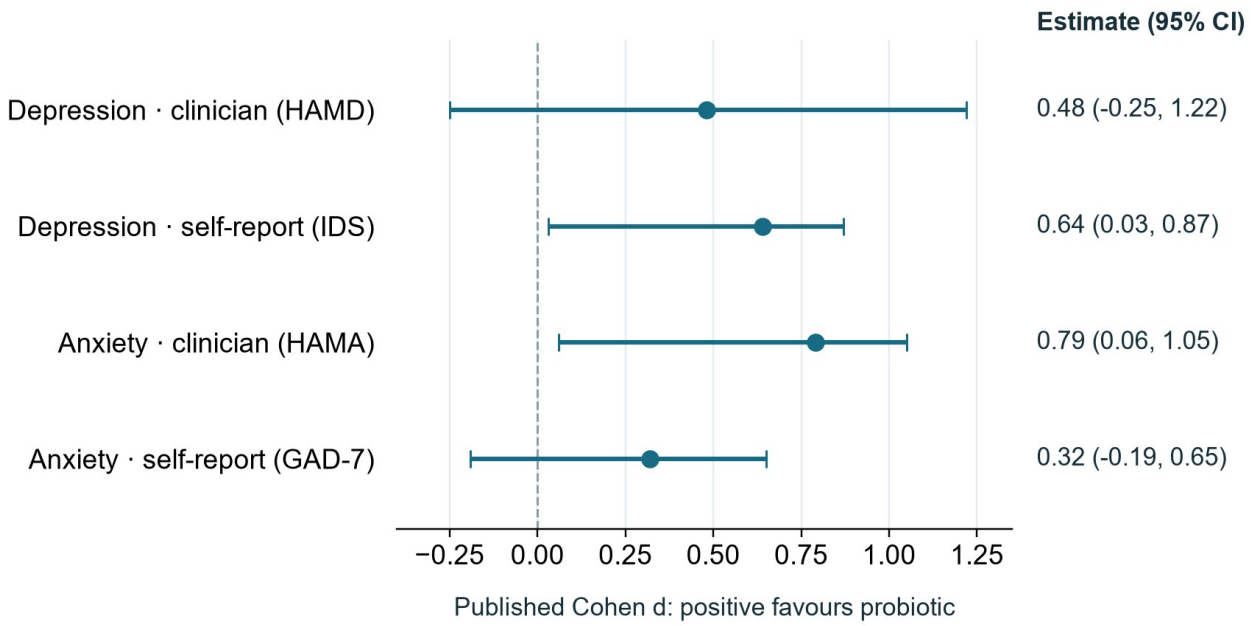


Figure 27. A promising pilot does not give one uniform answer. Published week-eight Cohen d estimates and 95% CIs from Table 2, before the small-sample correction; asymmetric intervals are retained as printed. HAMD/IDS measure depression; HAMA/GAD-7 measure anxiety. Positive values favour the probiotic. Small sample, multiple outcomes and adjunctive treatment limit conclusions; these are not ADHD treatment estimates. [109]

A larger gut-directed trial adds a provisional treatment signal. RECOVERY randomized 463 adults with post-COVID symptoms to the synbiotic SIM01 or placebo for six months. Reported alleviation favoured SIM01 for concentration difficulty (OR 2.64, 95% CI 1.69–4.14) and memory loss (1.97, 1.27–3.04). These were interview-based symptoms, not objective cognitive gains. Quality of life, physical activity and the cytokine analysis did not show significant between-group results. The single-centre design, symptom measurement, missing-data assumptions and investigators' commercial interests make independent replication valuable. The trial does not establish an inflammatory mechanism or ADHD efficacy; its historical outcome-registration changes were not independently audited here. [197]

A separate 12-week pilot involved 38 children assigned to an ADHD diagnostic group and 42 to an autism group, with overlapping diagnoses placed in the latter. Its adjusted overall comparisons did not show clear probiotic–placebo differences on the main behavioral or neuropsychological measures. Within-group improvements and exploratory age findings do not overturn that comparison. The formulation also included vitamin D in the active product, further limiting attribution to bacteria alone. [110]

The next trial should specify the strain or formulation, population, primary outcome and time point before recruitment. Symptom relief in the gut, a changed stool profile and improved attention would be separate outcomes. These studies do not justify antibiotics, restrictive diets, stool testing or microbiota transplantation as treatments for ADHD.

What newer gut studies add

A 2026 ADHD analysis examined stool from 147 participants at baseline and 106 at follow-up from an earlier synbiotic trial. Treatment changed microbial composition and some circulating short-chain fatty acids without changing overall within-sample diversity. Medication-associated differences in bacterial vitamin B12 synthesis genes were also reported. A microbial gene pattern is not a diagnosis of vitamin deficiency, and medication use was not randomized. The parent nine-week trial had 182 completers and found similar improvement in core ADHD symptoms with synbiotic and placebo, with no significant daily-function benefit. The newer biological measurements do not reverse that clinical result. [213, 214]

The PROMOTe trial gives a narrower positive cognitive example. Thirty-six twin pairs aged 60 or older were studied for 12 weeks, with one twin assigned prebiotic and the other placebo. Both received an amino-acid supplement and exercise advice. The primary chair-rise outcome was negative; a secondary cognitive factor favoured the prebiotic. More cognitive data were missing than physical data, and most participants were women. This is a small older-adult result, not established ADHD treatment or dementia prevention. A 2025 correction revised microbial feature counts, supplementary figures and ingredient units; it did not announce a changed cognitive treatment estimate. [215, 216]

Diet can change immune measurements without demonstrating a brain benefit. In a 2021 study with 36 analyzed participants, a fermented-food diet increased microbial diversity and reduced several circulating inflammatory proteins. The primary cytokine-response score was unchanged. The study compared two dietary programmes, lacked a usual-diet control, and two participants chose their allocation. It measured no cognitive outcome. “Gut health” needs a defined meaning: bowel symptoms, nutrient absorption, a microbial function or a measured metabolite. Diversity alone is not a health score, and these results do not supply one diet for everyone. [220]

Hookworms test whether immune change becomes clinical benefit

Hookworms are parasitic worms, not gut bacteria. Researchers have tested controlled infection because it can alter immune regulation. That makes them relevant to the hypothesis, but the outcome still has to improve. The trials below show why immune modulation cannot simply be called immune improvement. [217, 218, 219]

Table 4. Human hookworm trials answer different questions

Study	Primary result	Other findings and limits
Metabolic pilot 40 randomized Pierce 2023 [217]	Safety was primary; metabolic outcomes were secondary.	At 12 months, median HOMA-IR change was −1.1 with lower-dose infection versus +0.8 with placebo (P=.039); higher dose did not differ significantly. Per-protocol analysis, attrition and baseline imbalance.

Table sources (in row order): [217].

Study	Primary result	Other findings and limits
Relapsing MS 71 randomized Tanasescu 2020 [218]	MRI lesion primary outcome not significant (P=.19).	Regulatory T cells increased. Biological target engagement did not establish clinical efficacy.
Celiac disease 54 participants Croese 2020 [219]	Week-42 completion: placebo 57%, lower dose 37%, higher dose 44% (P=.61).	Fewer gluten-related symptoms did not establish sustained gluten tolerance or protection of intestinal tissue. Small, unequal groups.

Table sources (in row order): [218], [219].

HOMA-IR is an indirect fasting glucose–insulin estimate, not a brain measurement. The metabolic pilot reported gastrointestinal adverse events in 44% of treated participants and three early deworming treatments. Its secondary signal does not establish diabetes prevention or cognitive benefit. These studies do not justify self-infection, abandoning celiac treatment or adding hookworms to an ADHD protocol. The useful question is whether a specific immune effect predicts a meaningful outcome, with harms measured alongside benefits. [217, 219]

16.6 Environment is more than a toxin list

The controlled particle-exposure experiment discussed earlier found selective effects after four hours in 26 adults; some cognitive tasks did not change. [23] A 2026 double-blind crossover study, HIPTox, exposed 15 healthy adults aged at least 50 with a family history of dementia to four pollutant mixtures and clean air. Cognitive responses differed by source and task; some processing-speed comparisons improved. With only 13 completing all five visits, this is a small, mixed experiment, not evidence that pollutants help cognition or uniformly impair every domain. It cannot determine an ADHD-specific susceptibility. [112]

The protocol should also record noise, heat, light timing, shift work, housing disruption, financial strain and access to recovery as candidate contextual exposures. Their relevance is a proposal for measurement here, not a claim that each has a proven immune mechanism. A noisy room may impair performance through distraction without changing inflammation. Poor sleep may connect a difficult environment to the next day's attention. Both are legitimate explanations to test.

Environmental causes also have different time scales. A developmental exposure that changes baseline ability is not the same as a short-term trigger of fluctuation. Record dose or a defensible exposure proxy, timing, task conditions and the opportunity to avoid exposure. Do not turn unexplained symptoms into a claim about unspecified toxins, mold or a need for detoxification.

Brake dust is a distinct exposure

Brake wear adds particles even when there is no tailpipe exhaust. A 2025 experiment compared particles from four brake-pad types with diesel particles in cultured human alveolar cells. Copper-rich formulations produced greater oxidative stress, inflammatory signalling and metabolic disruption under the tested conditions. Removing available copper reduced several effects. These were laboratory lung-cell outcomes, not measured brain injury, cancer incidence or cognition in commuters. “More toxic than diesel” is too broad unless the particle dose, composition and endpoint are stated. [221]

A 2026 mouse study adds an important dose qualification: particle rankings changed when exposure was expressed by mass versus surface area. Particles were delivered by aspiration, which differs from normal inhalation. The result supports measuring composition and dose rather than treating all particulate matter as identical. The human pollution experiments already discussed in Section 8.1 and Section 16.6 concern different exposure mixtures. They cannot identify brake dust as the cause of their cognitive findings. [222]

Microplastics need measurement as well as mechanism

A 2025 postmortem study reported more plastic-associated material in brain than liver or kidney, with higher concentrations in 12 dementia cases. The authors explicitly did not infer causality. Disease could alter tissue volume, barrier function or clearance, and postmortem sampling cannot establish what came first. A subsequent correction addressed images, scale bars and preparation details. Published criticism raised analytical and quality-control concerns, including possible interference from lipid-rich tissue; the authors responded. The exchange means that exact brain-burden estimates remain contested. Particle detection deserves investigation, but it does not establish that plastics caused dementia, ADHD or someone’s current symptoms. [223, 224, 225, 226]

Exposure estimates are also being revised. A 2026 analysis compiled 2,782 atmospheric measurements from 76 studies and compared them with size-aligned models. The modelled global median concentrations exceeded observations by two to four orders of magnitude. That finding concerns atmospheric estimation, not a direct remeasurement of brain tissue or proof that plastics are harmless. Size detection limits, contamination controls and the distinction between particle number and mass matter at both ends of the proposed exposure–health relationship. [227]

The vascular evidence is prospective but still observational. Among 257 people followed after carotid surgery, the composite of heart attack, stroke or all-cause death occurred more often when plastic-associated material was detected in plaque. Figure 28 shows crude proportions; the adjusted hazard ratio was 4.53 (95% CI 2.00–10.27). The selected surgical population, residual confounding and contamination concerns limit causal interpretation. Cognition was not measured, and removing an exposure was not tested. [228]

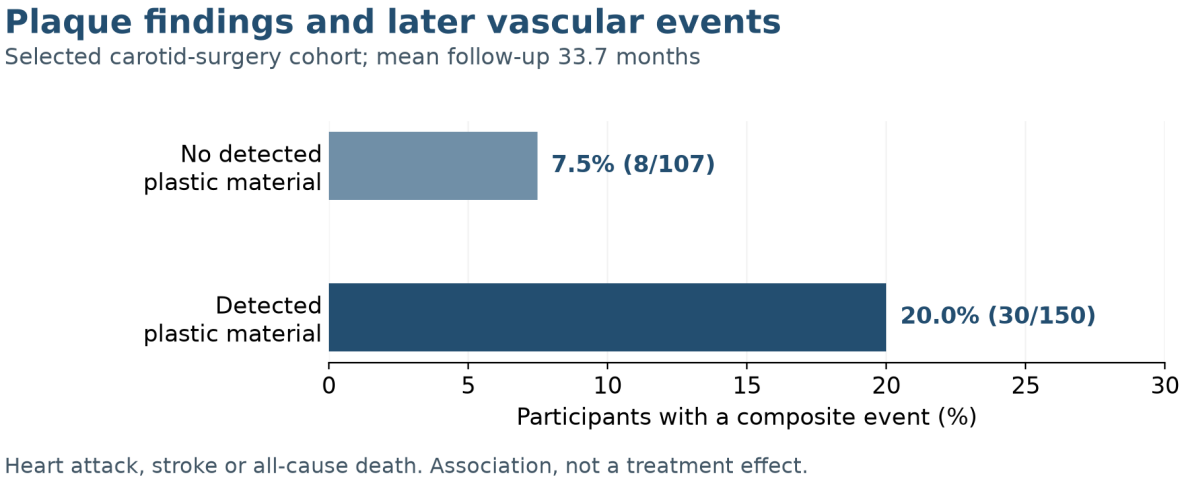


Figure 28. Calculated proportions from Marfella et al. Counts are participants with an event. Bars are unadjusted and do not depict the adjusted hazard ratio or a cognitive outcome. No confidence intervals are inferred from these bars. [228]

16.7 Physical health across the life course

The 1970 British Cohort Study connects childhood traits with later physical health. In 10,930 participants, high ADHD traits at age ten were associated with an estimated 42.1% probability of physical multimorbidity by age 46, versus 37.5% without high traits. These are model estimates based on reported conditions, not percentages diagnosed with ADHD. Smoking, psychological distress and BMI contributed indirect associations, without proving causal mediation. Missing-data selection and limited ethnic diversity matter. Figure 41 gives the uncertainty. The result supports attention to cumulative health and access to care; it does not identify one shared disease process. [184]

16.8 Cancer shows why the route and outcome matter

Cancer-related cognition is a boundary comparison for this framework. It lets us examine symptoms, objective performance, treatment and recovery in a setting where the illness and its treatment may both contribute. The purpose is to improve the questions we ask about change, not to imply that ADHD and cancer share a cause.

A large prospective study assessed 581 patients with breast cancer and 364 noncancer controls. Clinically important worsening on a self-report cognitive questionnaire was more frequent after chemotherapy and remained more frequent six months later. Figure 29 shows the published proportions. These are reported changes, not diagnoses of objective impairment. Without an untreated cancer comparison, the study cannot isolate chemotherapy from illness and other accompanying changes. [229]

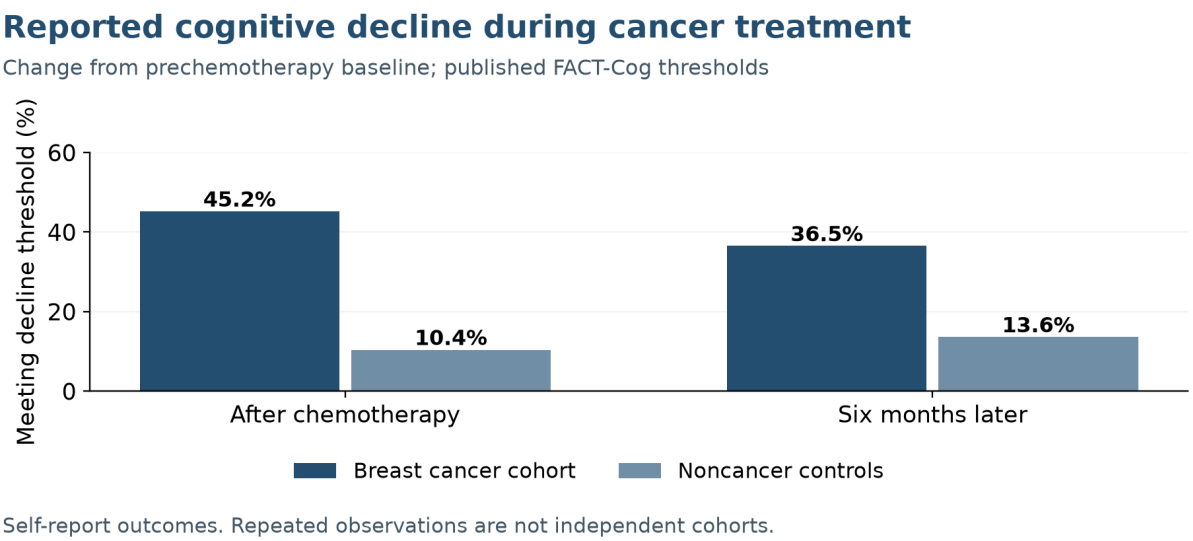


Figure 29. Janelsins et al. reported these percentages of clinically important self-reported decline. Cohort totals are 581 patients and 364 controls; no counts are reconstructed from rounded percentages. Observational comparison, not a randomized chemotherapy effect. [229]

A companion report from the same research cohort assessed objective performance in 580 patients and 363 controls. Visual-memory trajectories differed between groups, with a significant group-by-time effect. It is a complementary measurement, not an independent replication. Together, the reports support measuring both cognitive experience and performance, while leaving the responsible pathway unresolved. [230]

A 2024 study measured 77 patients before, during and after chemotherapy. Twenty-seven of 77 participants (35%) met the study threshold of reliable decline on at least two scores from the four-test battery. Lower microbial diversity tracked poorer objective cognition independently of the measured circulating cytokines. Subjective cognitive changes did not track the same microbial or inflammatory measures. Treatment can change several systems at once, so this does not prove microbial mediation or that increasing diversity restores cognition. A negative cytokine association also cannot exclude every inflammatory route. [231]

Microbiome intervention during immunotherapy

The 2026 TACITO trial directly tested a gut intervention alongside cancer therapy. Fifty people with metastatic kidney cancer were randomized; 45 received at least the first assigned treatment. Donor fecal microbiota transplantation (FMT) was compared with placebo FMT alongside pembrolizumab and axitinib. The primary 12-month progression-free outcome was 70% versus 41%, but did not meet the prespecified significance threshold (P=.053). A secondary progression-free survival result favoured donor FMT in the treated analysis; the all-randomized sensitivity analysis was inconclusive (HR 0.66, 95% CI 0.33–1.35). Much donor material came from one person. This is a promising, limited cancer-treatment experiment. It did not measure cognition and cannot validate FMT for ADHD. [232]

Removing an infection can help when the target is specific

A Korean trial randomized 1,838 people with Helicobacter pylori infection and a first-degree relative with gastric cancer. In the modified intention-to-treat population, gastric cancer developed in 10 of 832 treated participants versus 23 of 844 assigned placebo over a median 9.2 years: HR 0.45 (95% CI 0.21–0.94). This is direct evidence for a specific infection-targeted prevention strategy in a selected risk group. It measured no cognitive outcome. It does not justify antibiotics for unexplained fog, and it should not be combined with the hookworm trials into a claim that more or fewer microbes are always better. [233]

What these connections change

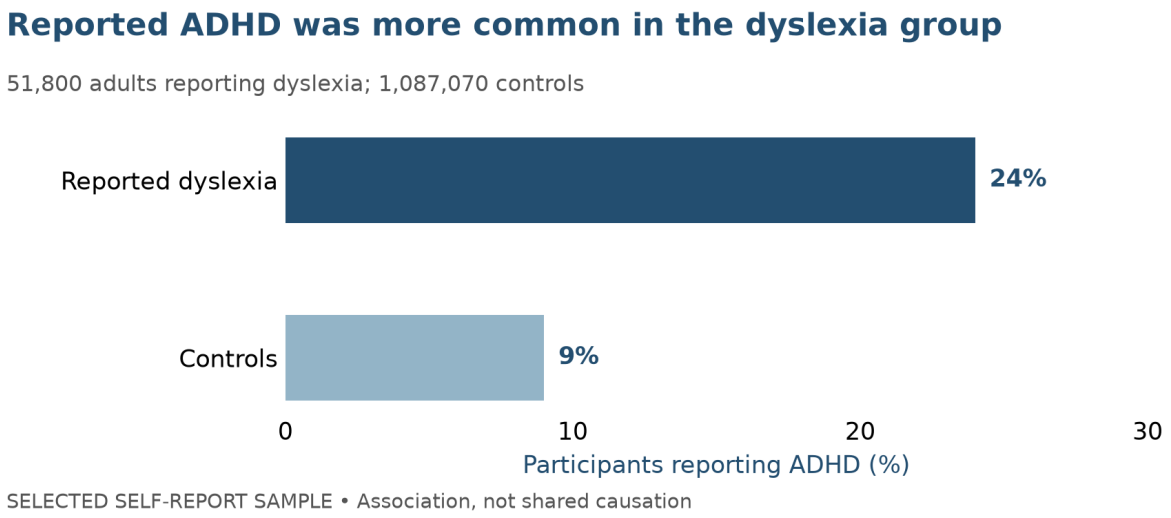
I want the study to reflect what is happening in someone's life while keeping each explanation testable. Lung exposure could affect sleep or breathing. A gut intervention could improve comfort while attention stays the same. Cancer treatment could change cognition through several routes at once. To distinguish those possibilities, a study has to specify what changes first, what follows and when the relationship should disappear.

16.9 Autism, dyslexia and other neurodevelopmental profiles

Autism and dyslexia belong in this discussion. The question is how a person's developmental profile interacts with changing bodily state and everyday demands. We should not turn a model of fluctuating attention into a claim that these conditions are temporary states, or that one immune or metabolic explanation accounts for all of them.

Dyslexia and learning differences

A large genetic study included 51,800 adults reporting a dyslexia diagnosis and 1,087,070 controls. ADHD was reported by 24% of the dyslexia group and 9% of controls. The study found genetic overlap with ADHD, but no statistically significant genetic correlation with autism in that analysis. These are selected, self-report data, not universal diagnostic prevalence estimates. Shared genetic liability does not identify histamine, gut inflammation or another daily physiological mechanism. [234]



Visual guide B. ADHD was reported by 24% of adults reporting dyslexia and 9% of controls in the cited genetic study. Group sizes describe the source groups; the rounded percentages are not reconstructed exact case counts. Self-report and selection limit generalization. This comparison does not identify a daily immune or metabolic mechanism. [234]

A 2025 Netherlands Twin Register study extended the comparison to dyscalculia, involving up to 19,125 twin children and 2,150 siblings assessed at ages 7 and 10. Across the conditions, having one was associated with a 2.1–3.1-fold likelihood of a second. Yet 77.3% of children with at least one had only one. The models generally favoured correlated genetic influences over one trait causing another, apart from a reading-to-spelling pathway. That interpretation depends on the modelling assumptions; it is not an experimental demonstration that environment or support cannot matter. [235]

For our framework, reading performance needs its own baseline. A difficult reading task can mix decoding demands with attention, effort and fatigue. A lower score after poor sleep would not show that sleep caused dyslexia. We should compare change within the same person and include a task with limited reading demands before calling the result a general loss of executive function.

Autism and ADHD overlap without being interchangeable

In a Swedish register cohort of 1,899,654 people, 13,793 had both recorded autism and ADHD: approximately 48% of the 28,468 autism cases and 17% of the 82,398 ADHD cases. The denominator changes the percentage. Family associations were consistent with shared genetic liability, while shared environmental contributions remained possible. Historical diagnostic rules, medication-based ADHD ascertainment and selection of people reaching services limit generalization. This is evidence of overlap, not evidence that autism is a form of ADHD. [236]

Sleep provides a more direct intervention example. A 13-week placebo-controlled trial randomized 125 children and adolescents with persistent insomnia after behavioural intervention; 96.8% were autistic and the remainder had Smith–Magenis syndrome.

Prolonged-release melatonin improved the primary caregiver-reported total-sleep-time outcome. The reported sleep-response rate was 68.9% versus 39.3%, using the trial's combined sleep-duration and/or

sleep-latency criterion. This supports treating a defined sleep problem in that population. It does not establish a change in autism, objective cognition or the proposed resilience index. [237]

The gut connection also needs a direction check. A stool-metagenomics study of 247 participants found little direct association between autism diagnosis and the microbiome. Its results supported a pathway involving dietary preferences, dietary diversity and microbial diversity. The observational design does not settle every gut–brain question, but it gives us a concrete alternative to assuming that a microbial difference caused the developmental condition. Gastrointestinal discomfort and nutrition can still deserve attention in their own right. A later erratum corrected a microbiome transformation and associated estimates and plots; the authors report unchanged overall conclusions. The affected numerical estimates are not reproduced here. [238, N06]

What this changes in the proposed study

Record autism, dyslexia and dyscalculia separately, including how they were identified. Developmental language disorder, developmental coordination disorder, tic disorders and intellectual disability are also relevant candidates for the intake profile; this section does not claim the same physiological pathway has been demonstrated in each. Describe communication, learning, sensory and motor requirements alongside diagnoses. Avoid a single combined category that hides those differences.

A testing room can overwhelm an autistic participant. Long written instructions can put a participant with dyslexia at a disadvantage before a timed task starts. The design could mistake an access barrier for poor resilience. Offer accessible instructions and a suitable environment, record accommodations, and keep them consistent across visits. Measure sleep, pain and illness alongside performance rather than inferring bodily dysfunction from a diagnosis.

The falsifiable question is whether a measured physiological change predicts later cognitive change within each profile, and whether those relationships differ across profiles. Prespecify a small number of comparisons and recruit enough participants to estimate them. A similar relationship across groups would support a broader state–cognition account while weakening an ADHD-specific claim. Helping someone sleep, communicate or manage pain is a worthwhile outcome without claiming to reverse their neurodevelopmental identity.

16.10 What the wider evidence changes

The wider evidence adds competing explanations that the model has to address. In a 60-person randomized trial, infliximab did not improve the overall depression outcome versus placebo. A baseline-inflammation interaction and exploratory subgroup were interesting, but did not establish general efficacy or a clinical CRP threshold for treatment. We cannot rescue the hypothesis by inventing a new subgroup after every null finding. [111]

Worked example: three routes to the same difficult morning

Imagine someone with ADHD and bowel disease who reports poor concentration after a painful, sleepless night. One possibility is a direct inflammatory effect; another is that pain disrupted sleep; a third is that medication or worry changed arousal. This is hypothetical. Measure the proposed sequence rather than naming the route in advance. If sleep and pain explain the later performance change and the immune measure adds no predictive value, the direct immune claim weakens even though the person's illness and difficulty are real.

Table 5. How the wider evidence changes the next study

Area	What to distinguish	Test that could weaken the claim
Mental health and trauma	Developmental history; current mood, trauma symptoms, sleep and cognition	Physiological measures add no prediction beyond these simpler measures.
Autoimmune and infection history	Specific diagnosis, severity and timing; symptoms versus cognitive tests	No within-person relationship despite reliable measurement and adequate change.
Gut microbiome	Composition, function, treatment exposure and clinical outcome	A microbial shift occurs without the prespecified cognitive benefit.
Environment	Measured exposure, timing, direct distraction and sleep disruption	Association disappears with exposure timing or task conditions properly controlled.

These comparisons need reliable measures, adequate exposure variation and enough precision for a null result to inform the next study. A well-tested route that fails narrows the hypothesis.

17. Candidate mechanisms and follow-up studies

17.1 Timescale and the missing causal links

Acute vigilance and perfusion operate over seconds to hours; recovery after a disturbed night can unfold over days; metabolic, immune and vascular adaptation may take weeks or months. These are useful study timescales rather than proven biological compartments. A disturbance can affect several.

The proposed sequence from sleep or immune state through vascular/CSF dynamics, solute transport, cognition and everyday function contains unestablished links. Selected experiments support individual arrows [200, U30, U48, U49, U50, U51], while anatomical ALPS dependence, conflicting tracer experiments and limited human transport measurement constrain the chain [U43, U52, U56]. Evidence for one arrow cannot substitute for the next.

17.2 Modules and the missing evidence each would address

Table 6. Proposed mechanism modules: a common core with separate questions.

Module	Paired measurements	The question it can answer
Sleep, timing and vigilance	Prior sleep, chronotype, repeated attention, sleepiness; optional waking EEG	Does vigilance instability add predictive information beyond sleep and medication timing?
Migraine and orthostatic state	Attack/recovery timing, pain, treatment, HR/BP; supervised carbon-dioxide and Doppler measures where justified	Are bodily response, cognitive response and practical consequence separable?
ADHD × endometriosis	Repeated endocrine measures or validated phase, pain, sleep, migraine and attention	Is the association additive, interactive, or carried through specified pain/sleep routes?
Allergy and histamine	Documented allergic state, symptom burden, sleep and preselected mediator measures when justified	Does an identified immune-linked change precede or explain cognition beyond allergy discomfort and sleep?
Post-infectious illness	Infection history, pre-existing ADHD ascertainment, low-burden repeated measures and delayed symptom sampling	Does pre-existing ADHD modify cognitive change or recovery, without conflating acquired symptoms with developmental ADHD?
Metabolic or vagal intervention	Clinically justified target, credible control, co-treatment and target measures, cognition and functioning	Does the intervention affect the specified outcome, and what evidence distinguishes the possible routes?

Each module belongs in a separate or nested study with a prespecified primary outcome, an appropriate comparator and a rationale for its ethics and safety decisions.

Prespecify a small set of measures for the wider modules. Record current depression, anxiety and relevant trauma symptoms, recent infection and diagnosed disease activity, gastrointestinal symptoms, diet, medication changes and selected timed environmental exposures. Record baseline history separately from changing states. Choose a primary comparison and outcome, account for missing data on difficult days and evaluate added predictive value on held-out observations. Exploratory microbial or immune subgroups need independent replication.

17.3 Design details that prevent a misleading answer

The timing/EEG substudy should counterbalance visits across days where possible and distinguish clock time from biological circadian phase. The orthostatic study needs repeated- task control sessions, planned recovery observations and separate models of physiology and cognition. It must not attribute task-induced cardiovascular change to posture. [42, 43]

The endometriosis module should observe more than one cycle where feasible and document diagnosis ascertainment, hormone treatment, irregular cycles, amenorrhea and reproductive stage. Calendar days alone should not stand in for measured endocrine state. The combined ADHD–endometriosis group having the worst average score is not sufficient evidence of interaction. Record bleeding history, clinically indicated iron assessment, fatigue and relevant treatment as possible contributors. Avoid a universal supplement or biomarker panel. Separate menopausal stage from calendar phase.

The allergy and post-infectious modules need ordinary-day sampling as well as flares. A preselected mediator panel should specify assay limitations, timing and handling. Biomarkers are not interchangeable with diagnoses, and a negative result should not be dismissed by invoking an unspecified hidden compartment. The post-infectious module must protect participants with post-exertional worsening and accommodate missed or shortened assessments. [46]

A later intervention trial may enrich for a relationship that already predicted independent observations. Randomization can estimate an intervention's effect; it does not automatically prove the pathway responsible. Blocking a receptor need not lower the amount of its mediator. Conversely, a blood-marker change may not reflect the target relevant to cognition. No personal response to several simultaneous treatments is used as evidence here.

A clinically indicated GLP-1 or other peptide treatment can be studied within this design, using the sequence specified in Section 14.17. Record discontinuation and tolerability, because analyses restricted to people who remain on treatment may select a different group. A mediator analysis should state its assumptions and compare pain, sleep and weight routes before attributing cognitive change to immune or neuronal action.

17.4 Testing endocrine, immune and metabolic connections

I would look for a change in a measured bodily state followed by a change in a particular outcome, then ask whether that relationship differs between people. PMDD findings leave the mast-cell question open. Improvement on a GLP-1 treatment leaves open what role insulin resistance played.

For the endocrine module, compare prospectively characterized ADHD, PMDD, both and neither. Record daily symptoms and a brief cognitive task across repeated cycles, with sleep, pain, migraine and medication timing. Hormonal measurements should test a prespecified timing or change hypothesis. If endocrine data add no reliable prediction beyond those simpler observations, that weakens the endocrine contribution to this version of the model.

For the metabolic module, distinguish diabetes from other populations and record treatment, meals and any clinically indicated glucose data. Test whether glucose state adds useful within-person prediction. We should not deliberately induce hypoglycemia in an ADHD pilot or infer brain insulin resistance from fatigue. For the immune module, define the disease and mediator measurement before testing its association with cognition.

A combined hormone–mast-cell–barrier hypothesis needs temporally aligned evidence for each step. If an immune marker changes without the predicted cognitive change, or the association is accounted for by pain and sleep, the proposed direct route loses support. Interaction terms should be limited and prespecified; dividing a small sample into ever narrower subgroups would make almost any story look plausible.

17.5 Testing gut and environmental routes without broadening every visit

The gut module should pair a limited set of stool and circulating measures with diet, bowel transit, gastrointestinal symptoms, medication and the existing cognitive core. A microbial change should precede the proposed metabolite and cognitive changes if that sequence is part of the claim. Replicate it before assigning a biomarker-defined subgroup or testing a formulation. Stool diversity should not be added to the Neurophysiological Resilience Index as an assumed benefit.

The environmental module should observe ordinary variation or a separately justified exposure-reduction intervention. Record particle size, source information, timing, noise, heat and sleep; a generic PM reading cannot uniquely identify brake wear. Microplastic assays require validated blanks, recovery testing and independent laboratory agreement before they become explanatory variables. Deliberate pollution exposure, parasitic infection and cancer treatment are not components of the proposed ADHD study.

Clinically indicated treatment cohorts can supply separate tests of recovery. Track fatigue, pain, sleep, mood, anemia and co-medication alongside cognition before and after treatment. If these simpler measures explain the change and a proposed microbial or immune measure adds no reliable prediction, that route loses support. Cancer or infection recovery may inform the model without validating it in ADHD.

17.6 Using AI to support research

AI is becoming useful at finding candidates that researchers could not realistically test one by one. For this hypothesis, genes, immune signals, microbial products and brain circuits generate far more possible relationships than any single study can examine. A pattern still needs testing. We still need to know whether it survives a new dataset, an experiment and, eventually, a meaningful test in people. The examples below were published in 2024–2026 and span very different stages of that process. None establishes an AI-discovered treatment for ADHD.

Predicting molecular interactions. AlphaFold 3 extended structure prediction to complexes containing proteins, nucleic acids and small molecules. This gives researchers a way to propose how a peptide or drug might fit a biological target. For our framework, that could help prioritize experiments on receptors involved in immune or metabolic signalling. A predicted fit does not establish receptor activation, safe exposure, access to the brain or a useful clinical effect. Those are separate measurements. [144]

Reading the regulatory genome. AlphaGenome, published in January 2026, accepts about one million DNA bases at a time and predicts several kinds of gene regulation. It matched or exceeded the strongest external models in 25 of 26 variant-effect evaluations. This is a result across benchmark tasks, not a 96% diagnostic accuracy rate. It could help choose which non-coding variants to investigate when looking at overlapping conditions. It does not show that ADHD, endometriosis and autoimmunity share one causal mechanism. That would require disease-specific genetic evidence, appropriate cell types and experimental validation. [145]

Designing antimicrobial peptides. A 2025 ProteoGPT study combined sequence generation with laboratory testing. Of 42 generated peptides synthesized for initial testing, 26 showed potential bacterial inhibition in a disk-diffusion assay. That is a selected laboratory sample, not a patient response rate or the success rate of every AI proposal. Further selected candidates were tested in mouse infection models. These experiments supply no evidence that those peptides treat chronic cognitive symptoms or restore a human gut

microbiome. Antimicrobial activity can be useful while its effects on a person's wider microbial community remain uncertain. [146]

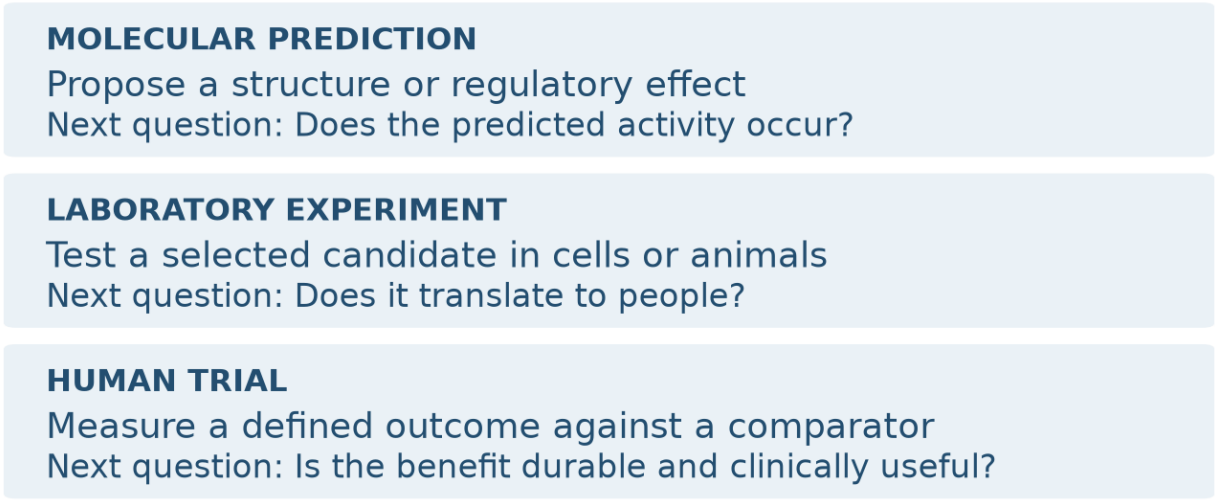


Figure 31. Conceptual guide to the stages represented by the AI examples. These are separate research programmes, not one demonstrated discovery pipeline. Success at an earlier stage does not establish ADHD treatment efficacy. [144, 145, 146, 147]

An AI-assisted candidate in a human trial. Rentosertib is a small molecule, not a peptide. Its target selection and molecular design used AI. A 2025 phase 2a trial randomized 71 people with idiopathic pulmonary fibrosis across three doses and placebo for 12 weeks.

Safety was the primary endpoint. In the 60 mg once-daily arm, mean forced vital capacity changed by +98.4 mL (95% CI 10.9 to 185.9), versus −20.3 mL (−116.1 to 75.6) with placebo. These are within-arm changes, not the confidence interval for a treatment contrast. Sixteen participants discontinued treatment before the 12-week endpoint; this is not the same as withdrawing from study follow-up. This is an early clinical signal that justifies further testing; it does not establish durable benefit, cognitive benefit or superiority of AI drug development over conventional discovery. [147]

Finding clinically useful subgroups. Machine learning also operates at the level of people. A 2024 brain-imaging study identified six circuit profiles in 801 people with depression and anxiety; treatment-related analyses involved 250 participants. The profiles differed in symptoms, cognitive performance and treatment response. These are candidate ways to divide a heterogeneous clinical population, not six settled diseases. We face the same problem with any proposed state-sensitive subgroup: it must work in new people, beyond the dataset used to define it. A trial must also test whether using the classification improves treatment decisions. [148]

An adjacent 2025 study gives a more direct connection to cognition and an ADHD-related medication. Seventeen participants with a prospectively defined cognitive subtype of depression completed at least six weeks of guanfacine immediate-release treatment. Circuit measures and cognitive performance improved, and 76.5% met the study's depression-response threshold. This was an open-label, single-site, per-protocol study without randomization or placebo. Expectation, repeated testing and selection of completers remain

plausible explanations for some change. It is a useful treatment hypothesis, not proof of an AI-selected therapy or evidence that depression and ADHD are the same disorder. [149]

I would ask whether repeated measures of sleep, pain, autonomic state, medication and cognition can predict a difficult day better than usual performance and recent sleep. Test later days in time order and hold out entire participants when testing prediction in new people. Fit preprocessing rules on training data alone. Report uncertainty and examine performance across sites, devices and demographic groups. If the added measures offer little improvement, keep the simpler model.

If a model can identify a day when someone is likely to struggle, the next experiment is to see whether acting on that prediction helps them finish an everyday task. I would specify the predicted sequence, the comparison and what would count as failure before seeing the result. AI can suggest a hypothesis. The controlled study has to test it.

17.7 Priorities for later studies

Follow people with and without established ADHD through documented infection and recovery, ideally with measurements from before infection. Compare starting performance, the size of the decline and the time course of recovery. Mono is a useful candidate because it has a recognizable onset. The reviewed studies leave that direct comparison open.

Measure the proposed chain in the same participants. An immune or histamine study needs a peripheral marker, a credible measure of the intermediate process and a later cognitive assessment. An autonomic study needs measures of circulation, symptoms and cognition collected together. Results from separate cohorts can help choose measures, but the sequence must be tested within a study.

Test how a treatment benefit arises. In a trial for an overlapping condition, measure cognition alongside disease severity, sleep, pain and medication exposure. If cognition improves, those observations can help distinguish symptom relief, a direct drug effect and the proposed pathway.

A later clinical-physiology subset may measure respiratory arousals, hypoxic burden, perfusion and independently validated transport markers. Personal wearable histories can motivate questions but cannot establish drug synergy, mast-cell dysfunction, neuroinflammation or impaired clearance. Consumer stage estimates, oxygen saturation and HRV require appropriate validation. Treatment changes are outside this observational proposal.

17.8 Future validation and transfer

A model has to predict a change large enough to matter on days it has yet to see. The measurements must be reliable, and the timing of the prediction has to fit the proposed sequence. I would compare it with a simpler account based on usual performance, context, sleep and medication. Set means, cutoffs and subgroups before using later observations to evaluate the prediction.

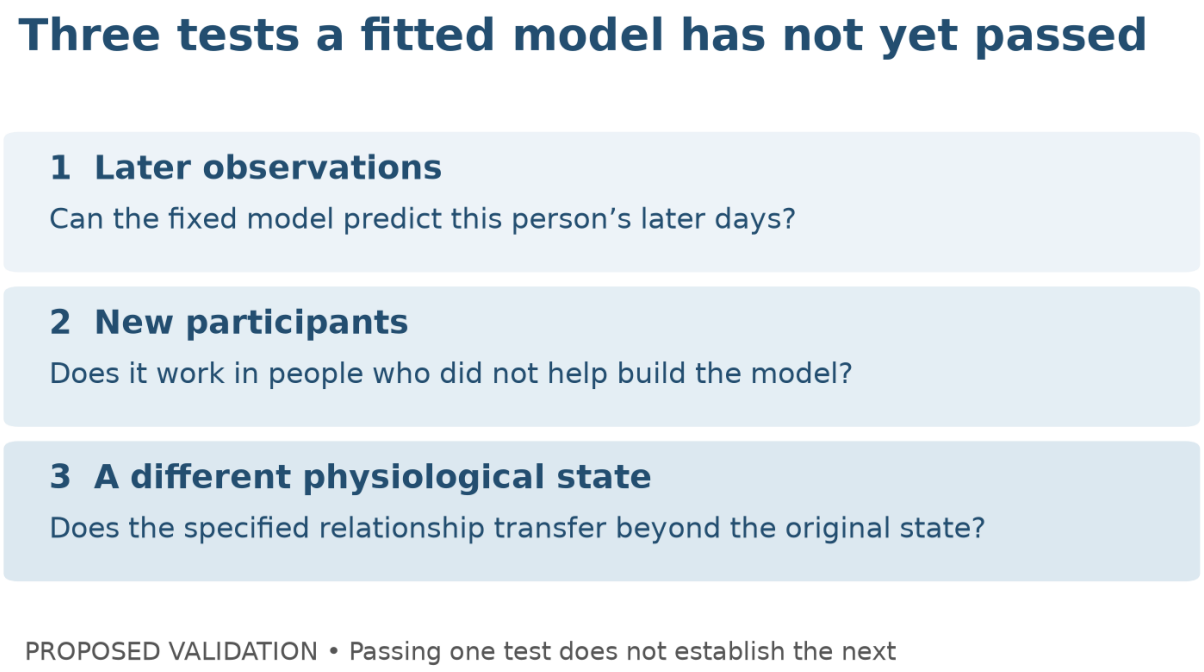
Table 7B. How to interpret later tests of the framework.

Question	Would support	Would weaken or reverse	Would leave unresolved
Added physiology	Extra measurements improve later prediction by a prespecified useful amount.	The simpler model predicts as well under adequately precise evaluation.	Sparse data, unstable measures or imprecise comparison.
Reactivity, sensitivity and recovery	A defined challenge yields repeatable differences in physiology, cognition or time to recovery.	A sufficiently precise challenge excludes the proposed difference in the corresponding outcome.	An unmatched challenge, incomplete follow-up or unreliable measurements.
Transfer to other people or states	A fixed model predicts on later days, in new participants or in another state, according to the test used.	Adequate testing fails at that level of transfer. Retain the narrower claim.	Unreliable measurements or insufficient variation in the validation setting.

How to evaluate prediction

A later study can ask whether added physiological measurements improve predictions beyond sleep, treatment timing, mood, pain, workload and prior performance. Compare errors and calibration on later observations with a prespecified improvement worth the extra measurement burden.

Appendix J sets out the provisional resilience index and the tests it would need before use.



Visual guide C. Later observations, new participants and a new physiological state test different claims. The model and preprocessing rules must be fixed using development data before each evaluation. These are proposed validation steps, not completed results.

The eTNS, antiviral, GLP-1 and H3-antagonist results and the neurofluid measurement limits constrain different routes [U04, U19, U27, U43, U52, U54, U55]. Any responder subgroup needs a prespecified definition and its own test. A validated change in transport with a precise, negligible cognitive effect would weaken that proposed clinical route.

17.9 The important tests still missing

The cited evidence does not demonstrate a complete peripheral immune-to-central histamine-to-ADHD cognition chain in the same people. It does not establish that GLP-1 or vagal interventions improve ADHD through immune modulation. Nor does it establish a general ADHD-specific response across infection, migraine, hormones and autonomic challenge. More confident wording will not fill those gaps. The added fatigue, sleep-surgery and lifestyle findings narrow the predictions; they do not fill the missing ADHD-specific causal tests.

Methods and sources

Appendix A. Technical model and measurement safeguards

A1. Separate within-person change from between-person differences

A multilevel model separates changes within a person from differences between people. A causal claim and a prediction for someone new each need their own evidence. [61, 62]

For participant i at assessment t, a proposed association model is:

$$g(\mu_{it}) = b0 + u_i + bA*A_i + (bW + v_i)*Xwithin_{it} + bB*Xmean_i + gamma*(A_i*Xwithin_{it}) + theta*C_{it} + f(time, practice)$$

For a logistic model, the ADHD-by-state coefficient is conditional on the link scale. The proposed sleep study instead uses model predictions to estimate the prespecified standardized absolute contrast in Section 5.2. A difference on the log-odds scale alone does not distinguish the two explanations in Figure 1.

Here mu is the expected outcome; A indicates ADHD ascertainment; Xwithin is deviation from the person's mean; Xmean captures between-person differences; and C contains prespecified context and treatment variables. Random intercepts and slopes provide partial pooling. For omissions, a binomial logit model requires known target opportunities.

Alongside the coefficient, report marginal contrasts, their uncertainty and how well exposure overlaps across groups. Limit the comparison to states observed in both groups. Use separate models for physiology, independently recorded daily function and a validated functional threshold.

To trace recovery, measure both physiology and cognition more than once after a change. Specify plausible delays or nonlinear effects before looking at the results and keep an additive model in the comparison.

A2. Prediction is not mediation

Decide what effect you want to estimate before choosing what to adjust for. Sleep could be part of the route from allergy to symptoms. For a different exposure, it could affect both the exposure and the outcome. Estimating a total effect and a direct effect takes different assumptions. Measuring two things at once leaves their order unclear, and order alone cannot establish cause.

Record treatment changes that follow symptoms. Learn centering, scaling, cutoffs and subgroups from the training data. Evaluate prediction on later observations for the same participants and in a separate cohort for new participants.

A3. Reliability, missingness and participant burden

Robust group effects do not guarantee reliable differences between people [63, 64]. Reliability of change within one person is a separate measurement question [194]. Quantify trial-level noise, practice, speed–accuracy trade-offs and slope uncertainty. Standardize device latency and presentation, and record interruptions. Calibration is helpful but does not automatically validate daily change.

People may skip an assessment when symptoms flare or work piles up. Treat non-wear as missing. Decide in advance what counts as a usable record, ask briefly why data are missing where possible and test how different assumptions affect the result. Keep the burden of recording low.

Estimate slopes with partial pooling before clustering. Require later prediction and replication before naming response patterns; continuous heterogeneity may fit better than subtypes. Do not build an unrestricted physiological-load score after inspecting outcomes.

A4. Repeated measurement can change what gets reported

In 90 adolescents treated for ADHD, self-rated symptoms declined across 17 days of questionnaires sent four times daily. Reports were also lower after completed rather than missed prompts. Participants were not randomized to symptom monitoring versus an appropriate control, so this does not establish that questionnaires treated ADHD. Changing circumstances, selective completion and the act of monitoring could all contribute. It is nevertheless directly relevant to a study that relies on repeated self-report. [176]

I would record how often participants are prompted, how many times they have taken the task and which day each assessment occurred. After familiarization, use independent outcomes where feasible. A later feasibility study could randomize monitoring intensity to see whether it changes performance or burden. Improvement during observation may reflect the act of being measured.

Appendix B. A short glossary

Table B1. Terms used in the manuscript.

Term	Meaning in this paper
Allopregnanolone	A progesterone-derived neuroactive steroid that modulates GABA-A signalling. Its effects depend on context; a blood level is not a direct measure of its action in the brain.
ATP / phosphocreatine	Molecules involved in cellular energy transfer. Their measured ratio does not directly reveal energy-production rate.
Autonomic regulation	Nervous-system regulation of involuntary bodily functions, including cardiovascular responses.
Biomarker	A measured biological characteristic. An abnormal value alone does not establish a diagnosis, causal pathway or cognitive impairment.
Blood–brain barrier	A regulated interface between circulation and brain tissue; not a diagnosis inferred from cognitive symptoms.
Brain fog	A subjective report of reduced mental clarity. It can be important even when a brief cognitive test is normal.

Term	Meaning in this paper
Cognitive sensitivity	The proposed change in cognition for a comparable measured physiological change; distinct from starting margin or bodily reactivity.
Confidence interval	A range produced by a statistical procedure. A 95% procedure covers the fixed target value in 95% of repeated samples under its assumptions; it is not a 95% probability assigned to this particular interval.
Confounding	An alternative explanation arising when other causes influence both the exposure and the outcome. Adjustment addresses only what the design and assumptions can support.
Credible interval	A Bayesian range containing a specified proportion of posterior probability, conditional on the data, model and prior assumptions.
CSF	Cerebrospinal fluid, which surrounds the brain and spinal cord and moves through connected spaces. Its flow is distinct from cerebral blood flow and BBB permeability.
Cytokine	An immune-signalling molecule. A blood level does not identify its source or establish what is happening in the brain.
Ecological momentary assessment (EMA)	Brief repeated assessment during everyday life, often through a phone. Between-person reliability does not establish reliable within-person change.
Estradiol	One of the estrogen hormones. A measured level, a change over time and sensitivity to that change are different questions.
Executive function	Processes involved in maintaining goals, directing attention, holding information in mind and controlling responses; not one universal ability score.
Fecal microbiota transplantation	Transfer of screened donor stool material to a recipient. Its risks, purpose and evidence depend on the clinical indication.
Functional margin	A proposed distance from an independently meaningful task requirement; not a validated ADHD biomarker or an intelligence measure.
Genetic correlation	A population-level correlation in genetic effects across traits. It is not an individual risk or the percentage of people who have both conditions.
GLP-1	Glucagon-like peptide-1, a metabolic signalling hormone. Receptor-agonist studies must be interpreted by the actual population and outcome tested.
Hazard ratio (HR)	A comparison of instantaneous event rates among people still at risk. It is not a ratio of cumulative probabilities.
HbA1c	A blood measure reflecting glycemic exposure over preceding months. It does not show the timing of short glucose excursions.
Held-out prediction	Testing a model on observations that were not used to develop it.

Term	Meaning in this paper
Helminth	A parasitic worm. Hookworms used in experimental trials are distinct from gut bacteria.
Histamine receptor	A molecular target through which histamine signals. H1, H2 and H3 interventions are not interchangeable, and effects depend on location.
HOMA-IR	An indirect estimate of insulin resistance calculated from fasting glucose and insulin; not a direct brain measurement.
HRV	Heart-rate variability: variation in intervals between beats. A cardiac measure, not a direct assay of brain inflammation or universal vagal engagement.
Hypoglycemia	Low blood glucose. Experimentally induced, clinically low values are different from ordinary variation after meals.
Immunometabolism	The interaction between cellular metabolism and immune-cell behaviour.
Insulin sensitivity	Responsiveness to insulin. Measures of peripheral metabolism and experimental brain responses are not interchangeable.
Interoception	Detection and interpretation of signals from inside the body.
Mast cell	An immune cell that can release histamine and other mediators. Its involvement in a tissue is not automatically MCAS.
Mastocytosis	A disorder involving abnormal accumulation of mast cells. It is distinct from an inference of mast-cell activation based on nonspecific symptoms.
Mediation	A proposed pathway through an intermediate variable. Statistical mediation alone does not prove that pathway is causal.
Microglia	Resident brain immune cells whose functions include responses to injury and changes in the local tissue environment.
Microplastics and nanoplastics	Small plastic particles classified by size; boundaries and detection methods vary. Particle count and mass are different measurements.
mTOR / AMPK	Cellular signalling systems involved in nutrient/energy regulation; named candidates here, not universal treatment targets.
Neurophysiological Resilience Index (NRI)	An unvalidated proposal summarizing the depth and duration of a specified task deficit over a fixed window. It is not a diagnostic or general health score.
Odds ratio (OR)	A ratio of odds between groups. Odds are probability divided by one minus probability; an OR is not a risk ratio or a percentage-point difference.
Orthostatic	Related to becoming or remaining upright.
Physiological reactivity	The size or pattern of the measured bodily response to a specified challenge.

Term	Meaning in this paper
Physiological state	A specified, measured aspect of current bodily functioning, such as prior sleep or an orthostatic response.
PMDD	Premenstrual dysphoric disorder. Diagnosis requires a recurring impairing symptom pattern and prospective daily confirmation; a screening result remains provisional.
Post-exertional malaise	Delayed or prolonged symptom worsening after activity; not simply ordinary tiredness during a task.
Premenstrual exacerbation	Premenstrual worsening of a disorder that is also present at other times. It should be distinguished from a more restricted cyclical symptom pattern.
Progesterone	A steroid hormone with reproductive and neural effects. Its metabolites can have effects distinct from the circulating parent hormone.
Progression-free survival	Time until disease progression or death as defined by a trial. A cancer endpoint, not a cognitive outcome.
Recovery	The observed return toward a prespecified reference after a disturbance. The observation window and outcome must be defined.
Synbiotic	A preparation combining live microorganisms with substrates intended to support microbial activity. Effects are specific to the tested preparation and population.
Vigilance	The ability to remain alert and maintain attention over time.
VNS / taVNS	Vagus nerve stimulation / transcutaneous auricular VNS. Route, device and protocol matter; implanted and ear-based interventions are distinct.

Appendix C. Quantitative evidence at a glance

These studies answer different questions. Do not rank diseases using incompatible statistics. HR compares instantaneous event rates among people still at risk; OR compares odds. Neither is an individual probability. SD and Hedges g are standardized differences, not percentages. Intervals are 95% confidence intervals unless labeled otherwise; see sources for full methods.

Table C1. Direct ADHD and hormonal-state evidence

Study and sample	What was measured and found	What it can establish
Sleep restriction Becker 2019 [9] 72 enrolled; 48 analyzed	6.5 vs 9.5 h in bed; measured sleep differed by 1.6 h. Parent-rated inattention d = 0.60.	Crossover symptom effect; adolescent inattention and continuous-performance tests did not show the same result. Adherence-selected analysis.
Sleep treatment Hiscock 2015 [12] 244 randomized	Parent ADHD score differences: 3 months −2.9 (−5.5 to −0.3); 6 months −3.7 (−6.1 to −1.2).	Imputed estimates from one trial. Teacher total scores did not show the same benefit; unblinded families and stimulant co-treatment.
Melatonin Van der Heijden 2007 [65] 105 randomized	Sleep onset 26.9 min earlier vs 10.5 min later on placebo after 4 weeks.	Improved sleep timing; no significant cognitive, behavioural or quality-of-life benefit. Selected insomnia population.
Menstrual symptom diaries Roberts [32]; Zaritsky [33]	32 community participants over 35 days; later pilot: 30 stimulant-treated participants over 35 days.	Within-person symptom associations. No validated universal percentage worsening, causal hormonal pathway or dose-adjustment rule.
Objective cycle cognition Jang 2025 [66]	102 articles; 3,943 participants; 730 comparisons. No systematic robust cycle-phase differences.	General-sample counterweight; does not exclude symptomatic subgroups or effort required to maintain performance.
H3-targeting drug Herring 2012 [27] 72 randomized	MK-0249 vs placebo AISRS difference −2.1 (−6.6 to 2.3) at 4 weeks.	Primary outcome null. Methylphenidate comparator improved symptoms. Histamine involvement alone does not establish treatment efficacy.

Table sources (in row order): [9], [12], [65], [32], [33], [66], [27].

Check how many people contributed to each endpoint. That group can be smaller than the enrolled cohort. The melatonin trials and the behavioural sleep intervention tested different treatments. AISRS is the Adult ADHD Investigator Symptom Rating Scale.

Cross-system overlap

Table C2. Co-diagnosis and shared liability

Study and sample	What was measured and found	What it can establish
Endometriosis Gao 2020 [34] 854,361 women	14,144 endometriosis diagnoses. Sibling-comparison ADHD HR 1.98 (1.34–2.93).	Later recorded ADHD diagnosis; not proven disease onset, impaired cognition or an immune cause. Within-family comparison reduces some confounding.
Endometriosis genetics Li 2026 [35]	ADHD–endometriosis genetic correlation: 0.21 using LDSC; 0.17 using HDL.	Two methods on source GWAS data, not two independent cohorts. Correlation is not individual risk or a percentage of shared causation.
Migraine cohort Hsu 2022 [67] 81,441 per group	462 vs 212 newly coded diagnoses. Adjusted HR 1.92 (1.64–2.34). Ages 18–29: 1.31 (0.58–2.98).	Overall association, imprecise young-adult estimate. No cognitive outcome; variable follow-up. Counts do not estimate lifetime prevalence.
Atopic dermatitis Strom 2016 [68]	354,416 children: adjusted ADHD OR 1.14 (1.03–1.26). 34,613 adults: OR 1.61 (1.25–2.06).	Cross-sectional surveys with healthcare-use adjustment. Itch, sleep and ascertainment remain possible routes; no mast-cell mechanism established.
Dermatitis synthesis Cheng 2024 [69]	ADHD pooled OR 1.28 (1.18–1.40). Review: 24 studies, 71.4 million subjects across ADHD AND autism analyses.	The overall denominator is not the ADHD-only denominator. Large sample size does not remove confounding or prove direction.

Table sources (in row order): [34], [35], [67], [68], [69].

An association between diagnoses leaves the size of any cognitive change open. Syntheses that reuse a dataset contribute overlapping evidence. LDSC (linkage disequilibrium score regression) and HDL (high-definition likelihood) estimate genetic correlation from genome-wide association study data.

Post-COVID cognition and physiology

Table C3. Different post-COVID measurements

Study and sample	What was measured and found	What it can establish
Community cognition Hampshire 2024 [70] 112,964 completers	Unresolved persistent symptoms: −0.42 SD (−0.53 to −0.31) vs no-COVID reference group.	Objective group difference; no personal pre-infection baseline. Not proof of a specific mechanism or irreversible IQ loss.
Brain barriers Greene 2024 [44] 32 MRI participants	10 recovered; 11 Long COVID without fog; 11 with fog. Greater leakage in the fog group.	Small selected imaging comparison. Association does not establish a universal cause or a sequence leading to brain fog.
Brain energy ratios Weber-Fahr 2026 [45] 27 vs 23	Authors report lower cingulate ATP/PCr and an anterior association with cognition.	Direct brain spectroscopy, but ratio ≠ ATP-production rate. Does not establish causality; exact regional estimates remain an independent verification gap.
Orthostatic testing Jamal 2022 [71] 24 referred patients	23 tilt-intolerant; 4 POTS; 15 provoked after nitroglycerin.	Highly selected sample and provocative protocol. Not 96% population prevalence or evidence that autonomic change caused cognition problems.
Famotidine Momtazmanesh 2023 [49] 60 / 50 / 45	Randomized / analyzed / completed. Week-12 endpoint MMSE +2.44 (1.47–3.41); MoCA +2.56 (1.69–3.43).	Mean enrollment about 29 days after onset. Early convalescence; low screening scores, early exclusions and carried-forward observations.
STIMULATE-ICP 2026 [50] 778 randomized	Famotidine–loratadine: adjusted FAS fatigue difference −1.48 (−2.88 to −0.08) at 12 weeks.	Open-label comparison with no study drug; supportive care in all groups. No significant group difference at week 24. Fatigue ≠ cognition.

Table sources (in row order): [70], [44], [45], [71], [49], [50].

BBB, ATP/PCr, tilt testing and cognitive outcomes came from different samples. The two drug trials differ in illness timing and outcome; one does not replicate the other. MMSE: Mini-Mental State Examination; MoCA: Montreal Cognitive Assessment; FAS: Fatigue Assessment Scale. MMSE and MoCA are cognitive screening tests.

Interventions and clinical boundaries

Table C4. Vagal and GLP-1 findings in their actual populations

Study and sample	What was measured and found	What it can establish
RESET-RA [57] 242 randomized	ACR20: 35.2% vs 24.2%. Adjusted difference 11.8 percentage points (0.6–23.1) at 3 months.	Arthritis composite response; one-sided primary test. Implanted stimulation, not ear stimulation. Related serious events 1.6%, perioperative.
Auricular VNS cognition [58] 30 pooled studies	Overall Hedges g 0.41 (0.30–0.53); working-memory/attention g 0.185 (0.043–0.327).	Mixed populations and protocols. Laboratory cognition signal, not established ADHD efficacy, mechanism or everyday durability.
Auricular VNS HRV [59] 16 studies	g 0.014; Bayesian shortest credible interval −0.103 to 0.132.	No reliable acute cardiac HRV effect under tested conditions. HRV is not a direct brain or inflammation assay.
SELECT [73] 17,604 randomized	Cardiovascular events 6.5% vs 8.0%; HR 0.80 (0.72–0.90).	Overweight/obesity plus cardiovascular disease, no diabetes. Vascular benefit; not a cognition or dementia outcome.
REWIND [72] 8,828 tested	Exploratory cognition HR 0.93 (0.85–1.02); post-hoc baseline-adjusted HR 0.86 (0.79–0.95).	Type 2 diabetes with cardiovascular risk. Initial estimate null; favorable result depends on stated analysis. Not diagnosed dementia.
evoke / evoke+ [56] 3,808 randomized	CDR-SB change differences −0.08 (−0.35 to 0.20) and +0.10 (−0.17 to 0.38).	Early symptomatic Alzheimer disease, 104 weeks. Neither showed slowed clinical progression. No inference about an ADHD-to-dementia trajectory.

Table sources (in row order): [57], [58], [59], [73], [72], [56].

These results cannot be averaged into a “brain health” effect. The targets, follow-up, populations and endpoints differ. Mouse receptor experiments [31, 51, 52, 55] supply mechanistic questions, not human efficacy estimates. ACR20 requires at least 20% improvement in tender and swollen joint counts and in at least three of five other clinical measures; it is not a measure of cognition.

Table C5. Medication benefit: four different outcomes

Study / population	What was measured	What the result permits
Symptoms, children Cortese 2018 [76]	Clinician-rated SMD: amphetamines −1.02 (−1.19 to −0.85); methylphenidate −0.78 (−0.93 to −0.62); atomoxetine −0.56 (−0.66 to −0.45).	Closest to 12 weeks; 10,068 children/adolescents in efficacy analyses. Negative means fewer symptoms. Network estimates include indirect comparisons.
Symptoms, adults Ostinelli 2025 [77]	Clinician-rated SMD: stimulants −0.61 (−0.71 to −0.51); atomoxetine −0.51 (−0.64 to −0.37).	Short-term placebo comparisons. Review included 113 trials across treatment types, not 113 medication trials. Not a measure of IQ or percentage improved.
Cognitive tasks Vertessen 2022 [79]	31 studies, 804 children. Methylphenidate effects d = 0.20 to 0.73 across domains.	A range of domain estimates, not one pooled effect or confidence interval. Executive-function benefit did not consistently increase with dose.
Classroom learning Pelham 2022 [80]	173 children: 37% more arithmetic per minute; 53% fewer rule violations. No extra curriculum-learning gain.	Randomized crossover. Author-reported percentages; structured summer setting. Productivity, behaviour and acquisition of new knowledge differ.
First health events Zhang 2025 [81]	Suicidal behaviour: 14.5 vs 16.9 per 1,000 person-years; adjusted IRR 0.83 (0.78–0.88). First accidental injuries: IRR 0.98 (0.96–1.01).	148,581 newly diagnosed people; two-year target-trial emulation. Observational, not randomized. Injury interval includes no difference; rates are not two-year percentages.

Table sources (in row order): [76], [77], [79], [80], [81].

Parentheses give 95% confidence intervals unless stated otherwise. SMD/d are standardized differences, not percentages. IRR is an incidence-rate ratio. These endpoints cannot be combined into a single treatment-benefit score.

Table C6. Medication effects on the rest of the state

Study / population	What was measured	What the result permits
Objective sleep Kidwell 2015 [82]	9 studies, 246 young people. Adjusted standardized effects: latency +0.54; efficiency −0.32; total sleep time −0.59.	Stimulant versus control conditions. No intervals reproduced here. Small evidence base; not every person experiences all three effects.
Pulse and pressure Farhat 2025 [83]	Adult methylphenidate: pulse +4.37 bpm (3.16–5.59); systolic pressure +1.66 mmHg (0.38–2.93).	102 trials; 13,315 young people and 9,387 adults overall. These adult estimates had very low confidence; median trial duration 7 weeks.
Autonomic context Green 2013 [84]	27 POTS patients, crossover. Standing heart rate 121 ±17 vs 105 ±15 bpm at 2 hours after atomoxetine vs placebo; p = .001.	± values are standard deviations. Acute experiment, not ADHD treatment efficacy, long-term risk or proof of a cognitive mechanism.
Longer-term CVD Zhang 2024 [85]	10,388 cases / 51,672 matched controls. >5 years cumulative use vs nonuse: adjusted OR 1.23 (1.12–1.36).	Nested case-control analysis, base cohort 278,027. Median follow-up 4.1 years. Residual confounding; odds are not absolute risk.
Migraine prevention Topiramate label [88]	Cognitive adverse reactions: 22% at 100 mg/day vs 10% placebo in 6-month trials.	Pooled reported adverse reactions, not an ADHD-specific cognitive scale. Trial dose is descriptive, not advice.
Drug exposure Atomoxetine label [86]	Fluoxetine/paroxetine: atomoxetine exposure about 6–8 times as high in normal CYP2D6 metabolizers.	Specified inhibitors and metabolizer context. Exposure is not a symptom effect size; does not apply equally to all antidepressants.

Table sources (in row order): [82], [83], [84], [85], [88], [86].

Intervals are 95% confidence intervals unless otherwise labeled. bpm: beats per minute; CVD: cardiovascular disease. Short-term physiological changes and long-term clinical events are different evidence. Official US labels are cited for specific interactions, not as jurisdiction-specific treatment guidance.

Appendix D. Who makes it into the data

The two pie charts show who entered each study and whose data made it into the analysis. The slices describe participation.

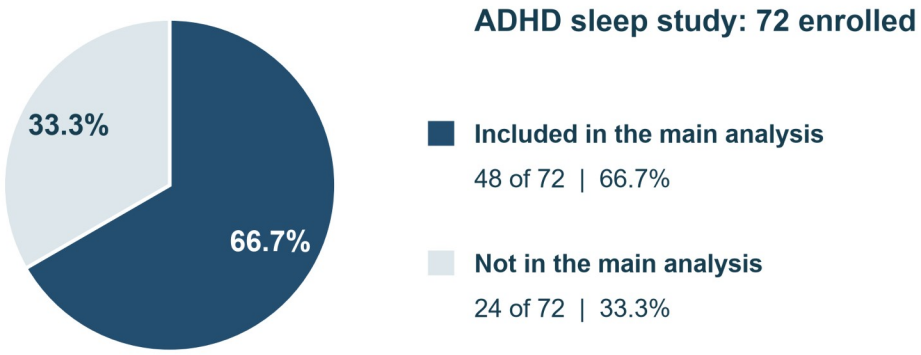


Figure 32. The sleep experiment's analysis sample. Of 72 adolescents who entered the protocol, 48 met the analysis requirements for actigraphy data and the prespecified sleep-duration contrast. The remaining 24 are calculated as 72 minus 48; this category should not be read as 24 dropouts. Percentages are calculated and rounded. [9]

The analysis covers adolescents who produced the required sleep contrast. To estimate the effect of assigning the protocol to everyone enrolled, we would also need to account for missed nights and incomplete measurements. We should expect both in a study at home.

Completion is another selection step

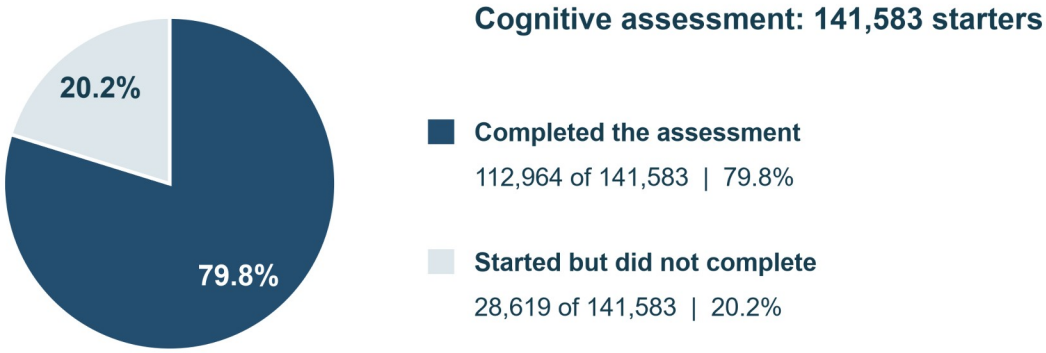


Figure 33. Completion among people who started the online cognitive assessment. Hampshire and colleagues reported 141,583 starters and 112,964 completers. The remaining 28,619 are calculated by subtraction. Percentages are conditional on starting the assessment; they are not population participation rates, infection rates or the prevalence of cognitive impairment. [70]

Even in a large sample, the people who finish may differ from those who stop. The chart leaves three questions open: why people stopped, how their cognition compared and which way any bias would run.

We will count invitations, study starts and usable assessments separately and record reasons for missing data. Where records allow, compare people who finish with those who stop. Set the sensitivity analyses before seeing the results.

Appendix E. Two outcomes from the same trial

SELECT tested semaglutide against placebo in 17,604 adults aged 45 or older with established cardiovascular disease, BMI at least 27 and no diabetes. These waffles put two outcomes on the same visual scale.

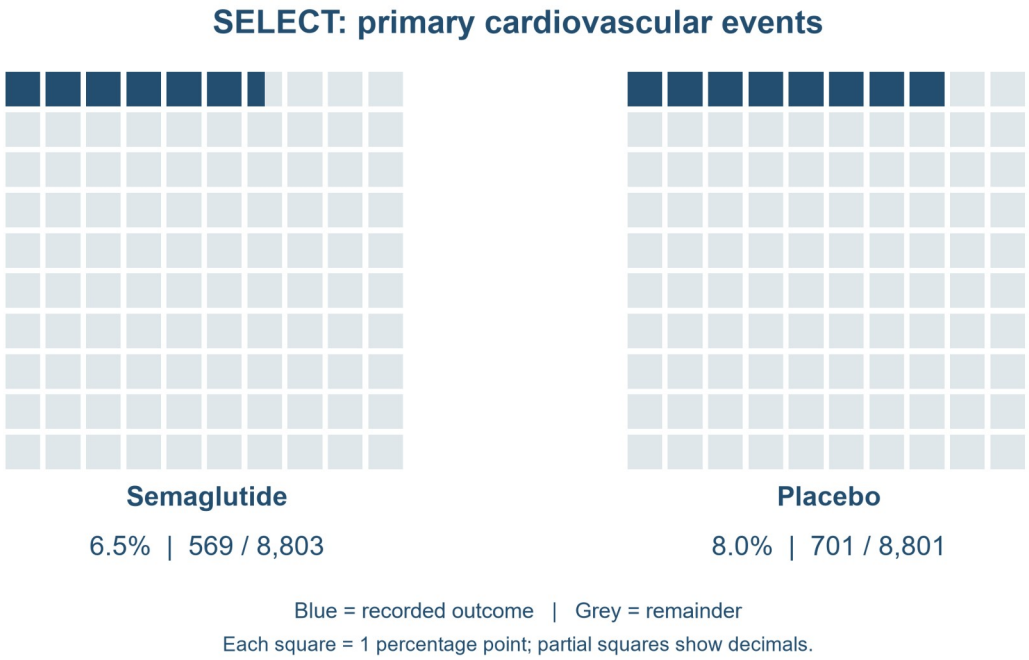


Figure 34. Cardiovascular death, nonfatal heart attack or nonfatal stroke: 6.5% versus 8.0%. Mean follow-up was 39.8 months. These are reported participant proportions, not annual risks; the time-to-event estimate was HR 0.80 (95% CI 0.72–0.90). [73]

This supports cardiovascular benefit in the studied population. It does not establish an ADHD or cognitive benefit, or identify which biological pathway produced the result.

Stopping treatment because of adverse events

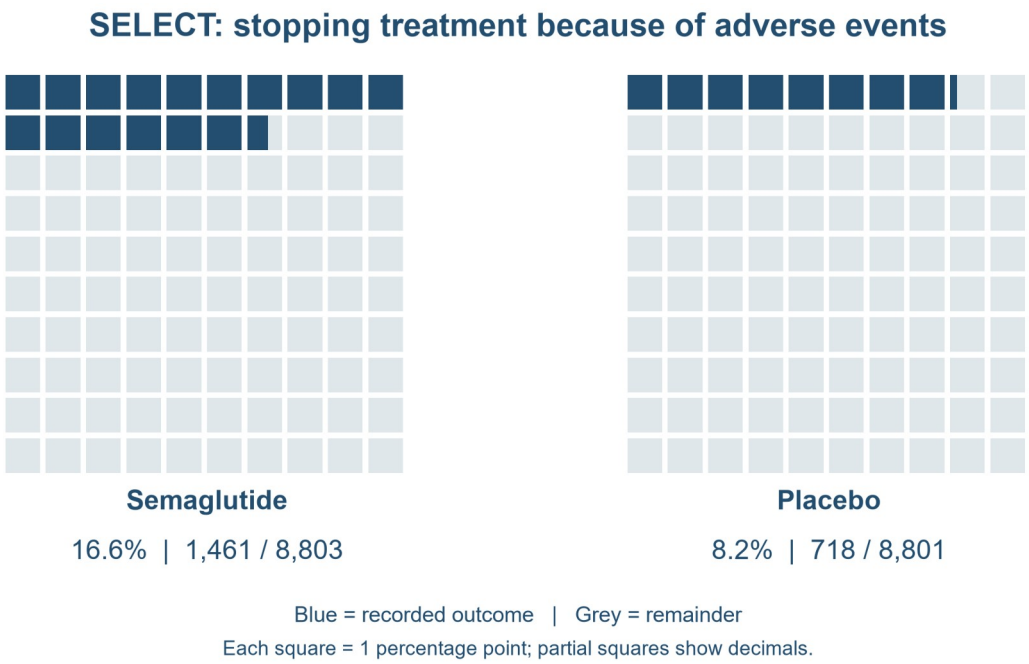


Figure 35. Adverse events leading to permanent discontinuation of trial treatment: 16.6% versus 8.2%. This is a treatment-discontinuation outcome, not withdrawal from study follow-up or the frequency of all adverse events. [73]

Grey marks the share without the specified recorded outcome. Each square represents part of a proportion. Both charts use the publication's rounded percentages.

A person can experience both outcomes, and their severity can differ. A single subtraction would lose that information. I would report functioning, benefit and tolerability separately.

Appendix F. Reading the overall result and the subgroups

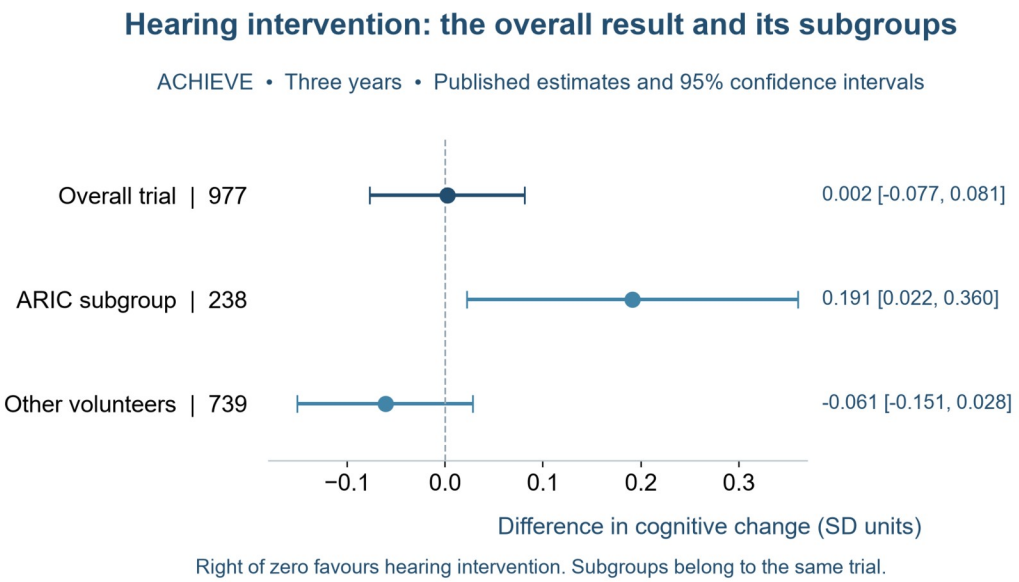


Figure 36. ACHIEVE: intervention minus control in three-year global cognitive change, in standard-deviation units, with 95% confidence intervals. Positive values favour hearing intervention. The two recruitment subgroups are components of the overall trial, not separate replications; their interaction was p = 0.010. A confidence interval crossing zero leaves the direction uncertain. [155]

Read each subgroup estimate beside the overall result. A difference can suggest the next test. Pooling these three rows would count some participants twice.

Appendix G. Mono outcomes and follow-up availability

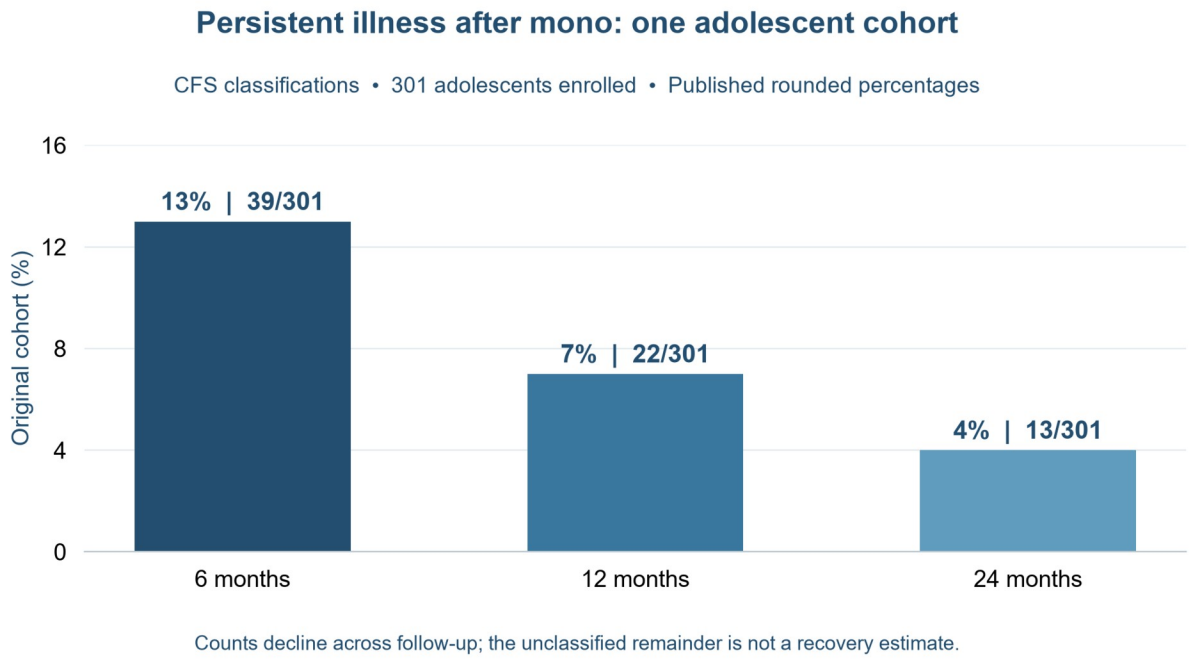


Figure 37. Chronic fatigue syndrome classifications after mono in an adolescent cohort. Labels show identified cases out of the original 301 participants; bar heights use the paper's rounded percentages. These are repeated observations from one cohort, not independent studies. Follow-up losses mean the remaining percentage cannot be labelled recovered. This chart measures clinical classifications, not cognitive impairment or ADHD risk. [160]

The chart answers a narrow question: how many participants were classified with CFS at each follow-up? It does not tell us which mechanism maintained illness, whether an individual's course was steady, or whether ADHD changed their recovery. The different college-cohort definitions and the selected seven-year follow-up belong in the text, not pooled into these bars.

Appendix H. Seven-year mono outcomes and follow-up

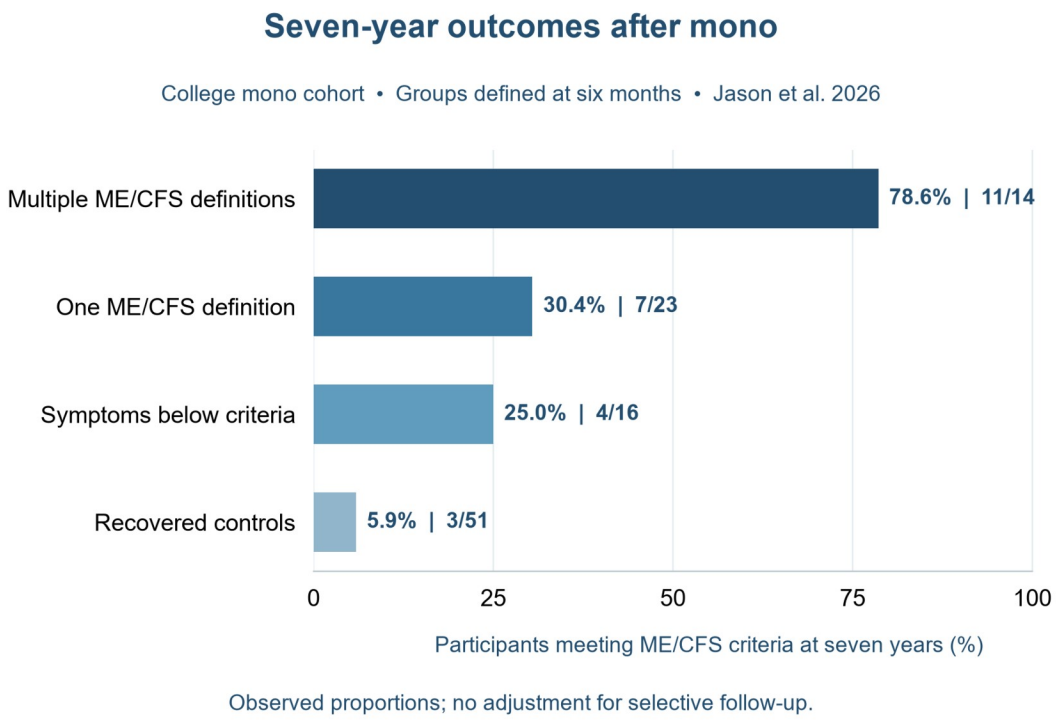


Figure 38. ME/CFS at seven years among returning participants, grouped by six-month status. Percentages are calculated from the labelled counts. These selected groups do not estimate risk for everyone with mono; group names replace the paper's definition-based "severe" and "moderate" labels. [172]

Follow-up behind the outcome estimates

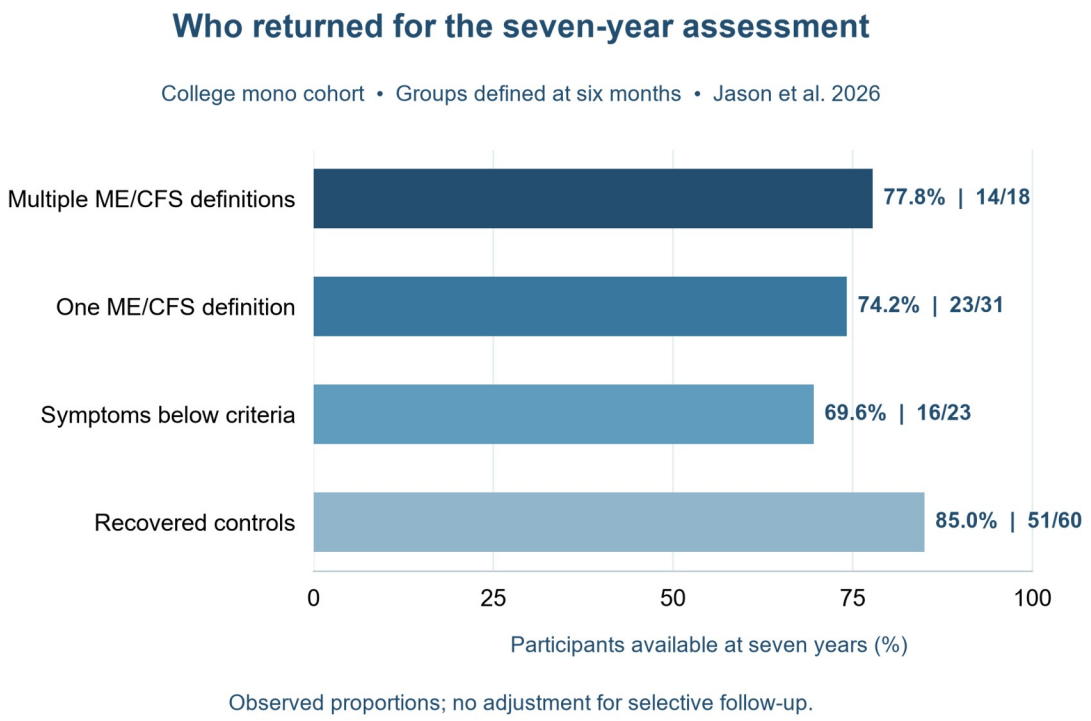


Figure 39. Follow-up availability in the same study, by six-month status. Denominators are the eligible groups with baseline data in the follow-up report, not all 238 original mono cases. Missing participants could have different outcomes. These bars show retention, not recovery. [172]

Appendix I. Source verification and corrections

The audit dated 6 September 2026 covered the manuscript's 177 reference entries, internal citation targets, 12 tables and 30 figure captions, with selected original-source checks. Europe PMC matched 161 entries to publication records. That confirms source identity, not the accuracy of every claim or method. Previous primary-source checks were retained where a fresh full-text check was not completed.

The changes are specific: the auricular VNS confidence interval now follows the paper's main text; the famotidine account states its reported registry limitation; and the EBV–MS estimate identifies its comparator, analysis sample and wide interval. Bibliographic details and the chart workbook's outdated guide were corrected. The displayed chart values did not change.

Eleven cited papers had indexed correction notices. Ten notices were checked against notice text or a documented correction. The 2026 ADHD dose-effect synthesis corrected a mislabeled adult curve in its fifth figure from guanfacine to modafinil. A copy of the corrected journal article confirms the label [78]. Full notice text for REWIND remains an access gap [72]. No retraction notice was flagged in the 161 matched records; this is not an exhaustive retraction search.

Three checks remain open: the adult waking-EEG methods [15], exact regional spectroscopy estimates [45] and an independent comparison with the famotidine registry [49]. These are gaps in the audit. The proposed pathways still need testing, and this narrative review still needs independent scientific review.

The November integration brings the 238 numbered references, the U01–U57 evidence register and the targeted gap and methods sources into the same argument. The prose and structure were consolidated on November 9, 2026. Selected consequential claims were checked against original reports or official records; Appendix I records the scope and unresolved access limits. This editorial integration is not a new independent verification of every claim or source.

The famotidine report gives the sample flow and mean enrollment timing: 60 randomized, 50 analyzed, 45 completed, with enrollment about 29 days after symptom onset. The authors report that its registry did not explicitly distinguish primary and secondary outcomes; an independent registry comparison remains outstanding. The adult waking-EEG report remains supported here at abstract level rather than by a complete methods audit; exact regional spectroscopy effects also need fuller review. Appendix I records the scope of this audit and two correction notices whose full contents remain unchecked. The paper still needs independent scientific review, and the ADHD-specific experiments still need to be done.

Later source checks

Checks of the expanded literature covered gut interventions, hookworms, brake wear, microplastics and cancer, including negative primary outcomes and corrections. Several findings were checked through original indexed abstracts. The full microplastics correspondence was inaccessible; its discussion is limited to the stated analytical dispute. Earlier EEG, spectroscopy and registry-access limits remain.

The developmental-profile review checked five original reports: dyslexia genetics and the autism register study in full text; twin, melatonin and autism microbiome findings through indexed original abstracts. Earlier references were not fully re-audited.

The illustrated-edition checks covered semaglutide in Alzheimer's disease, vagal stimulation in rheumatoid arthritis, EBV–ADHD and endometriosis–ADHD associations, GLP-1 safety, tirzepatide contraception advice, MCAS, meningeal mast cells and dyslexia–ADHD co-reporting. Accessible primary text, original abstracts and official notices were used; this was not a full methods audit. The 2023 Doust correction [234] restores a genomic-control Methods paragraph without changing the chart's 24% and 9% values. Earlier unresolved checks remain unresolved.

The November source inventory contains the 238 numbered entries and 57 U identifiers. Record-check counts reported during that workflow progressed from 33 to 42 of 295; these describe that particular review workflow, not a new exhaustive full-text audit. They should not be combined with the earlier source-identity checks as though they were one uniform review.

The SURMOUNT-OSA original report is DOI 10.1056/NEJMoa2404881; reference [113] links both that report and correction NEJMx240005. Reference [119] likewise links the SYNERGY-NASH original report, 10.1056/NEJMoa2401943, alongside correction NEJMx250014. The latter concerns a baseline AST value in Figure 2B. The "1985" in J Appl Physiol (1985) is part of the journal title, not the publication year of reference [42], which is from 2014.

The publisher correction for RECOVER-VITAL adds Marta Cerda as an author and leaves Figure 42's estimates and intervals unchanged [N02]. Original reference 190 and U19 describe one trial. Read the exenatide Parkinson's trial alongside its expression of concern [N03]. The PROMOTe and brain-microplastics notices remain relevant to their discussions [N04, N05]. The diet/microbiome erratum corrects a transformation, estimates and plots; its authors report unchanged overall conclusions. Affected values are not reproduced here [N06].

The 34-man sleep-deprivation experiment is a 2021 issue publication, online in 2020 [10]; the erroneous 2026 date in some working summaries does not apply to the study. The 21-person allergy time-series study included only 16 participants with sufficient data for individual VAR models [29]. The 105-child melatonin trial improved sleep timing/duration without significant cognitive or behavioural gains [65]. These sample and outcome distinctions constrain interpretation throughout the paper.

Source identity, bibliographic correction and claim-level validity are different checks. The integrated manuscript remains a targeted narrative perspective; primary methods, numerical outcomes and access-limited reports require independent review before scientific submission.

Original-study audit: 9 November 2026

The earlier audit covered 322 citation identifiers at varying access levels. The cross-session integration adds 20 separately sourced entries. Of 125 previously abstract-only entries, 105 now have targeted full-report comparisons, including the cited endometriosis and operations preprints. Full reports remain access-limited for 20 entries. The published vagus-stimulation meta-analysis has now been checked beyond its earlier preprint. This is not verification of every analysis or underlying dataset. Related identifiers are not independent studies.

Values in 36 evidence-bearing numbered figures agreed with accessible reports, abstracts or publisher-indexed results. Six figures are hypothetical or conceptual. No figure values changed. Figure 26's

learning-and-recall estimate used the publisher's result passage because full-text download failed. Captions identify confidence levels, units, non-independent follow-ups and calculated percentages.

The audit distinguishes preplanned secondary cognition from the null primary endpoint in the semaglutide depression trial [U54], clarifies analysis denominators in the vascular studies [U29, U30], and separates the pupil study's different variability measures and behavioral outcomes [U01]. The exenatide expression of concern is now stated beside the result as well as in the source notes [126, N03]. Corrections to the psoriasis-combination trial's SD/SE labels, the target-trial study's original curve labels and an author name in the cardiovascular review did not alter the contrasts plotted here [141, 81, 83].

Unresolved checks include ACX-02 versions [U57], EEG methods, spectroscopy and the famotidine registry [15, 45, 49], and the full REWIND correction notice [72]. The microplastics criticism/reply checks used publisher summaries [224, 225]. Sponsor announcements remain preliminary [U05, U12]. Published-value agreement does not validate raw data or the proposed framework.

Appendix J. A provisional resilience score

Future validation and the resilience index

The proposed Neurophysiological Resilience Index (NRI) summarizes how far a specified outcome falls below its expected course, and for how long, within a fixed window. Its interpretation depends on the outcome and reference trajectory chosen for that study. Clinical use would require separate validation.

Existing resilience research shows why the definition matters. A study of 170 older adults after knee replacement compared recovery trajectories with outcomes relative to model predictions. These approaches sometimes described the same person differently. A study in 248 adults combined cortisol, blood pressure and heart-rate responses to a laboratory stressor; that index is a precedent for measurement, not a validated ADHD score. Neither study validates the formula proposed here. [178, 180]

I would report the curve and all four measurements alongside the score: functional margin, physiological reactivity, cognitive sensitivity and recovery. Each answers a different question. A strong bodily response may be appropriate, and steady task performance may still require more effort or leave someone with less capacity later in the day.

First ask whether the score is reproducible and predicts later functioning measured independently. Compare it with baseline performance, the largest decline and a model using sleep and pain alone. The name describes the hypothesis. A physiological explanation needs evidence beyond the task score.

J1. What the measurement research adds

The studies below help choose how to measure the proposed NRI, which still needs validation. They use different populations, disturbances and outcomes. The ADHD study needs its own sample-size calculation.

Table J1. Original studies informing the proposed measurement

Study and sample	What was measured	What it tells us
Whitson 2026 [178] 170 older adults	Pain interference before knee replacement and at 1, 2, 4 and 6 months.	Trajectory groups and outcomes relative to predictions were not interchangeable. The reference defines the question.
SPRING 2025 [179] 79 older adults	Multisystem responses before major clinical stressors; cardiac and cortisol measures.	Feasibility and associations with resilience surrogates support further testing, not an ADHD classification.
Liang 2026 [180] 248 adults	Cortisol, blood pressure and heart-rate responses to the Trier Social Stress Test.	A physiological index and distress associations in a selected sample; internal modeling is not external validation.
Malinen 2026 [181] 26 trained runners	Repeated running tests. Change ICCs: peak speed 0.81; relative VO_2max 0.36; first ventilatory threshold 0.07–0.25; second threshold 0.46.	Reliability depends on the endpoint. Reliable absolute values do not guarantee reliable changes. ICC (intraclass correlation) measures reliability, not treatment benefit.
Kolk 2024 [183] 109 eligible; 30 enrolled	Thirty days of monitoring after an emergency visit; 28 participants completed followup.	Completion among volunteers does not establish representative coverage or complete daily observations.

Table sources (in row order): [178], [179], [180], [181], [183].

Source notes: All five reports were checked in full text. The runner study's Table 5 gives second-threshold ICC 0.46, outside its abstract's 0.07–0.36 range; we follow Table 5. COSMIN distinguishes reliability, error, validity and responsiveness for health-status instruments; it does not certify this score. [178, 179, 180, 181, 182, 183]

J2. Define the score before collecting outcomes

Start with one task outcome, one defined disturbance and one observation window. Orient the task so higher values mean better performance. Estimate the reference from repeated observations under the comparison condition, accounting for practice, time of day and the prespecified medication context. A person's single best day is not a reliable baseline.

$$B = (1 / (T \times \Delta)) \times \int_0^T \max(0, C_{ref}(t) - C_{obs}(t))\, dt$$

NRI = 100 × max(0, 1 − B)

Cref(t) is estimated reference performance at time t. Cobs(t) is the observed performance trajectory estimated with a prespecified model. T is the fixed follow-up duration, and Δ is a positive scaling change in the task's own units. The integral sums the deficits over time. Dividing by T × Δ gives the average deficit relative to Δ. Performance above the reference contributes zero deficit.

If there is an independently justified functional anchor, use it for Δ. Otherwise, report Δ as a provisional scaling choice and check how the score changes under other choices. A 20-point example is a scaling

illustration. To compare scores, use the same outcome scale, T and Δ, with comparable disturbances and reference conditions.

An NRI of 100 means the observed window had no estimated deficit. Assign it only when exposure and observation are adequate. An NRI of 0 means B is at least 1. Retain B and the raw area so that larger deficits remain distinguishable at the floor. Adding time after recovery can raise an intermediate score by spreading the same deficit over a longer window. The score stays at 100 when B is zero and at 0 while B remains at least 1.

This reference asks how performance differs from the specified comparison condition. It is not the expected recovery of a clinical prediction model, and it is not automatically a causal estimate of what would have happened without the disturbance. Sleep, pain and medication may be confounders, components of the state or mediators. Their roles must be specified before adjustment; adjusting away a post-disturbance pathway can change the question. [178]

J3. A worked example with invented values

All three examples start from an expected task score of 80. Set T to 24 hours and Δ to 20 points solely to show the arithmetic. Assume an immediate decline followed by straight-line recovery, with complete, noiseless observation. These are not patient data, clinical cutoffs or expected ADHD responses.

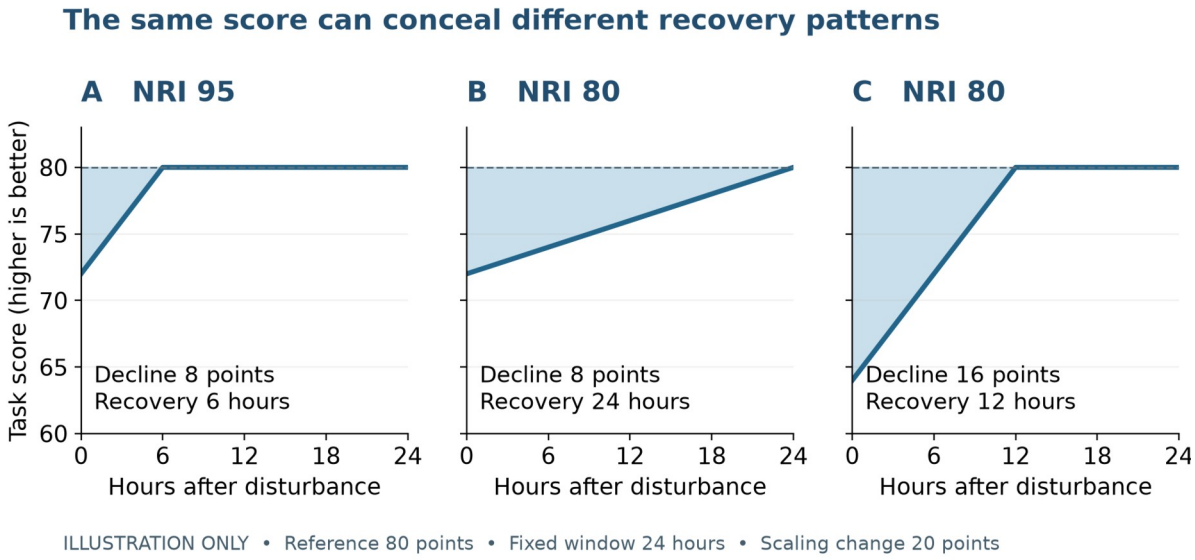


Figure 40. Depth and duration both affect the proposed index. Shading is the accumulated deficit below the reference line. B and C have equal areas and equal scores, despite different maximum declines and recovery times. All trajectories and scaling choices are hypothetical.

Table J2. The denominator is 24 hours × 20 points = 480 point-hours in every row.

Example	Deficit area	B	NRI
A: 8 points; 6 hours	½ × 8 × 6 = 24 point-hours	24 / 480 = 0.05	95
B: 8 points; 24 hours	½ × 8 × 24 = 96 point-hours	96 / 480 = 0.20	80
C: 16 points; 12 hours	½ × 16 × 12 = 96 point-hours	96 / 480 = 0.20	80

The score cannot tell us which person had more pain, a larger physiological change, less practical margin or a harder time completing ordinary tasks. Those outcomes need their own measurements. This is why the response curve and profile remain part of the result.

J4. Rules that keep the number interpretable

Measurement noise. Taking only positive deficits creates apparent burden even when a stable person fluctuates randomly around the reference. Prespecify the trajectory model and test the complete scoring procedure on stable-condition repeats and simulated null data. Report uncertainty from both reference and response estimation. If false burden is too large, revise or reject the score.

Sampling and missingness. Specify assessment timing and the longest acceptable gap. Numerical integration can use trapezoids on the estimated curve where coverage is adequate; it cannot establish what happened during an unobserved flare. Missing a substantial part of the window should produce an unscorable result or explicit bounds, not an assumption of recovery. Report scheduled and completed observations and reasons for missingness.

Recovery. Define return to reference using a prespecified tolerance and sustained observations, rather than one favorable reading. With no detectable decline, recovery time is undefined. If recovery is not seen by the end, report “not observed within follow-up” and treat the time as censored. A later relapse or delayed worsening needs enough follow-up to be observable.

Exposure and context. A smaller disturbance can produce a better score without greater resilience. Compare defined, reasonably comparable conditions or model exposure separately with held-out evaluation. Do not divide by an improvised physiological-load score. A low heart-rate or cortisol response is not inherently healthier.

The accompanying profile. Report baseline function and practical margin; the measured bodily response; the relationship of cognition to that response; and recovery. Also retain the peak deficit, raw deficit area, B, T, Δ, medication context, effort and independent daily function. Cognitive sensitivity requires repeated joint physiological and cognitive data, not just this area calculation.

The enrollment and monitoring losses in the feasibility study illustrate why missingness belongs in the design. A neat response curve from the people who can complete monitoring may leave out those with the hardest days. [183]

J5. What would make the proposal earn its place?

First, establish repeatability under reasonably stable and comparable conditions. Estimate measurement error and uncertainty at the individual level. In the runner study, some changes were much less reliable than absolute performance. That lesson transfers to measurement design; the exercise protocol and its thresholds do not transfer to ADHD or post-infectious illness. [181]

Second, lock the scoring rules and test later observations and new participants. Evaluate whether the score predicts an independently collected functional outcome beyond baseline ability, peak decline, recovery time alone and a simpler sleep–pain–medication model. Keep those outcomes separate from the measurements used to construct the index. Report calibration and useful predictive improvement, not just a statistically significant association.

Third, test generalization rather than assuming it. Start with a defined state and population. Compare ADHD and non-ADHD participants directly, distinguish an interaction from a lower starting level, and examine measurement behavior across relevant treatment and demographic groups. A score from one infection or hormonal state is not yet a stable personal trait. Performance in a new physiological state is a separate test.

Finally, ask whether a change in the score corresponds to a meaningful change in life. Reliability, validity and responsiveness are separate questions. The four-part profile is not a set of interchangeable questionnaire items; a high internal-consistency coefficient would not validate it. Nor does a component explaining considerable physiological variance prove clinical usefulness. [179, 180, 182]

Use naturally occurring variation or clinically justified observations. Do not induce illness, withdraw medication, restrict sleep or provoke exertional worsening just to obtain a score. No cutoffs, treatment choices or labels such as normal, fragile or highly resilient are justified by this proposal.

What would count against it? Apparent deficits generated by noise; poor repeatability; dependence on arbitrary timing or scaling; systematic failure on the worst days; or no useful prediction beyond simpler measures. These are reasons to change or abandon the metric, not reasons to keep adding biomarkers. The broader state hypothesis must remain testable even if this particular score fails.

Appendix K. Two scales of evidence

K1. A long-term association is not a daily mechanism

The paper asks about changes within a person, but it also needs to account for health across decades. This cohort result belongs to that longer timescale. It cannot tell us why cognition changes from one morning to the next.

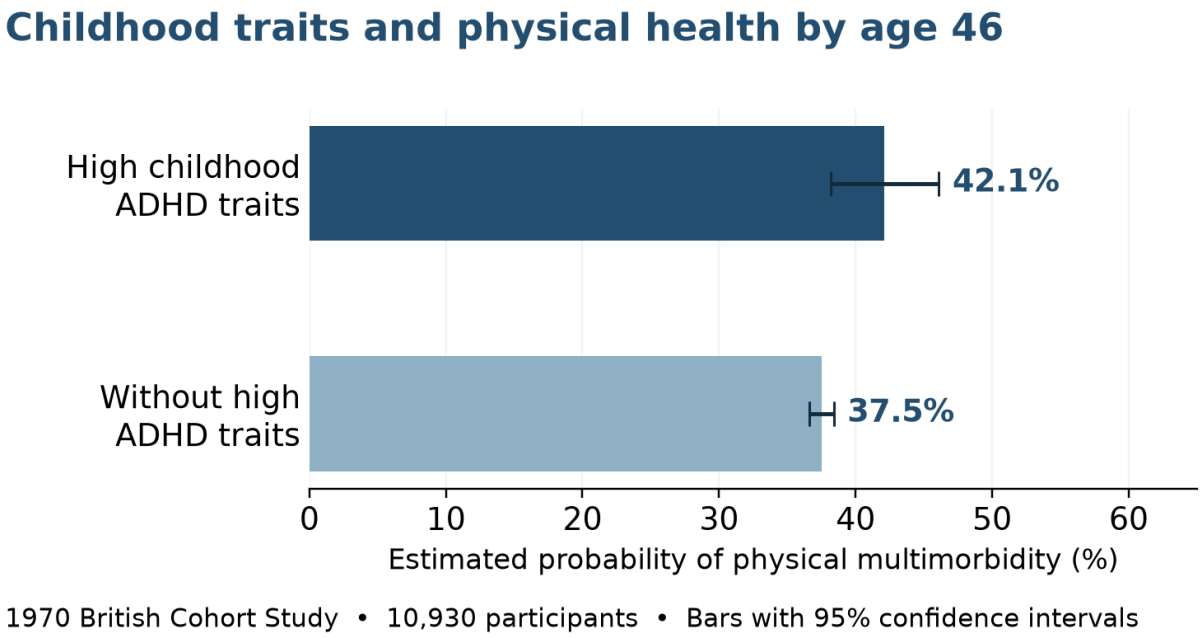


Figure 41. Model-estimated probabilities and 95% CIs for two or more reported physical conditions by age 46: 42.1% (38.2–46.1) with high childhood ADHD traits and 37.5% (36.6–38.4) without. The difference is 4.6 percentage points. These are not raw case counts or a causal treatment contrast. [184]

How to use this result

Keep the levels separate. A population association can identify a group whose health deserves attention. A repeated-measures study can test whether a particular change predicts a later lapse. An intervention can test whether changing a target helps a defined outcome. Evidence from one level can motivate the next test, but cannot substitute for it.

A biological explanation should also leave room for the conditions in which people live. Treatment access, workload and practical support belong in the research design. Recording these factors does not diminish bodily symptoms; it helps avoid assigning every difficulty to an unmeasured molecule.

K2. What a null cognitive result leaves open

The trial estimated differences between treatment groups, rather than counting anyone who felt better as evidence that the drug worked. The intervals show why “not statistically significant” is different from proof of exactly zero effect.

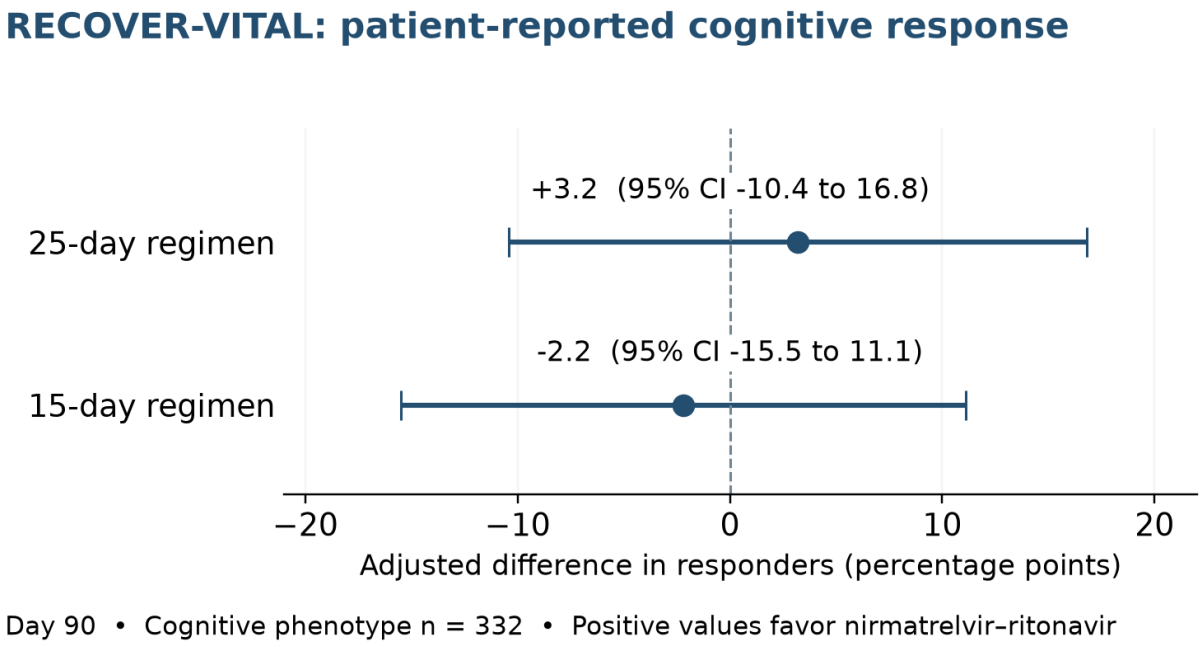


Figure 42. Adjusted responder differences versus ritonavir plus nirmatrelvir-matching placebo, with 95% CIs. Response meant at least a five-point improvement in the PROMIS Cognitive Function 8a T-score at day 90. Both intervals cross zero. This is a self-reported outcome, not a percentage cured. [190]

The implication for our hypothesis

When a trial finds little benefit, narrow the claim to the drug, population, regimen and outcome it actually tested. A proposed mechanism or a subgroup explanation needs a specific prediction and another test.

Someone may meet a goal more reliably after changing how they do the task. That improvement can matter while other cognitive difficulties persist. Test the proposed resilience index against a separate outcome from the one directly practised during treatment.

Source-checking note

The nine reports added during this stage of the source review received targeted source checks. RECOVER-VITAL was checked through its original abstract, investigator record and trial registry; the adult sleep study through its original abstract and publisher appendix. The sleep abstract lists 33 in its third arm, whereas the appendix lists 23, consistent with the total of 70; this paper uses the total and primary-comparison counts. Other additions used accessible original full text. The historical audit gaps in Appendix I remain unresolved. No new full audit of all references is claimed.

Appendix L. Prospective study methods and validation

STATE-ADHD-01 is an observational proposal with no enrolled participants, ethics approval or registered final protocol. The 14-day feasibility pilot would determine whether the proposed attention measure can support the conditional 28-day study. Document the endpoint, analysis and decision criteria before the relevant stage begins.

L1. Confirmatory question and independent decisions

Plan a sleep-only run-in before the 28-day main study. For each tested night, estimate usual sleep from the mean of at least 10 valid main sleep episodes in the 14 nights before it. Mark the exposure as missing when fewer than 10 nights are valid. Compare usual sleep with one hour below usual.

Use one additional miss per 60 valid targets as the provisional bound for the ADHD-minus-comparison contrast. This is exactly 100/60 percentage points, about 1.67. Apply symmetric positive and negative bounds. Ask participants and clinical and measurement reviewers what difference would justify a controlled follow-up given its burden. Review pilot reliability, floors and sleep coverage separately from that judgment. If the task cannot resolve the agreed bound, revise the task or design and recalibrate. A change in endpoint requires its own bound.

Decision rule for locking the attention bound

The panel receives participant feedback and blinded feasibility summaries. Keep all pilot estimates, plots and tests of the ADHD-by-sleep contrast withheld. Before reviewing any pilot contrast or recruiting for the main study, record the endpoint, exact bounds, rationale, reviewers, date and protocol version. L6 uses those locked bounds. L7 gives the handoff's separate prediction criterion.

Keep the two-to-four-hour post-rising window and set a local clock-time window before recruitment. Record actual test and wake times, study day and practice, treatment and dose timing, caffeine, pain, mood and workload. Section L6 separates primary adjustment from sensitivity analyses according to when each factor occurs.

Write the endpoint-selection rubric before the pilot. Distinguish invalid targets, invalid sessions and missed sessions and report them by group, device and prior sleep. If the panel selects PVT, freeze a compatible outcome, meaningful difference and analysis before the main study.

The main study would follow provisionally 200 adults for 28 observation days, evenly divided between the ADHD and comparison groups. After the feasibility pilot, the panel chooses the endpoint using the independent rubric and evidence about measurement quality. Keep the pilot contrast withheld during that choice.

Table L1. Endpoint decisions

Decision	Prospective consequence
Retain custom omissions	Require adequate nonzero variation, technical reliability and repeated-day interpretability. Pilot counts do not establish a 28-day effect size.
Select a brief PVT measure	Amend the protocol to fix one outcome, denominator, scoring rule, model family and missingness rule. Re-simulate with the new endpoint.
Neither task qualifies	Redesign the platform or task and repeat feasibility work before the main study.

A PVT lapse and a missed target from the custom task are distinct. The 355 ms threshold belongs to a studied brief-PVT implementation; it is not a universal threshold for every task or device. A reciprocal-response-speed endpoint requires a suitable continuous repeated-measures model, rather than an inherited binomial(60) likelihood. Brief and ten-minute PVT versions require agreement testing [M01, M02, M03].

L2. Instrument validity and physical timing

Technical verification, analytical validity, clinical relevance and user comprehension address different properties [M04]. A study of 26 smartphones reported combined display/touch delays of approximately 35–140 ms [M05]. Uncalibrated device differences could therefore mimic participant or day effects. A PVT calibration simulation proposed residual bias near ±5 ms and jitter SD ≤10 ms as stringent targets [M06]. These targets have not been demonstrated by the proposed platform.

Check when the stimulus actually appears and when a physical input registers. Use an external reference such as a photodiode and an instrumented response actuator. A high-speed camera can help if its measurement error is known. Use the app’s timestamps alongside those checks.

Table L2. Bench verification

Step	Required evidence
Supported device class	Device, CPU/GPU, display refresh, OS, browser/app version, input method, power and background settings.
Stimulus and response	Reference timestamps for physical screen change and button/touch actuation, linked to the app's recorded events.
Operating conditions	Repeat across alert/idle states, slow/fast intervals, background load, power conditions and orientations.
Timing distribution	Mean/median bias, SD, 95th/99th-percentile absolute errors, dropped frames, missing/duplicate inputs and version-specific anomalies.
Correction and lock	Apply only independently verified device-specific corrections; publish residual error, freeze builds and repeat checks after changes.

Use ±5 ms residual bias and ≤10 ms SD only as provisional PVT-specific acceptance targets, with long-tail errors reported separately. Prespecify bench counts, acceptable dropout and correction methods with the technical validator. A failing device class may be excluded prospectively or used only for activities that do not rely on reaction time. A pooled grand-average offset cannot solve variable latency across devices.

Assess comprehension, motor and screen access, accommodations, motion sensitivity, task fatigue and burden in adults with and without ADHD. Log these factors without treating difficulty using the device as a cognitive deficit. A platform that cannot establish timing suitability cannot support confirmatory reaction-time claims.

L3. Scoring and raw data

Table L3. Trial and session classification

Event	Required classification
Valid response within the task deadline	Hit; retain latency and trial ID.
No target response by deadline	Behavioural omission only when stimulus and input timing are technically valid.
No displayed stimulus or unusable timing	Technical failure, not a cognitive omission.
Early or repeated press	Separate false-start/duplicate categories under frozen task rules.
Response to a non-target	Commission error, distinct from omission.
Backgrounding, focus loss, lock or stall	Flag interruption and apply prespecified validity rules while retaining raw events.
Response precedes verified stimulus	Technical anomaly unless a valid early press meets the defined false-start rule.
No completed session	Missing observation; never encode as zero errors.

The custom task and PVT require separate scoring dictionaries and denominators. Verify valid-response thresholds, early presses, false starts, the selected brief-PVT lapse threshold, timeouts and duplicates against the chosen implementation. If all 60 custom targets were not validly presented, use a prospective session/exclusion rule rather than quietly changing a fixed binomial denominator.

The source protocol describes an eight-minute, 60-target custom task. The earlier scoring checklist includes a reaction time of at least 100 ms as a candidate valid-response boundary and a 355 ms brief-PVT lapse boundary. These are implementation-specific rules to verify against the selected task, not interchangeable thresholds or evidence that the platform has been validated [M01, M03].

Retain study and pseudonymous participant IDs, assessment day, session and trial IDs, trial type, task/software/algorithm versions, device and calibration IDs, OS/browser, UTC and local offset, sleep-episode label, scheduled and actual test time, intended and physically verified onset where available, app/response timestamps, raw reaction time, scoring status and technical-failure code. Keep raw events apart from derived scores and version the analysis. The final schema requires privacy and engineering review.

L4. One proposed feasibility schedule

The 14-day pilot has 40 adults, 20 with clinician-confirmed ADHD and 20 comparison participants, after two familiarization sessions. Participants continue usual medication, supplements and sleep-apnea care. Sleep variation is observed under usual conditions.

ADHD confirmation and comparison-group recruitment

For the ADHD group, verify a diagnosis documented by a qualified clinician or have a study clinician confirm it through a structured adult assessment covering developmental history and impairment. A questionnaire may guide screening. For the comparison group, require no known ADHD diagnosis or history and use a standardized ADHD screen and history review. Refer positive or unclear screens for qualified assessment before assigning either group.

Recruit both groups through overlapping channels where feasible, with the same age, consent and monitoring requirements. Record how each participant was recruited, their usual care, sleep disorders, work or shift schedule, habitual sleep and planned testing window. Report these characteristics by group so readers can judge whom the results represent. Stable comorbidities and ongoing treatment are documented as part of usual care.

At baseline, obtain consent and diagnostic/symptom information, medication and sleep-disorder history, device information and task practice. Confirm instructions and timestamps. The ten-minute reference subset may include up to 20 participants with counterbalanced sessions.

Table L4. Proposed observation schedule

Measurement	Timing and purpose
Actigraphy and sleep diary	Each main sleep episode: total sleep time, midpoint, wake after sleep onset, naps, wear quality and episode ambiguity. Record algorithm versions and manual resolutions; actigraphy does not diagnose apnea.
Morning state diary	Within one hour of final rising: grogginess, headache/neck pain, alertness, acute illness, medication/caffeine and work context. Update medication and caffeine timing at the test if either changed after the diary.
Three-minute PVT-B	Daily, two to four hours after rising, with scoring fixed before the pilot.
Custom 60-target task	Days 1, 3, 5, 7, 9, 11, 13 and 14; counterbalance order with PVT-B and allow appropriate rest.
Ten-minute reference PVT	Two preassigned sessions in the calibration subset; compare homologous metrics and their uncertainty.
Evening diary	Independent daily function, perceived effort and contextual updates. Before intended lights-out, record workload or stress, pain or illness, caffeine and departures from the usual medication schedule.

Eight custom sessions let the reviewer judge how precisely we can estimate zero scores, score variation and participant burden. Apply the same rule to the first valid session in the window. Flag tests outside it for sensitivity analysis. Record UTC, local offset, time awake and the participant's sleep episodes across midnight, daylight-saving changes and travel. Set the rules for naps and multiple sleep episodes before collecting data.

Aim for at least 16 of 20 people in each group to complete ten or more valid brief PVT sessions, and for at least 75% of actigraphy nights to be usable. For the custom task, aim for at least 16 of 20 per group with six valid sessions paired with prior-night actigraphy. L5 gives the floor and overlap rules.

L5. Measurement properties and the decision gate

Brief-PVT reliability and sensitivity have been examined under sleep loss and alcohol [M10]. Naturalistic virtual-reality work and cognitive momentary-assessment reliability research provide complementary measurement settings [M11, M12]. Their task-specific findings cannot validate the proposed custom omission instrument.

Table L5. Pilot reporting

Property	Report
Usability/adherence	Valid sessions by group, day and device; burden, adverse events and comprehension.
Floors/ceilings	Zero-omission sessions, saturation and score distributions by group, device, task variant and day.
Within-person variation	Person-centred distributions, stable-state repeatability, practice/day trends and within-/between-person variance.
Reference agreement	Paired short/long PVT metrics, difference plots and uncertainty; correlation alone does not establish interchangeability.
Responsiveness	Descriptive change with naturally varying sleep and timing, without selecting the strongest ADHD interaction.
Data quality	Timing anomalies, drift and missingness related to sleep, symptoms, burden or hardware.
Functional meaning	Independent evening function and participant feedback; vigilance does not measure every executive domain.

The reliability paradox concerns how consistently a task distinguishes people [M07]. Tracking one person over time has its own measurement error [194]. Assess repeatability when conditions are reasonably stable and responsiveness when conditions change [182]. Report variation in sleep, medication and caffeine timing, workload, time awake and adherence. Keep missing events marked as missing.

Choose an endpoint for the main study only after reviewing technical suitability, the score distribution, adherence, variation across repeated sessions and construct validity. A task may describe pilot performance while still requiring a revised endpoint or model. Redesign if scoring is unstable, timing cannot be defended or the floor is severe. Keep the pilot's group-by-sleep estimates hidden during endpoint selection.

With complete records and no run-in, pilot day 11 is the first day with enough prior nights to form a sleep reference. Report how many participants have that reference by day 14. For the descriptive pilot check, compare each eligible night with the participant's other valid pilot nights. Count participants with a pair within 30 minutes of usual sleep and another within 30 minutes of one hour below usual. Assign the shared boundary to the shorter band. The pilot check can use later nights. The main-study reference uses prior nights.

Decision rules for the custom attention task

Zero scores. Report zero-omission sessions out of valid custom sessions overall and by group. Within each group, show the participant-level distribution, the number who always score zero, and the zero rate by device and sleep band. Give denominators and participant-clustered uncertainty. If more than half of valid sessions score zero overall or in either group, revise the task and recheck before using omissions as the sole primary outcome. Passing this screen still leaves precision and score stability to be checked.

Sleep overlap. In each group, report participants with both a near-usual and a near-one-hour-short sleep–attention pair out of all enrolled participants, alongside the number of usable nights in each band. Retain the provisional target of at least 10 of 20 per group and report whether coverage depends on a few people, devices or testing times. Review the leave-one-night-out pilot check separately from coverage under a prior-nights-only reference. If the bands are poorly covered, revise monitoring or the run-in and recheck. If shared support remains inadequate, pause the one-hour comparison or prespecify a different contrast and recalibrate before recruitment.

Proceed only if the task clears timing, adherence and floor checks, both groups support the intended contrast and the planned estimator recovers known contrasts under pilot-informed simulations. Freeze tolerances for bias, interval coverage and decision precision before calibration. At a true difference of one extra miss per 60, check how often the interval excludes zero in the predicted direction. At zero, check how often it lies inside the negligible range. Use larger effects to check how often the interval clears the meaningful bound. Keep the pilot interaction hidden during endpoint selection. If the task or analysis changes, repeat calibration before choosing a sample size.

L6. Analysis of the sleep and attention association

Fit a mixed-effects logistic model to the count of omissions out of valid targets per session, with a participant intercept, ADHD status, nightly deviation from usual sleep, the group-by-deviation interaction and usual sleep duration. Use the prespecified evening covariates and scheduled study day described below. Apply the same valid testing windows in both groups. Use pilot simulations to check whether the estimator needs a random sleep slope or extra-binomial variation before freezing the model. Keep the conditional log-odds coefficient as a secondary diagnostic. [61, 62]

For each group, average model-predicted omission probabilities at usual sleep and at one hour below usual over the same prespecified covariate distribution. Give participants equal weight in that distribution.

Restrict it to the region supported by both groups at both exposure levels. Report sleep distributions, usable nights and participant coverage near each level. Subtract the comparison group's absolute change from the ADHD group's change. Report both changes, the contrast in percentage points and expected additional misses per 60, and a 95% interval from participant-level resampling within each group.

With adequate measurement and overlap, an interval entirely above the positive bound supports a practically meaningful difference in the predicted direction. An interval entirely within the symmetric bounds weakens that prediction for the measured task. A wide interval that spans negligible and meaningful differences leaves practical importance unresolved. An interval entirely below the negative bound supports a meaningful difference in the opposite direction. The contrast describes an observational association.

What we adjust for and why

Table L5A. Evening-to-morning timeline.

When	What we record	How it enters the analysis
Before sleep	Evening workload or stress, pain or acute illness, caffeine and changes from the usual medication schedule.	Primary adjustment. Each could influence both that night's sleep and next-day attention.
Overnight	Main sleep episode and deviation from usual duration, estimated from prior nights.	Exposure. Compare usual sleep with one hour below usual.
After waking	Grogginess, wake time, actual morning medication or caffeine taken before testing, and work demands that arise after waking.	Separate analyses. Some may be responses to the night's sleep.
At the attention test	Omissions out of valid targets, scheduled study day and actual test time.	Score valid tests in the planned windows. Adjust for scheduled study day to capture practice. Check actual timing separately.

A late deadline can shorten sleep and still affect what someone has to do the next morning. We record it before sleep and account for it in the primary comparison under usual care. We will fix the evening variables, windows and coding before the main-study analysis. Model comparisons use the same participant weights and the same range of observations supported by both groups.

On the same days, fit the model with and without the evening measures and report both estimates with their intervals. If the estimate gets smaller after adjustment, circumstances present before sleep may explain some of the association. A wider interval means we have less precision about the size of the difference.

One sensitivity model adds the test's actual clock time and time since waking. The next adds morning symptoms, medication, caffeine and demands recorded before the test. Refit the primary model on the same days for each comparison. Someone might reach for caffeine earlier after a short night, so adjusting for it can remove part of the morning response or distort the association. These comparisons show how much the estimate depends on those choices. A direct effect or mechanism would require a different design.

Sensitivity check for comparable schedules

Before reviewing attention contrasts, define joint overlap by habitual sleep duration and wake time, work or shift schedule, and the planned local testing window. Use schedule information from enrollment and the sleep-only run-in. Specify the schedule bands and minimum usable participant and night counts in each group at both sleep levels. Apply the same validity rules and repeat the primary contrast on the jointly supported subset, using one shared covariate distribution and equal participant weighting. Actual test time remains a separate sensitivity check.

Show who remains in the comparable-schedule analysis and how many nights contribute near each sleep level. Report its estimate and interval beside the primary result. A meaningfully smaller estimate with adequate precision would limit the claim for that subset. Sparse overlap or a wide interval leaves the schedule-specific question open.

Compare the prespecified linear sleep term with a smooth alternative within the supported range.

Tipping-point check for skipped morning tests

Use scheduled tests as the denominator. For each group, report counts and rates by prior-night sleep band and reason. Separate valid tests, skipped tests, invalid or out-of-window attempts, and technical failures. When a test is skipped, offer a brief reason field for illness or fatigue, time or competing demands, another reason, or no reason supplied. Report withdrawals separately. Include a separate row for nights with an unavailable sleep duration or usual-sleep reference. Those nights cannot be assigned to a sleep band from the observed data.

For scheduled mornings with known sleep exposure, predict expected misses per 60 from the primary model. On skipped mornings, add a hypothetical number of extra misses, keeping the adjusted expectation between 0 and 60. Vary this addition separately by group and by usual versus one-hour-short sleep, first one at a time and then jointly. Start with 0, 0.5, 1, 2, 3 and 4 additional misses and refine the grid near a change in conclusion. Use the estimated skip frequencies at the two supported sleep levels under the same participant weighting as the primary contrast. Technical failures and unknown reasons warrant separate scenarios. Assess missing sleep separately because the skipped-score check assumes the exposure is known.

Recompute the ADHD-minus-comparison contrast and its 95% interval at each setting, resampling participants within group and refitting the outcome and skip-frequency estimates. Report the smallest addition that changes the original interpretation under each one-at-a-time scenario, and the combinations that do so when assumptions vary jointly. Use the same meaningful and negligible bounds as the primary analysis. Before examining the main-study contrast, document which additions are plausible given pilot score variation, available test records and reasons for skipping. If any plausible scenario changes the conclusion, classify the result as inconclusive.

Hypothetical example. Suppose the 95% interval runs from 1.20 to 1.80 extra misses per 60, and 20% of the comparison group’s shorter-sleep mornings lack a score because the test was skipped. One extra miss on those mornings raises that group’s mean change by 0.20 and lowers the contrast by 0.20. Holding the interval width and skip fraction fixed for illustration, the lower limit reaches 1.00, so the interval is no longer wholly above the meaningful bound. The actual analysis would re-estimate the interval for each assumption.

L7. Secondary analysis of functional margin

Offer eligible participants the option to track one low-stakes task they already submit online, such as a regular work or class check-in. Use an existing deadline and a platform-generated submission timestamp. Record when the handoff becomes available and when it is due, with the morning attention assessment completed before the handoff opens. Collect timing and outcome metadata only. Classify on-time submissions, late submissions and confirmed non-submissions separately from cancelled tasks and unavailable logs.

For the optional handoff, aim for at least ten eligible people per group, 75% verifiable outcomes among scheduled handoffs after confirmed cancellations, and at least ten late or missed handoffs across at least five people. Inspect pairing with morning attention and available completion history. The core study can proceed independently of the handoff module.

To estimate time to spare, subtract the person's typical time from task opening to submission from the time available before the deadline. In the pilot, use the median of at least three earlier comparable, verified submissions, including late ones. Track earlier missed submissions separately. Keep a verified outcome even when the history is too short to estimate margin. A confirmed missed submission counts as an outcome. An unavailable log stays missing.

On held-out later days, test whether the same increase in morning omission probability is associated with a larger rise in late or missed handoffs at smaller margins. Compare a flexible baseline using morning attention, prior task performance, time available, sleep, medication and context with a model that explicitly represents margin and its interaction with attention. Use identical held-out days and history available before each prediction. Report calibration and Brier score, the mean squared error of the predicted probabilities. Prespecify the smallest worthwhile improvement in Brier score before the main study. A prediction gain would support margin as a useful summary of the same inputs. Keep this outcome exploratory if eligible handoffs or verified misses are too sparse.

L8. Historical simulations and current calibration

These synthetic checks used different assumptions and estimators. They show how the models behave under those assumptions. Estimating an observed ADHD effect or final study power requires separate evidence. The first used 140 replicates per cell, autoregressive sleep, participant intercept and slope variation, daily perturbations and roughly 73% session completion. Its cluster-robust binomial regression served as a GEE-like proxy for the planned GLMM.

Table L6. Initial proxy-model simulation

Total N	Baseline omissions	Interaction OR per +1 h	Detection
100	1%	0.80	64.3%
200	1%	0.90	37.9%
200	1%	0.80	88.6%
200	5%	0.90	47.1%
200	5%	0.80	98.6%
300	1%	0.90	54.3%

Null significance rates ranged from 5.7% to 8.6%. A subsequent non-adaptive-quadrature implementation exceeded 10% in one null scenario; its apparent power cannot support recruitment. The adaptive analysis used nine-point Gauss–Hermite quadrature for a binomial random-intercept logistic model and a nested-model likelihood-ratio test. Random sleep slopes and day-level extra-binomial variation remained unresolved.

Table L7. Adaptive-quadrature model checks

Scenario	Replicates	$p < .05$	Monte Carlo interval
Null, 1% omissions	1,000	4.7%	0.0352–0.0614
Null, 5% omissions	1,000	6.8%	0.0536–0.0849
OR .90, 1% omissions	1,000	63.6%	0.6058–0.6654
OR .90, 5% omissions	1,000	99.8%	0.9936–0.9996
Null plus daily noise	500	8.4%	0.0621–0.1107

The earlier hypothetical interaction odds ratios of .90 and .80 do not calibrate this absolute contrast. Use pilot estimates of floors, sleep overlap, within-person variability and missed sessions to simulate the locked analysis. Include zero group difference with unequal baseline omission rates, one extra miss per 60 and larger differences. Report coverage and the chances of ruling out or supporting a meaningful difference. Keep 200 as a planning placeholder until these checks inform the sample size.

The source analyses identify STATE_ADHD_01_AGHQ_Model_Checks_Nov8_2026.csv and STATE_ADHD_01_AGHQ_Simulation_Nov8_2026.py as their reproducibility files. Their reported outputs are preserved here; they have not been rerun or independently validated in this editorial pass.

Table L8. Required calibration work

Area	Required analysis
Independent implementation	Check data generation, quadrature, likelihood, parameter recovery, gradients, convergence and the inflated null rates.
Variance/dependence	Examine random intercepts/slopes, day effects, overdispersion and serial dependence.
Inference	Compare methods for estimating uncertainty on the standardized absolute contrast, including resampling at the participant level.
Missingness	Vary adherence, sleep-dependent missingness and missing-not-at-random assumptions; report bias and interval coverage.
Endpoint	Use a compatible likelihood for reciprocal reaction time or for the observed lapse opportunities and dispersion.
Sample size	Agree on the meaningful contrast before seeing subgroup effects; use pilot nuisance parameters and calibrated scenarios.

Before interpreting power, report null rejection, interval coverage, nonconvergence, bias, zero-error sessions and missing-data patterns. For each major null scenario, aim for at least 2,000 independent replicates for precision, up from the earlier minimum of 1,000. Keep the seeds and report Monte Carlo uncertainty. Obtain independent statistical and sleep-clinical review. Poor calibration means the analysis needs further work before confirmatory recruitment.

Synthetic check of unequal starting scores and sleep overlap

Each scenario uses 2,000 synthetic 28-day datasets with 100 people per group and 60 independent targets per valid session. At usual sleep, omission probabilities of 0.4% and 4% yield about 79% and 9% zero-omission sessions, respectively. The two-band check averages each paired participant’s change from usual to shorter sleep and compares group means with a Welch 95% interval. Risks are constant within each group and sleep band. Near-usual and shorter nights each occur on 25% of days, except in the sparse scenario, where only 0.6% of ADHD days are in the shorter band. Its purpose is to isolate the score scale and loss of overlap. The planned repeated-measures model still needs calibration using pilot estimates.

Table L8A. Synthetic contrasts in additional misses per 60 targets.

Synthetic scenario	True difference	Mean estimate	ADHD with both bands	95% interval above zero
Same odds multiplier, unequal starting risks	1.21	1.21	99.9%	100.0%
One extra at a common starting risk, lower ADHD starting risk	-1.10	-1.10	99.9%	0.0%
Zero absolute difference, unequal starting risks	0.00	-0.00	99.9%	1.8%
One extra, both groups have overlap	1.00	1.00	99.9%	100.0%
One extra, few shorter nights in ADHD	1.00	1.01	15.6%	44.8%

An identical 1.6-fold increase in odds produced a gap of about 1.21 misses per 60 because the ADHD group started at 4% and the comparison group at 0.4%. That is a real difference on the chosen absolute scale in the simulation. It does not distinguish differences in physiological sensitivity. In the second scenario, the stronger ADHD odds response would give a one-miss gap at a shared 4% starting risk. At the actual simulated ADHD starting risk of 0.4%, the absolute ADHD-minus-comparison gap instead reversed to about −1.10 misses.

With a true one-miss absolute gap and good pairing, every simulated interval excluded zero in the predicted direction. When shorter nights were rare in the ADHD group, only about 16% of that group contributed both bands and the interval excluded zero in the predicted direction in about 45% of runs. The average estimate stayed close to one. An effect exactly at the decision bound seldom produced an interval entirely above it, even with good overlap. These checks illustrate why the pilot must examine floors and shared sleep coverage before sizing the study. The reproducibility files are simulate_pilot_floor_overlap.py and STATE-ADHD-01-Synthetic-Floor-Overlap.csv.

L9. Governance, consent and reporting

Pilot recruitment requires appropriate research ethics review, informed consent, secure data handling, accessible instructions and a plan for incidental clinical disclosures and symptom referral. The Canadian TCPS 2 guidance addresses review of human pilot studies [M08]. Preparatory bench work without participants is a separate activity. STROBE provides reporting guidance for the observational study [M09].

Table L9. Decisions required before the main study

Responsible review	Evidence required; current status
Scientific and patient review	Meaningful question, endpoint and dated bound lock before pilot contrast review; open.
Clinical lead and ethics board	Eligibility, consent, privacy, referral and pilot oversight; not approved.
Technical measurement lead	Independent timing report, calibration IDs, QA and build lock; unverified.
Pilot investigators	Completed feasibility and technical report under approved procedures; not started.
Psychometric reviewer	Construct, score, stable-state reliability, floors and task-selection rubric; open.
Independent statistician	Endpoint-compatible model, dependence, type-I error and coverage; unresolved.
Protocol custodian	Dated schedule/endpoint amendment, analysis, exclusions, final sample size and preregistration; not completed.

Each decision record needs an owner, date, version and supporting evidence. No institutional or professional sign-off is implied here. The pilot cannot establish ADHD-specific physiological sensitivity, glymphatic impairment, treatment benefit or clinical validation. Its purpose is to establish whether a later test can be interpreted.

References

The 238 references link to original reports, evidence syntheses, official records or prescribing information. Published correction notices are linked alongside the relevant entries. Appendix I records the source checks and unresolved limits; Appendix J develops the provisional score; Appendix K compares long-term associations with trial outcomes. Literature approach and limits describes the review method. Matching a citation to a publication does not verify every claim made from it.

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Evidence register U01–U57

These identifiers retain their original meanings. Study populations and reports may overlap; 57 identifiers do not mean 57 independent experiments. The register preserves design, finding, limitation and model consequence for each source.

U01 Trial-level pupil dynamics reflect attention-related variability and medication-associated modulation in ADHD

Design: Direct ADHD | 9 Sep 2026; small observational/within-person ADHD study | Weight: moderate, exploratory. Finding: Pupil-response variability was associated with accuracy; a medication-condition comparison suggested lower variability on methylphenidate. Limitation: Small cohort; pupil responses are not specific to LC-norepinephrine, histamine or inflammation. Implication: Adds trial-level physiological variability as a candidate observable state measure.

Source: https://doi.org/10.1038/s41598-026-68871-2

10.1038/s41598-026-68871-2

U02 Transcranial photobiomodulation improves prefrontal oxygenation and impulse control in adults with ADHD

Design: Direct ADHD | 14 Jul 2026; randomized sham comparison | Weight: preliminary randomized. Finding: Some acute attention/inhibition outcomes and prefrontal fNIRS oxygenation measures favored stimulation. Limitation: Single-blind and short-term; changes in oxygenation do not establish mitochondrial mediation or lasting efficacy. Implication: Motivates a direct neurovascular/metabolic target-engagement test, without claiming clinical utility.

Source: https://doi.org/10.1186/s12888-026-08391-5

10.1186/s12888-026-08391-5

U03 Individualised Structural and Functional Frontostriatal Connectivity Mapping in Adult ADHD

Design: Direct ADHD | 1 Oct 2026; cross-sectional imaging, 40 ADHD and 36 controls | Weight: moderate observational. Finding: Frontostriatal functional differences were more apparent under working-memory demands than at rest. Limitation: Small and cross-sectional; structural findings less robust; no peripheral physiology manipulation. Implication: Makes task context a candidate state gate rather than treating resting connectivity as sufficient.

Source: https://doi.org/10.1002/hbm.70652

10.1002/hbm.70652

U04 Effects of External Trigeminal Nerve Stimulation on Neurocognitive Functioning in Children and Adolescents With ADHD

Design: Direct ADHD - negative | 28 Oct 2026; multicentre, double-blind sham-controlled RCT, n=150 | Weight: high for tested protocol. Finding: Four weeks of eTNS did not outperform sham on measured cognition, HRV or other autonomic outcomes. Limitation: Secondary analysis from a previously negative parent trial; specific device and protocol only. Implication: Directly constrains generalized nerve-stimulation claims: target engagement must be shown, not assumed.

Source: https://doi.org/10.1176/appi.ajp.20260002

10.1176/appi.ajp.20260002

U05 ALKS 7290 orexin-2 agonist ADHD Phase 1b company results

Design: Direct ADHD - early signal | 21 Sep 2026; 50 adults with ADHD, sponsor-reported phase 1b | Weight: company topline - provisional. Finding: Sponsor reported symptom and translational EEG/cognitive signals after two weeks. Limitation: Primarily a safety study; unreported full placebo-adjusted symptom and EEG outcomes; no peer-reviewed efficacy paper. Implication: Orexin is a candidate upstream arousal regulator; no established ADHD efficacy.

Source: https://alkermes.gcs-web.com/news-releases/news-release-details/alkermes-announces-positive-phase-1b-results-demonstrating

Source record

U06 The Association Between ADHD and Gastrointestinal Symptoms: A Systematic Review and Meta-Analysis

Design: ADHD comorbidity | 10 Sep 2026; 23 studies; >1.9 million participants | Weight: large synthesis, associational. Finding: ADHD was associated with greater functional GI symptom burden (overall OR 1.48). Limitation: Observational inputs; medication, behavioural and ascertainment pathways remain plausible. Implication: Measure GI/visceral symptoms as potential confounders or modifiers, not a microbiome diagnosis.

Source: https://doi.org/10.1177/10870547261469626

10.1177/10870547261469626

U07 Attention-Deficit/Hyperactivity Disorder and Late-Life Cognitive Impairment

Design: ADHD long-term boundary | 7 Oct 2026; EHR/genomic cohort, n=187,341 older adults | Weight: large cohort, associational. Finding: ADHD diagnosis and polygenic liability were associated with later MCI/dementia recording. Limitation: Residual confounding, reverse causation and diagnosis timing; no state-change or reserve mechanism tested. Implication: Keeps long-term reserve separate from within-day regulation, avoiding causal overinterpretation.

Source: https://doi.org/10.1001/jamapsychiatry.2026.3165

10.1001/jamapsychiatry.2026.3165

U08 Voluntary attention regulates acute immune responses in humans

Design: Bidirectional neuroimmune | 17 Aug 2026; three preregistered within-person histamine skin tests, total n=57 | Weight: human experimental, limited transfer. Finding: Directing attention inward altered the wheal response and some autonomic measurements. Limitation: Exogenous local histamine in healthy volunteers; no evidence of altered mast-cell degranulation or ADHD cognition. Implication: Allows a reciprocal attention-to-peripheral-response arrow, conditional on context.

Source: https://doi.org/10.1038/s41562-026-02541-1

10.1038/s41562-026-02541-1

U09 Histamine H2 receptor in substantia nigra pars reticulata parvalbumin positive neurons mediates risk-taking behavior

Design: Histamine circuit - animal | 19 Sep 2026; mouse neuronal and projection perturbations | Weight: preclinical. Finding: Local H2R function altered risky exploration, with rescue in a DAT+/- mouse model. Limitation: Not an ADHD human trial; no link from systemic allergy/mast cells to the circuit. Implication: Central histamine must be distinguished by receptor, cell type and circuit from peripheral immune histamine.

Source: https://doi.org/10.1038/s41380-026-03892-x

10.1038/s41380-026-03892-x

U10 Deletion of N-ethylmaleimide-sensitive factor in dopamine D2 receptor-expressing cells

Design: Catecholamine circuit - animal | 15 Sep 2026; D2R-lineage conditional-knockout mouse experiment | Weight: preclinical. Finding: NSF deletion disrupted striatal development and produced hyperactivity/impulsivity-like phenotypes. Limitation: Severe developmental manipulation; no dynamic metabolic-state or clinical ADHD test. Implication: Receptor/circuit integrity is distinct from neurotransmitter amount; do not extrapolate NSF to illness-state ADHD.

Source: https://doi.org/10.1038/s41386-026-02526-8

10.1038/s41386-026-02526-8

U11 High-resolution spatial transcriptomics reveals fibroblast and neuroimmune microenvironments in endometriosis lesions

Design: Endometriosis neuroimmune | 9 Oct 2026; human lesion mapping and patient-derived models | Weight: human mechanistic tissue. Finding: Neural, fibroblast and immune cell populations formed spatially distinct microenvironments. Limitation: Does not measure executive function or systemic state effects on cognition. Implication: Supports localized neuroimmune and pain signalling as distinct candidate inputs, not generalized mast-cell causation.

Source: https://doi.org/10.1038/s41467-026-78453-5

10.1038/s41467-026-78453-5

U12 LUNA phase 3 cycle-timed atogepant for menstrual migraine

Design: Menstrual migraine - early trial | 10 Sep 2026; 468 participants, sponsor topline of double-blind trial | Weight: company topline - provisional. Finding: Perimenstrual migraine days improved more than placebo (1.20 vs 0.40 reduction); cognitive secondary improvement reported. Limitation: Cognitive effect size not yet fully published; not an ADHD/endometriosis trial. Implication: Treat cycle-linked migraine as a defined CGRP pathway and potential repeated-state test.

Source: https://news.abbvie.com/2026-09-10-AbbVie-Extends-Migraine-Leadership-with-Positive-Phase-3-Atogepant-Results-in-Menstrual-Migraine

Source record

U13 Latent brain-cognition dimension indicative of neuroinflammatory basis of Long COVID

Design: Long COVID - MRI | 19 Sep 2026; cross-sectional quantitative MRI/cognition, n=77 Long COVID and 32 controls | Weight: moderate observational. Finding: Free-water and neurite-density features contributed to a joint brain-cognition group axis. Limitation: Not a direct measure of inflammation or BBB leakage; small controls, no longitudinal causality. Implication: Neurovascular/extracellular state is a candidate intermediate, not a proven cause.

Source: https://doi.org/10.1093/braincomms/fcag344

10.1093/braincomms/fcag344

U14 Loss of vesicular monoamine transporter 2 in striatum of Long COVID and relationship to neuropsychiatric symptoms

Design: Long COVID - dopamine PET | 2026; PET case-control study, 24 Long COVID and 24 matched controls | Weight: small human imaging. Finding: Lower striatal VMAT2 radioligand binding correlated with selected motivation, motor and memory symptoms. Limitation: VMAT2 PET cannot distinguish permanent terminal loss from functional downregulation; cross-sectional. Implication: Supports a plausible downstream neuromodulatory interface; not evidence that Long COVID is ADHD.

Source: https://doi.org/10.1016/j.ebiom.2026.106339

10.1016/j.ebiom.2026.106339

U15 A cognitive architecture for Long COVID: the interplay of brain fog, fatigue, attention and memory

Design: Long COVID - large cognitive cohort | 24 Sep 2026; community cognitive cohort, n=24,095, with 548 ongoing Long COVID | Weight: large, observational. Finding: Fatigue and attention/memory abnormalities made partially independent contributions to brain-fog reports. Limitation: Observational statistical pathways, no biomarkers or causal interventions. Implication: Separates subjective fatigue, objective capacity and their interacting routes.

Source: https://doi.org/10.1093/braincomms/fcag361

10.1093/braincomms/fcag361

U16 Somatosensory processing in long COVID fatigue and its relations with physiological and psychological factors

Design: Long COVID - negative mechanism | 25 Sep 2026; preregistered-style cognitive/physical exertion comparison, n=44+44 | Weight: moderate negative observational. Finding: Long COVID showed more fatigue and lower resting HRV, without worse immediate objective fatigability; HRV did not predict fatigue. Limitation: May miss delayed PEM; HRV is an incomplete autonomic measure. Implication: Do not equate low HRV, reported fatigue and immediate task performance.

Source: https://doi.org/10.1113/EP093600

10.1113/EP093600

U17 Comparison of new and recurring post-acute infection syndrome – German DigiHero cohort

Design: Long COVID - recovery history | 12 Sep 2026; cross-sectional retrospective survey of 2,801 PAIS cases | Weight: large observational. Finding: A new episode during incomplete recovery was associated with more symptoms and impairment. Limitation: Recall/selection bias and nonrandomized exposure; no biomarker of accumulated load. Implication: Requires pre-challenge recovery status to be measured prospectively, not inferred after the fact.

Source: https://doi.org/10.1038/s41598-026-70451-3

10.1038/s41598-026-70451-3

U18 Timing of Long COVID improvement in prospectively followed RECOVER-Adult participants

Design: Long COVID - longitudinal history | 3 Nov 2026; prospective cohort, n=1,019, 83 US sites | Weight: large prospective cohort. Finding: Approximately half had sustained symptom-index improvement by three years and ~27% complete index remission. Limitation: Self-reported symptom index; does not directly measure objective attention or physiological recovery. Implication: Treat trajectory and endotype as separate measurable outcomes.

Source: https://doi.org/10.7326/ANNALS-25-05275

10.7326/ANNALS-25-05275

U19 RECOVER-VITAL extended nirmatrelvir-ritonavir for Long COVID

Design: Long COVID - negative RCT | 31 Aug 2026; randomized double-blind trial, n=959 analysis | Weight: high randomized, intervention-specific. Finding: Neither 15 nor 25 days of antiviral improved cognitive, autonomic or exertional phenotype outcomes on average. Limitation: Participants not selected for drug-susceptible viral persistence; does not rule out reservoirs or subgroups. Implication: Separates initiating infection from downstream maintainers; no universal antiviral-rescue prediction. Deduplication note: ALREADY IN BASE MASTER: reference [190]; index only

Source: https://doi.org/10.1016/S1473-3099(26)00406-8

10.1016/S1473-3099(26)00406-8

U20 SARS-CoV-2-specific monoclonal antibody AER002 in Long COVID: outSMART-LC

Design: Long COVID - negative RCT | 30 Sep 2026; randomized placebo-controlled phase 2a, n=36 | Weight: small randomized negative. Finding: No significant overall advantage on reported cognitive or functional outcomes. Limitation: Small trial, variant-dependent potency and post-hoc subgroup signals. Implication: Adds independent negative treatment evidence, not a proof that viral persistence never occurs.

Source: https://doi.org/10.1038/s41467-026-77925-y

10.1038/s41467-026-77925-y

U21 Autoantibody landscapes in Long COVID with neurological symptoms show heterogeneity without a shared disease signature

Design: Long COVID - negative immunity | 29 Sep 2026; two independent cross-cohort CSF/serum analyses | Weight: moderate mechanistic negative. Finding: No reproducible CNS-directed autoantibody signature separated neurological cases or cognitive subgroups. Limitation: Limited sensitivity for certain antibodies and subtypes; cannot exclude all autoimmunity. Implication: Weakens a universal autoantibody explanation, favors explicit heterogeneity.

Source: https://doi.org/10.1172/jci.insight.207122

10.1172/jci.insight.207122

U22 Microtesla magnetic therapy for cognitive impairment in Long COVID

Design: Long COVID - exploratory treatment | 28 Sep 2026; sham-controlled randomized pilot, n=30 | Weight: exploratory randomized. Finding: Selected objective cognitive endpoints favored active treatment in exploratory analysis. Limitation: Very small, multiple outcomes and uncertain target mechanism; independent replication required. Implication: Suggests a downstream neural intervention point, not proof of inflammatory correction.

Source: https://doi.org/10.1093/braincomms/fcag364

10.1093/braincomms/fcag364

U23 EBV reactivation priming of the peripheral immune system in multiple sclerosis relapse

Design: MS - longitudinal immune state | 16 Sep 2026; longitudinal pre-relapse observational immune profiling | Weight: mechanistic observational. Finding: A monocyte/B-cell transcriptional state associated with EBV activity preceded relapses by months. Limitation: Does not prove EBV triggered the event or establish an ADHD analogy. Implication: Supports prospective sampling of transitions rather than reliance on one blood draw.

Source: https://doi.org/10.1038/s41591-026-04665-3

10.1038/s41591-026-04665-3

U24 Intracellular complement factor H protects neurons during CNS inflammation

Design: MS - cellular resilience | 2 Sep 2026; human donor tissue plus cellular and animal experiments | Weight: human/preclinical mechanistic. Finding: CFH-related programs were associated with neuronal stress tolerance during inflammation. Limitation: Clinical cognitive consequences and short-timescale ADHD state effects not tested. Implication: Treat cellular stress tolerance as possible intermediate modifier distinct from systemic load.

Source: https://doi.org/10.1038/s41586-026-10981-y

10.1038/s41586-026-10981-y

U25 Fibronectin mediates APOE4-driven blood-brain barrier dysfunction in Alzheimer disease

Design: BBB - glial interface | 11 Sep 2026; human tissue plus experimental BBB models | Weight: human tissue and preclinical. Finding: Astrocyte-derived fibronectin contributed to barrier dysfunction and experimental rescue. Limitation: AD-specific context; no living-human intervention on moment-to-moment cognition. Implication: BBB interface is actively regulated; do not equate this with evidence of ADHD barrier leakage.

Source: https://doi.org/10.1038/s43587-026-01204-0

10.1038/s43587-026-01204-0

U26 Central nervous system effects of GLP-1 receptor agonists: systematic review of human neuroimaging

Design: GLP-1 brain state | Online 19 Sep 2026; systematic review of 37 human studies | Weight: moderate synthesis. Finding: Distributed neural effects appear to vary with metabolic phenotype and exposure duration. Limitation: Heterogeneous, often small studies; neural signal does not imply cognitive benefit. Implication: Treat incretin modulation as state-/time-dependent, not a universal brain enhancer.

Source: https://doi.org/10.1016/j.neubiorev.2026.106990

10.1016/j.neubiorev.2026.106990

U27 GLP-1 receptor agonists in Alzheimer disease and MCI: a meta-analysis

Design: GLP-1 - negative cognition | 15 Oct 2026; 7 RCTs, n=4,132 | Weight: high randomized synthesis, disease-specific. Finding: No significant average benefit on cognition or CDR-SB in Alzheimer/MCI. Limitation: Dominated by semaglutide Alzheimer trials; indirect to ADHD and short-timescale attention. Implication: Places a strong negative constraint on broad GLP-1-to-cognition claims. Deduplication note: NEW SYNTHESIS; largely overlaps base reference [56], not independent trial evidence

Source: https://doi.org/10.1002/trc2.70334

10.1002/trc2.70334

U28 CREST-2 revascularization of asymptomatic carotid stenosis: cognitive results

Design: Vascular - negative cognition | 16 Sep 2026; randomized secondary cognitive outcomes, n=2,165 | Weight: high randomized, population-specific. Finding: Revascularization added no measurable cognitive advantage over medical management. Limitation: Chronic carotid disease in older adults differs from dynamic ADHD perfusion changes. Implication: An upstream flow change does not guarantee improved cognition.

Source: https://doi.org/10.1001/jama.2026.14703

10.1001/jama.2026.14703

U29 CREST-H hemodynamic impairment substudy of CREST-2

Design: Vascular - negative moderator | 16 Sep 2026; prespecified randomized substudy, 392 enrolled, 291 with complete data; 51 in the baseline-impaired primary subgroup | Weight: moderate randomized substudy. Finding: Baseline hemodynamic impairment did not detectably modify cognitive benefits of revascularization. Limitation: Small cognitively impaired subgroup; indirect to orthostatic/cerebrovascular state. Implication: Do not infer cognitive rescue from a perfusion proxy alone.

Source: https://doi.org/10.1001/jama.2026.13384

10.1001/jama.2026.13384

U30 Vasomechanic treatment is associated with increased efflux of human brain proteins

Design: Neurovascular fluid - human intervention | 3 Nov 2026; crossover human physiological study, 29 overall; 26 in the combined condition | Weight: moderate mechanistic human. Finding: NO plus patterned vibration altered brain-fluid imaging and circulating Aβ/GFAP. Limitation: No demonstrated brain amyloid reduction, cognition benefit, or direct proof of net clearance; NO lowered BP. Implication: Broaden neurovascular state to include pulsatility and fluid transport as candidate processes, not treatment advice.

Source: https://doi.org/10.1038/s41467-026-78894-y

10.1038/s41467-026-78894-y

U31 Joint impact of pathological burden and cognitive resilience on Alzheimer disease risk

Design: Cognitive resilience boundary | 11 Sep 2026; longitudinal older-adult cohort, n=3,119 | Weight: large longitudinal, indirect. Finding: A residual-defined resilience score modified dementia risk alongside pathology. Limitation: Statistical residual does not identify a molecular mechanism; aging timescale, not ADHD fluctuations. Implication: Distinguishes pathological load from functional outcome without assuming a common cognitive-reserve scale.

Source: https://doi.org/10.1038/s41591-026-04635-9

10.1038/s41591-026-04635-9

U32 Emotion regulation capacity lacks a shared neural signature

Design: Measurement - strong negative | 28 Oct 2026; preregistered 40-dataset consortium, n=2,175 | Weight: large preregistered synthesis. Finding: Common reappraisal measures did not converge; canonical prefrontal activation failed to predict capacity. Limitation: Emotion regulation is not identical to attention or autonomic resilience. Implication: Biomarker change, neural engagement and effective regulation require separate validation.

Source: https://doi.org/10.1038/s41586-026-11203-1

10.1038/s41586-026-11203-1

U33 Magnetoencephalography oscillation-based functional connectivity identifies depression phenotypes

Design: Measurement - neural heterogeneity | 31 Aug 2026; cross-sectional MEG study, 263 MDD and 75 controls | Weight: exploratory adjacent condition. Finding: Data-derived oscillatory brain-symptom dimensions suggested clinically heterogeneous configurations. Limitation: Clusters not independently validated; not longitudinal and not ADHD-specific. Implication: Illustrates subgroup-discovery risk; do not name new permanent ADHD types from a single cluster analysis.

Source: https://doi.org/10.1038/s44220-026-00723-4

10.1038/s44220-026-00723-4

U34 Functional architecture of the human insula

Design: Interoception circuit | 21 Oct 2026; intracranial recordings/stimulation in 87 epilepsy-care participants | Weight: human mechanistic, indirect. Finding: Four functional insular territories and directed interactions help map bodily sensation and salience. Limitation: Neural mapping is not proof of changed peripheral inflammation or ADHD functioning. Implication: Adds specificity to brain-body signal processing without collapsing perception into physiology.

Source: https://doi.org/10.1038/s41586-026-11175-2

10.1038/s41586-026-11175-2

U35 Cognitive state dependent enhancement of cognitive control with transcranial magnetic stimulation

Design: State-gated intervention | 3 Oct 2026; 23 human participants with prefrontal rTMS and EEG | Weight: small human experimental. Finding: Cognitive-control improvement occurred when relevant circuits were engaged during stimulation, not low-demand control. Limitation: Small non-ADHD experiment; no immune, vascular or metabolic state manipulation. Implication: Intervention response should be modelled as interaction with engaged circuit state.

Source: https://doi.org/10.1016/j.brs.2026.103227

10.1016/j.brs.2026.103227

U36 Fatigue and processing speed predict subsequent fatigue override

Design: Dynamic compensation | 18 Sep 2026; coordinated six-study EMA, n=1,088 | Weight: large within-person observational. Finding: Higher-than-usual fatigue and unusually fast performance predicted reduced later fatigue override. Limitation: Associational, small effects, one processing-speed family; cannot prove energy depletion. Implication: Observed performance may be preserved temporarily despite rising perceived effort.

Source: https://doi.org/10.1038/s44271-026-00537-1

10.1038/s44271-026-00537-1

U37 Neuropharmacology of language: ERP evidence for catecholaminergic modulation of syntactic processing

Design: Catecholamine computation | 22 Oct 2026; human pharmacology/EEG experiment | Weight: experimental human, indirect. Finding: Methylphenidate altered task-specific syntactic-processing EEG responses and their dependence on capacity. Limitation: Not an ADHD or systemic-physiology challenge; task-specific readout. Implication: Emphasizes context- and capacity-dependent neuromodulation rather than global neurotransmitter levels.

Source: https://doi.org/10.1073/pnas.2606806123

10.1073/pnas.2606806123

U38 Gene-stress-epigenetic regulation of FKBP5: clinical and translational implications

Design: Stress regulation - historical | 2015 online review; brought into this project during autumn 2026 | Weight: contextual, not new evidence. Finding: FKBP5 literature links glucocorticoid-response sensitivity and some human cortisol recovery differences. Limitation: Old review, mixed observational and mechanistic evidence; no demonstrated ADHD histamine bridge. Implication: Provides historical mechanism for regulatory feedback and recovery, not a newly discovered cause. Deduplication note: HISTORICAL 2015 REVIEW; contextual, not a new 2026 study

Source: https://doi.org/10.1038/npp.2015.235

10.1038/npp.2015.235

U39 Fang et al. (2025), adult ADHD perivascular MRI

Design: Direct ADHD; cross-sectional, 41 ADHD / 108 controls | Weight: Moderate observational

Finding: Lower ALPS in ADHD; ALPS related to visual-memory performance; other MRI indicators not consistently different.

Limitation: ALPS is indirect; medication not fully controlled; no within-person state test.

Implication: Retain as suggestive neurovascular phenotype, not proven clearance deficit.

Source: https://doi.org/10.1016/j.jad.2025.03.059

10.1016/j.jad.2025.03.059

U40 Tian et al. (2026), pediatric multimodal MRI

Design: Direct ADHD; cross-sectional, 80 ADHD / 110 controls | Weight: Moderate observational

Finding: Lower ALPS (1.42 vs 1.51); gBOLD-CSF coupling did not differ significantly.

Limitation: Cross-sectional; divergence between imaging markers weakens a single-factor explanation.

Implication: Prespecify distinct ALPS, coupling and clinical endpoints.

Source: https://doi.org/10.1016/j.pscychresns.2026.112198

10.1016/j.pscychresns.2026.112198

U41 Hu et al. (2026), ADHD neurovascular coupling

Design: Direct ADHD; cross-sectional, 66 ADHD / 41 controls | Weight: Moderate observational

Finding: Reduced CBF-fALFF coupling and ALPS; associations with executive measures and sleep duration.

Limitation: Small cohort, imaging surrogates, no direct experimental transport or clinical causation.

Implication: Prioritize coupling and perfusion as separate candidate features.

Source: https://doi.org/10.1186/s12888-026-08233-4

10.1186/s12888-026-08233-4

U42 Zhang et al. (2026), pediatric ADHD frontothalamic MRI

Design: Direct ADHD; cross-sectional, 159 ADHD / 163 controls | Weight: Moderate observational

Finding: Lower ALPS and fronto-thalamic alterations with exploratory statistical mediation of cognitive measures.

Limitation: Regression mediation does not show causal mediation; ALPS specificity unresolved.

Implication: Replicate with longitudinal individual physiology and anatomical controls.

Source: https://doi.org/10.1093/cercor/bhag099

10.1093/cercor/bhag099

U43 Wright (2026), anatomical determinants of ALPS

Design: Human measurement methods, n=1,000 across two HCP cohorts | Weight: Strong methodological constraint

Finding: Region selection, ventricular anatomy and tissue deformation substantially influenced ALPS; median left-right asymmetry 12.5%.

Limitation: Healthy reference cohorts do not settle disease-specific transport mechanisms.

Implication: Downgrade ALPS from clearance biomarker to anatomically sensitive diffusion surrogate.

Source: https://doi.org/10.1162/IMAG.a.1399

10.1162/IMAG.a.1399

U44 Acute stimulant neuroimaging (2026)

Design: Human acute medication study, 36 ADHD / 42 controls | Weight: Human experimental confounder

Finding: Stimulant exposure changed apparent structural/diffusion MRI measures within 1-2 hours; HR/BP rose.

Limitation: Not an ALPS study; imaging changes may have several physiological or technical explanations.

Implication: Record medication timing and physiological covariates for imaging and daily models.

Source: https://doi.org/10.1002/brb3.71606

10.1002/brb3.71606

U45 Lee et al. (2025), sleep apnea and four-year change

Design: Human prospective cohort, n=1,110, PSG/MRI/cognition at two time points | Weight: Large longitudinal, indirect

Finding: AHI change tracked ALPS change and visual-memory change; modeled indirect associations.

Limitation: Observational mediation, residual confounding and surrogate marker; not ADHD.

Implication: Supports repeated physiology-cognition study but not a glymphatic causal pathway.

Source: https://doi.org/10.1164/rccm.202411-2221OC

10.1164/rccm.202411-2221OC

U46 Bezerra et al. (2026), apnea and attention

Design: Human PSG / attention study, 84 OSA / 22 controls | Weight: Moderate cross-sectional

Finding: Sleep-stage shifts predicted reaction-time slowing in final model; AHI and hypoxemia measures were not retained.

Limitation: Cross-sectional; small sample; backward selection and different cognitive endpoints may matter.

Implication: Measure fragmentation, hypoxic burden and sleep opportunity separately.

Source: https://doi.org/10.3390/jpm16020117

10.3390/jpm16020117

U47 APPLES arousal-threshold secondary analysis (2025)

Design: Secondary moderator analysis of sham-controlled RCT, n=833 OSA | Weight: Randomized parent; secondary moderator

Finding: Higher estimated arousal threshold modified CPAP benefit on executive function (interaction p=0.042); not attention/memory.

Limitation: Post hoc phenotype assessment; not ADHD; moderator not yet prospectively validated.

Implication: Explicitly model physiological response heterogeneity, with correction for multiple tests.

Source: https://doi.org/10.1183/13993003.01183-2024

10.1183/13993003.01183-2024

U48 Sleep-to-blood amyloid/tau human crossover (2026)

Design: Randomized crossover, n=39, normal sleep versus deprivation | Weight: Human intervention, mechanistic

Finding: Morning plasma A-beta/tau patterns and model consistent with sleep-dependent transport.

Limitation: Plasma ratios and compartment modelling are not direct quantification of net parenchymal removal.

Implication: Separates transported proteins from the clinical cognitive endpoint.

Source: https://doi.org/10.1038/s41467-026-68374-8

10.1038/s41467-026-68374-8

U49 Levitt et al. (2026), sleep slow waves and CSF

Design: Human closed-loop auditory stimulation during EEG/fMRI | Weight: Human mechanistic experiment

Finding: Phase-timed auditory stimulation increased slow waves and CSF flow-wave responses.

Limitation: Measures CSF motion, not clearance or cognition; small mechanistic study.

Implication: Establishes manipulable flow-related physiology without implying clinical benefit.

Source: https://doi.org/10.1126/scitranslmed.aea5469

10.1126/scitranslmed.aea5469

U50 Hauglund et al. (2025), NE-driven vasomotion

Design: Animal mechanistic sleep study | Weight: Strong animal mechanism

Finding: Norepinephrine oscillations, blood-volume and CSF rhythms covaried; manipulations changed transport.

Limitation: Mouse work; no immediate ADHD mechanism or treatment extrapolation.

Implication: Add oscillatory catecholamine timing as candidate sleep-state mechanism.

Source: https://doi.org/10.1016/j.cell.2024.11.027

10.1016/j.cell.2024.11.027

U51 Peng et al. (2026), intermittent hypoxia / adenosine

Design: Male-mouse intermittent-hypoxia intervention | Weight: Animal intervention

Finding: Chronic hypoxia altered adenosine transport, AQP4 and tracer dynamics; ENT-pathway manipulation modified effects.

Limitation: Species/sex and exposure model; does not demonstrate benefit in human apnea or ADHD.

Implication: Add adenosine as an exploratory link between hypoxia, energy and fluid regulation.

Source: https://doi.org/10.1093/sleep/zsag182

10.1093/sleep/zsag182

U52 Miao et al. (2024), contrary sleep-clearance finding

Design: Animal experimental fluorescent-tracer clearance study | Weight: Material contradictory evidence

Finding: Reported decreased rather than increased clearance under sleep and anesthesia.

Limitation: Tracer, method and compartment specificity may explain conflicts; actively debated.

Implication: Do not treat the direction of sleep-related net clearance as universally settled.

Source: https://doi.org/10.1038/s41593-024-01638-y

10.1038/s41593-024-01638-y

U53 Pataky et al. (2026), insulin resistance and brain function

Design: Human randomized metformin/placebo component, n=40; separate mouse mechanism | Weight: Small human RCT + animal

Finding: Metformin was associated with preserving regional brain glucose uptake/connectivity and improved processing-speed/working-memory composite.

Limitation: Small older metabolic sample; brain ATP mechanism from mice; not ADHD or glymphatic.

Implication: Strengthen metabolic-state pathway separately from neurofluid speculation.

Source: https://doi.org/10.1172/JCI207388

10.1172/JCI207388

U54 Badulescu et al. (2025/26), semaglutide cognition

Design: 16-week randomized placebo-controlled MDD trial, n=72 | Weight: Randomized negative primary

Finding: Did not improve the prespecified executive-function primary outcome; a preplanned secondary analysis favored semaglutide on global cognition. This secondary result does not reverse the primary null result.

Limitation: Different clinical population; not evidence of no effect in all endotypes.

Implication: Preserve negative primary outcomes when discussing GLP-1 and cognition.

Source: https://doi.org/10.1016/j.medj.2025.100916

10.1016/j.medj.2025.100916

U55 Bavisant H3 antagonist for adult ADHD (2012)

Design: Six-arm double-blind RCT, 430 randomized | Weight: Historical negative ADHD RCT

Finding: H3 antagonist not significantly superior to placebo on ADHD-RS; active treatments separated.

Limitation: One compound/dose range; central H3 pharmacology not equivalent to peripheral H1 allergy treatment.

Implication: Use as direct counterexample to simple histamine-targeting narratives.

Source: https://doi.org/10.2165/11631990-000000000-00000

10.2165/11631990-000000000-00000

U56 Baser et al. (2026), 105 human neurofluid studies

Design: Scoping review through July 2026; 105 studies | Weight: Broad human evidence map

Finding: Only five studies used direct/near-direct transport measures; most used diffusion/structural proxies.

Limitation: No formal risk-of-bias synthesis or pooled estimate; categories overlap.

Implication: Add explicit outcome-measurement hierarchy and standardized reporting.

Source: https://doi.org/10.1111/ejn.70682

10.1111/ejn.70682

U57 ACX-02 combined pharmacology (2026 preprint)

Design: Sponsor-linked human preprint; small sequential placebo/active conditions | Weight: Preprint: provisional

Finding: Reports changes in slow-wave activity and modeled plasma markers during monitored sedation.

Limitation: Not peer-reviewed; fixed treatment order, model-dependent outcomes and financial conflicts.

Implication: Track for independent replication only; do not classify as established brain clearing.

Source: https://www.medrxiv.org/content/10.64898/2026.03.10.26348048v1

Source record

Additional evidence and methods references

These identifiers preserve the source drafts' citation scheme rather than enumerate independent studies. M07 repeats original reference [63]; N04 repeats [216]; N05 repeats [226]. RECOVER-VITAL also appears under [190] and [U19]. Related reports, corrections and pooled analyses must be distinguished from independent participant samples.

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Author and research transparency

Author: Alex Yesilcimen. Article type: hypothesis/perspective with targeted narrative synthesis. No original participant dataset, new meta-analysis or conducted intervention is reported. Proposed studies are not registered protocols. The NRI is an unvalidated research proposal; its worked example contains invented values. Figures 1, 2, 7, 13, 30, 31 and 40 and Visual guides A, C and D are hypothetical or conceptual. Other figures show attributed study results or labeled calculations. Figures 32 and 33 describe sample inclusion and completion, not clinical outcomes. AI assistance supported literature organization, source checking, editorial development and figure production. This is not independent peer review.

The source audit distinguishes verified numerical claims from retained narrative material and remaining access limits. The adult waking-EEG methods and exact regional spectroscopy effects still require fuller review. Funding, conflicts of interest, affiliation and correspondence require author confirmation before submission. No clinical or institutional endorsement is implied.

This full integrated edition was prepared on November 9, 2026. It incorporates the illustrated foundation, the November evidence register, the targeted evidence-gap review and the study-methods work into one reading sequence. Research dates remain attached to sources. Dated insertion labels and repeated version instructions have been removed; substantive competing results and provisional methods are retained.